Light Collection in a Polished Plastic Scintillator Read Out by Side-Mounted SiPMs: a Geant4 Matrix Simulation Study

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Abstract

We describe a Geant4 (v11.3.2) simulation of a 250 mm×250 mm×10 mm polished plastic scintillator (PVT, UPS-92S/EJ-200 class) read out by side-coupled silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)], and ask how far it reproduces the *measured* light-collection behaviour of four surface treatments.

The physics question is sharper than it first appears, because the four treatments differ in *how* they return light to the plate: the paint returns it diffusely, the two mirrors specularly, and the bare plate returns none at all. A polished plate facing a black holder traps light by total internal reflection over a cone of half-angle 39.3°, but every photon outside that cone dies on the black wall, whereas the two mirrors keep the same cone and hand the escaped fraction back with a reflectivity of 0.90–0.96. Even in the idealised model the bare plate is therefore dimmer than both mirrors—only the diffuse paint is dimmer—and in the measurement it is the dimmest of the four. We resolve that inversion with two calibrated surface parameters—a mean-field plate–holder contact fraction f and the optical backing of the aluminium foil—and then show that the measured ordering is recovered for every directly-coupled layout.

With $f = 0.045$ and an air-backed foil, a 48-file matrix scan (4 SiPM layouts × 4 treatments × 3 beam positions, 2500 events each) and a whole-plate irradiation (25 000 events per configuration and treatment) give, for all four layouts,

$$\text{bare} < \text{TiO}_2 < \text{Al} < \text{glossy},$$

with glossy paper brightest, air-backed aluminium within 92 % of it, and the contact-limited bare plate weakest. The ordering is explained by a single mechanism: the response tracks the *distance from the interaction point to the nearest sensor* whenever the cavity is not a perfect light mixer, so the diffuse paint (contrast $\kappa_{2D} = 1.91$–10.90) and the contact-limited bare plate (1.56–4.14) are position dependent, while the specular wrappers (1.07–1.37) are not. That same mechanism is what makes the eight channels of config 1 a position detector: using a symmetry-augmented nearest-neighbour regression that combines the integrated charges with a 12.5 % optical constant-fraction (CFD) wavefront timing observable, the diffuse paint localises the muon to 12.3 mm per axis from light sharing alone (11.9 mm combined), whereas the specular wrappers carry the information in the arrival-time wavefront instead and reach 15.4 mm–16.2 mm per axis (19.4 mm for the bare plate). We also report per-event photon yields, photoelectron spectra, wavelength and arrival-time distributions, and a systematic comparison with the published measurements, including the one wrapper for which the simulation and the measurement disagree on the sign of the position trend.

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1 Introduction and motivation

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers and veto systems for cosmic-ray muon tomography [18, 19]. The amount of scintillation light that reaches the photosensor is governed almost entirely by how the non-instrumented faces of the plate are treated: a polished face traps light by total internal reflection (TIR) over a cone of half-angle 39.3°, and what happens to the light that leaves that cone—handed back by a mirror, re-emitted diffusely by a paint, or absorbed by a black wall—decides the yield. Understanding this trade-off quantitatively is the central design question, and it is precisely where simulation is most useful, because the photon transport is a multi-bounce random walk that is difficult to model analytically [7].

Of the four surface treatments compared in Ref. [1]—no reflective layer at all, a white TiO_2 paint, aluminium foil and high-gloss paper—the bare plate is the only one that returns no light to the plate at all. It is a polished TIR guide, but every photon outside its 39.3° cone crosses the air gap and is absorbed by the black holder, whereas the two mirrors keep exactly the same cone and hand the escaped fraction back with a reflectivity of 0.90–0.96. The naive expectation, that a polished plate in a black holder should be at least as good as a diffuse paint, is the opposite of what the experiment finds, and the reason is not obvious: in the idealised model ($f = 0$) the bare plate collects 305 N_{pe}/event at the config-1 centre—more than the diffuse paint (125) but less than air-backed aluminium (482) and glossy paper (521)—whereas in the measurement it collects the *least* of the four. A clamped plate touches its holder at scattered high spots, and each contact frustrates the total internal reflection that the whole design relies on. Reproducing the measurement therefore requires modelling not only the reflectors but also the *non*-reflector, and the size of that hidden loss channel has to be calibrated against data.

This report documents a Geant4 (v11.3.2) [2, 3] simulation built to address that question for the specific geometry and surface treatments of Ref. [1]. The deliverable is a self-contained "matrix scan": every combination of four SiPM layouts and four reflector treatments, at three beam positions on the plate, plus a uniform whole-plate irradiation. The goal is not to fit a single number but to test how far the simulation reproduces the *ordering* and the *position dependence* of the light-collection efficiency that the experiment observes, to expose the physics behind both the parts it reproduces and the parts it does not, and to quantify the residuals.

Table 1 collects the whole study in one place, so that the individual sections can be read as the derivation of the numbers in it. The three quantities are the mean number of photoelectrons per muon crossing (whole-plate irradiation, 25 000 events per cell), the position contrast $\kappa_{2\mathrm{D}}$ of the full-plate efficiency field (the ratio of the brightest to the darkest 50 mm cell), and, for config 1, the per-axis resolution with which the eight-channel pattern localises the muon.

Table 1: The study at a glance. *Left*: mean number of photoelectrons per muon crossing, from the uniform whole-plate irradiation (25 000 events per cell). *Right*: the position contrast κ_{2D} of the full-plate efficiency field, i.e. the ratio of the brightest to the darkest 50 mm cell. *Last column*: the per-axis resolution σ_x with which the eight SiPM channels of config 1 localise the muon (Sec. 8).

	photoelectrons / event				position contrast κ_{2D}				
Treatment	C1	C2	C3	C4	C1	C2	C3	C4	σ_x [mm]
Bare	79.9	40.4	20.5	19.5	1.56	2.08	4.14	3.35	19.4
TiO_2	186.1	94.1	47.9	36.9	1.91	2.56	8.41	10.90	11.9
Al	480.8	253.8	129.2	82.8	1.07	1.14	1.30	1.37	15.7
Glossy	520.7	275.1	140.7	87.0	1.07	1.13	1.28	1.36	15.9

The remainder of this document is organised as follows. Section 2 states precisely what Ref. [1] measured, which of its statements this simulation can be tested against, and where the model deliberately departs from the experimental geometry. Section 3 describes the detector model: the scintillator, the SiPM read-out, the four read-out layouts (including the wavelength-shifting fibre clamp of config 4), the reflector treatments, the tabulated optical inputs that the transport actually consumes, and the optical-physics implementation. The line-by-line verification of the Geant4 boundary process against the Geant4 source is collected in Appendix A. Section 4 describes the simulation campaigns and fixes the two unmeasured surface inputs first: the beam and patch setup, the bare-holder contact scan, and the aluminium grease-versus-air comparison. These select the nominal contact fraction and Al backing used everywhere below. Section 5 presents how much light is collected and why the wrapper ordering comes out the way it does; Sec. 6 presents how uniform the response is over the plate and what sets it; Sec. 7 presents the spectral and temporal signature of the detected light. Section 8 turns the whole-plate data set into a position-reconstruction study, and Sec. 9 compares the whole set of results with the published measurements. Section 10 documents the open items, and Sec. 11 summarises.

2 The measurement being reproduced

Before presenting results it is worth stating what is being reproduced, in the terms of the experiment, and where the model and the experiment are known to differ. Everything in this section is quoted from Ref. [1].

2.1 What was measured

A 250 mm×250 mm×10 mm plate of UPS-92S organic scintillator is read out by CEFLA S.C. ASD-NUV4S-P-40 SiPMs, each with a 4 mm×4 mm active area, mounted on the *narrow* side faces of the plate. That choice is deliberate: because the area of the narrow faces is small, a large fraction of the light is transported by total internal reflection off the large faces and then collected towards the narrow ones, so the detected yield is higher on the narrow faces than on the large ones. Configurations with the SiPM and the fibre clamp on the large faces were explicitly not considered, for the same reason.

Four *surface treatments* were applied to the non-instrumented faces, in addition to the bare plate:

1. EJ-510 white reflector paint with TiO_2 suspension (Eljen Technology);
2. commercial aluminium foil, shiny side out, with small amounts of optical grease applied selectively to ensure contact between the scintillator and the reflective layer;
3. no reflective material at all — the polished plate simply *placed* in a black 3D-printed holder;
4. super high-gloss resin-coated photo paper (300 g m^{-2}, Print4Life), cut to size and placed on each surface *without* grease.

These are the four treatments modelled here, one-to-one.

Four *read-out configurations* were then assembled (Table IV of Ref. [1]), and are the four layouts of this report:

Config 1 eight SiPMs, two on each of the four narrow faces (`TBLR1`, `TBLR2`), 128 mm^2 of active area in total;

Config 2 four SiPMs, one on each narrow face (`TBLR1`), 64 mm^2;

Config 3 two SiPMs, both on the same narrow face (`T1`, `T2`), 32 mm^2;

Config 4 one SiPM plus a wavelength-shifting fibre clamp, whose four fibres provide an effective collection area of 2500 mm^2.

The efficiency was measured with a muon coincidence detector (MCD) system of two 50 mm×50 mm×10 mm reflective-painted reference scintillators read out by photomultiplier tubes. The MCD was placed over one of three *regions of interest* on the plate — the central region, a middle-edge region and a corner region — and coincidences between the two MCD arms provided the reference muon sample. The reported observable is the detection efficiency

$$\mathrm{DE}(u_{\mathrm{thresh}}) = \frac{N_{\mathrm{det}}(u_{\mathrm{thresh}})}{N_{\mathrm{ref}}}, \tag{1}$$

i.e. the fraction of reference muons that produce a trigger signal above the discriminator threshold u_{thresh}, together with the pulse-height (detected photon yield) distribution of the trigger itself.

2.2 The statements this simulation is tested against

The experiment reports a small number of clear, qualitative statements, and these are the yardstick used in Sec. 9:

- configurations *without* a reflective layer consistently produce the lowest detected photon yield at the SiPMs;
- the signal-to-noise ratios are highest and nearly identical for the aluminium foil and the high-gloss paper;
- the lowest detected photon yield distributions occur for all configurations when the MCD is positioned at the *corner* of the scintillator plate;
- all configurations except those without a reflective layer reach a detection efficiency of 80 % or more at $u_{\mathrm{thresh}} = 400\,\mathrm{mV}$, whereas the bare configuration is the only one that falls short;
- the WLS-fibre configuration (config 4) produces a significantly higher detected photon yield distribution than config 3 at all measurement positions, and reaches a detection efficiency of 90 % at $u_{\mathrm{thresh}} = 400\,\mathrm{mV}$ across its full active area;
- the converted (wavelength-shifted) photons still produce electrical signals with approximately 30 % efficiency.

The first three concern the *ordering* of the treatments and the *sign* of the position trend, and are the statements this simulation reproduces, fails to reproduce, or contradicts; the fourth and fifth are threshold statements that the simulation can address only indirectly, since it contains no electronics (Sec. 10); the sixth quantifies the optical efficiency of the WLS path and is the natural yardstick for the one configuration that the simulation does *not* reproduce.

2.3 Where the model deliberately departs from the experiment

Two geometric departures are worth stating explicitly, because both affect the comparison and neither is accidental.

The config-3 and config-4 faces. In the experiment, config 3 reads out the *top* narrow face and the config-4 fibre clamp sits on the *left* narrow face — two adjacent, not shared, faces. In this simulation both assemblies are placed on the same narrow face of the model (Sec. 3), which has the advantage of keeping the four layouts on a common geometric footing and of making the two single-sided layouts directly comparable to each other, at the cost of removing the adjacency that the experiment realises. Any statement about the relative performance of config 3 and config 4 in

this report therefore refers to the shared-face geometry and cannot be read as a statement about the experimental pair.

The position of the irradiated patches. The three regions of interest are defined by where the MCD is placed, and in the experiment they are chosen so that the MCD (which is 50 mm square) sits entirely on the plate. Here the same requirement is imposed, and the edge and corner patches are placed on the side of the plate *opposite* the instrumented face of the single-sided layouts, so that the three patches sample the far side of the cavity for configs 3 and 4.

Finally, the experiment itself warns that the four configurations "cannot be directly compared with one another", because the number of parallel SiPMs changes the total input capacitance, the effective gain and the mean probability of a photon hitting the sensitive area. The same caveat applies here, and it is the reason the results below are presented per layout rather than pooled.

3 The detector and its optical model

3.1 Scintillator plate

The active volume is a rectangular PVT scintillator $250 \times 250 \times 10$ mm, mirroring the UPS-92S plate of Ref. [1]. It is assigned the Geant4 NIST base material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical properties table built from the EJ-200 and UPS-923A data sheets [11, 12, 13]. The optical constants—the refractive index ($n = 1.58$, essentially flat across the visible), light yield, emission peak, bulk attenuation length, scintillation decay and rise times, and the Birks constant [14, 15]—are collected in Table 3 and are not repeated here. Two entries are deliberate modelling choices rather than quoted numbers: the bulk attenuation length is peaked at 3.8 m but falls steeply in the UV to model self-absorption of the short-wavelength part of the emission, and the emission spectrum (Fig. 4) is essentially zero below 380 nm and above 550 nm. Both scintillation and Čerenkov photons are produced and tracked; the optical grid spans 310 nm to 700 nm so that the Čerenkov continuum, which extends well below the scintillation band, is covered.

3.2 SiPM readout and coupling

Each SiPM [8] is modelled mechanically as a $4 \times 4 \times 0.6$ mm box of "SiPM epoxy" ($n = 1.55$), coupled to a side face of the plate through a 0.1 mm silicone grease pad ($n = 1.46$). The coupling chain is therefore *scintillator* $\rightarrow$ *grease* $\rightarrow$ *SiPM*, with Fresnel transmission at each dielectric interface and photodetection at the final grease/SiPM boundary. This describes the side-mounted sensors of configs 1–3. In config 4 the read-out SiPM instead sits at the $-y$ end of the WLS fibre bundle, and its 4 mm$\times$4 mm grease pad overlaps the slab's $-y$ face: the pad centre is at $(x, y) = (-125.55, -125.05)$ mm, so 1.45 mm of its width crosses the slab edge and 5.8 mm^2 of the 16 mm^2 pad is in direct optical contact with the scintillator. That contact is modelled explicitly and is on by default (Sec. 3.4), so the config-4 SiPM receives both direct blue light through the contact and green wavelength-shifted light from the fibre ends.

Four layouts are studied (schematic in Fig. 1):

Config 1 (8 SiPMs) two SiPMs per side face, at $\pm 50\%$ of the half-length.

Config 2 (4 SiPMs) one SiPM per side face, at the *same* $\pm 50\%$ offsets as config 1—i.e. config 1 with one device removed per side, the removed slot rotated from side to side so the layout remains four-fold symmetric.

Config 3 (2 SiPMs) both SiPMs on the *same* narrow face—the $-x$ face in the simulation frame, which is the report plan view's top edge (Sec. 4)—at $\pm 50\%$, matching T1/T2 of Ref. [1].

Config 4 (1 SiPM + WLS clamp) a single SiPM reading out a wavelength-shifting (WLS) fibre clamp along the $-x$ narrow face (Sec. 3.4).

SiPM footprints are clamped so they cannot overhang the slab face.

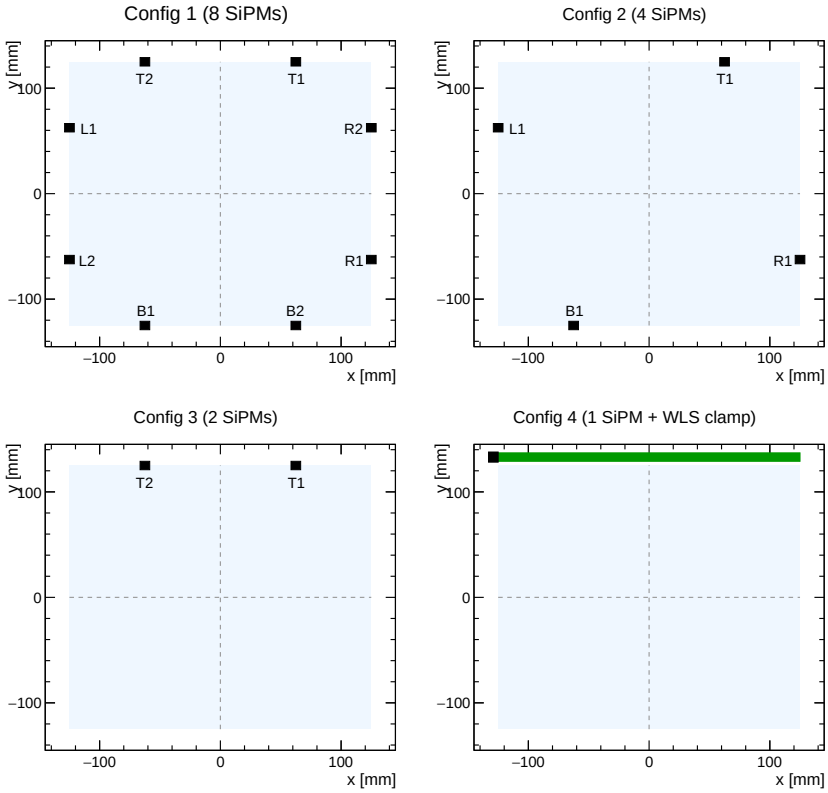


Figure 1: Schematic plan view of the four detector layouts on the 250 mm×250 mm plate footprint. Black squares mark the side-mounted SiPMs (drawn oversized for visibility; the active area is 4 mm×4 mm), and the dashed grey lines are the $x = 0$, $y = 0$ axes. Config 1 places two SiPMs on each of the four edges; config 2 places one SiPM per edge at the same offsets as config 1 (one device removed per side); config 3 places both SiPMs on the top ($+y$, report frame) edge; config 4 uses a single SiPM reading out a WLS fibre clamp on the same top edge, with the read-out SiPM at its left end (green bar), following Fig. 3(b) of Ref. [1]. In the simulation frame both configs 3 and 4 sit on the $-x$ narrow face (the report plan view maps $(x, y) \to (y, -x)$). The inwards-offset labels name the eight side-mounted sensors (T/B/R/L = top/bottom/right/left); on each edge "1" marks the sensor nearer the corner reached when travelling clockwise around the perimeter, so "1" sits at the top-right (T1), bottom-right (R1), bottom-left (B1) and top-left (L1) corners. This is the scheme used in the light-sharing discussion below. The same black-square markers are overlaid on the collection-efficiency maps of Figs. 18–21.

3.3 Reflector treatments

Four treatments are applied as a *logical skin* over the non-instrumented surfaces of the plate; the grease/SiPM patches override the skin locally with dedicated *border* surfaces. The mapping from each physical treatment to the Geant4 optical-surface parameters is the heart of the model—it determines the light-collection ordering—so it is given in full. Table 2 lists every property that is actually attached to each surface, and the subsubsections below describe each reflector physically, state how it is modelled, and assess how faithfully the model represents the real object. (A fifth finish, PTFE tape, exists in the code for completeness but is not part of the reference study.) The strong sensitivity of light collection to the wrapping and reflector choice is well established experimentally [21].

Table 2: Geant4 optical-surface parameters for the four reflectors, exactly as attached in `src/GeometryMuonScint.cc`. $R(\lambda)$ is tabulated on the shared ten-point 360 nm to 650 nm grid and quoted here as its range; the full arrays are in the code. A dash marks a property that is neither set nor used for that finish.

	Bare (black holder)	**TiO_2** (EJ-510 paint)	**Al foil (grease/air)** (backing choice)	**Glossy** (photo paper)
Physical contact	1.2 mm air gap	direct (optical)	grease film / air gap	air gap
Geant4 surface object	none*	skin	skin	skin
Type	`dielectric_dielectric`	`dielectric_dielectric`	`dielectric_dielectric`	`dielectric_dielectric`
Model	`glisur`*	`unified`	`unified`	`unified`
Finish	`polished`*	`groundfront-painted`	`polished-backpainted`	`polished-backpainted`
`sigma_alpha`	—	0.10 rad	0	0
`REFLECTIVITY` $R(\lambda)$	—	0.955–0.983	0.904–0.925	0.900–0.955
Backing `RINDEX`	—	—	1.46 grease / 1.0003 air	1.0003 (air)
`SPECULARSPIKE` f_{ss}	—	0	1	1
`SPECULARLOBE` f_{sl}	—	0	0	0
`BACKSCATTER` f_{bs}	—	0	0	0
Total internal reflection	yes (Fresnel/TIR)	no (Lambertian)	grease/air ($\leq 67.5°$ grease) (full cone air)	yes (full cone)

*The bare plate carries *no* surface object: Geant4 then applies its built-in `glisur/polished dielectric_dielectric` default, with Fresnel and TIR taken from the material refractive indices. The black holder shell, in contrast, does carry an absorbing skin (`MuonScintHolderAbsorber`, a `polished dielectric_dielectric` surface with `REFLECTIVITY` = `TRANSMITTANCE` = 0), so the sub-critical light that refracts across the air gap is killed at the wall—and the holder bulk, with its 1 µm absorption length, would absorb it anyway. The bare model also carries an optional mean-field contact fraction f (production default $f = 0.045$; the clean air-gap reference is $f = 0$); for $f > 0$ every hit on a bare face is absorbed with probability f ($R = 1 - f$, $T = 0$), representing partial plate–holder contact.

3.3.1 Bare plate in its black holder

Physical object. The polished UPS-92S plate is placed in a black 3D-printed ABS enclosure, its faces simply resting near the wall.

Model. No wrapping skin is attached. The plate keeps its native polished scintillator ($n = 1.58$) against air ($n = 1.0003$) across a real 1.2 mm gap; light below the critical angle ($\arcsin(1/1.58) = 39.3°$) refracts across the gap and is absorbed in the holder, whose black ABS has an attenuation length of 1 µm, while light above it is returned by total internal reflection. Fresnel reflection and TIR are taken from the material `RINDEX` tables.

Fidelity. This is the least faithful configuration and the source of the report's central modelling choice. A real clamped plate touches the holder at scattered high spots, and each contact forms a frustrated-TIR sink that removes trapped light; surface scratches do the same. Those loss channels are absent in the clean air-gap model ($f = 0$), where the plate keeps the full TIR cone but loses the whole sub-critical fraction at the black wall: it is brighter than the diffuse paint but dimmer than the two mirrors in every layout (305 N_{pe}/event at the config-1 centre, against 482 for air-backed aluminium and 521 for glossy paper), whereas the reference measurement gives it the *lowest* signal of all four treatments. The code therefore offers an optional mean-field contact fraction, `/dcs-monitor/det/muonScint/setBareContactFraction <f>` ($0 \leq f \leq 1$): for $f > 0$ every hit on a bare face is absorbed with probability f (a polished `dielectric_dielectric` skin with `REFLECTIVITY` $= 1 - f$ and `TRANSMITTANCE` $= 0$), representing frustrated TIR over an effective contact area. The loss is applied to every bare layout, including config 4, whose bare faces carry the same skin (only the clamp face is exempt, being coupled to the plate through

the grease film). The dedicated scan (Sec. 4.2) gives a nominal $f = 0.045$, the smallest sampled fraction that places all nine config/position cells below the corresponding TiO_2 reference; $f = 0$ is kept as the clean-air reference and lower bound on the loss.

3.3.2 Titanium-dioxide paint (EJ-510)

Physical object. A white, diffuse-reflectance paint applied directly to the plate (three coats, ≈ 0.11 mm), in optical contact with the scintillator.

Model. `groundfrontpainted` with a purely Lambertian reflection: a random draw against `REFLECTIVITY` returns the photon with a cosine-law distribution and *no* specular or TIR component ($f_{ss} = f_{sl} = f_{bs} = 0$). The reflectance rises from 0.955 in the UV to a plateau of 0.983 near 470 nm.

Fidelity. The Lambertian/no-TIR treatment is the physically correct description of light emerging from a thick, strongly scattering paint, and is this wrapper's strongest feature. The *absolute* reflectance curve is its weakest: no public spectral-reflectance data exist for EJ-510, so $R(\lambda)$ is a fitted representative white-paint curve (≈ 0.96–0.98), not a measurement of the actual coating, and no angular dependence of the diffuse lobe is modelled. The `sigma_alpha` value is set for bookkeeping but is inert at this finish (`groundfrontpainted` reflects diffusely regardless of the facet normal).

3.3.3 Aluminium foil

Physical object. Commercial shiny aluminium foil wrapped around the plate; per Ref. [1] optical grease is applied in places so that the foil contacts the scintillator rather than leaving an air gap.

Model. `polishedbackpainted` with a selectable backing refractive index, chosen pre-init with `/dcs-monitor/det/muonScint/setAlBacking air|grease`. The production choice is `air` ($n = 1.0003$), which leaves the full TIR cone intact and lets only the sub-critical fraction cross to the foil, modelling a foil separated from the plate by a trapped air layer. The alternative, `grease` ($n = 1.46$), represents a foil applied over the coupling grease, as in Ref. [1]; the scintillator/grease interface is a normal Fresnel/TIR boundary, but since the index step is small the critical angle is wide ($\arcsin(1.46/1.58) = 67.5°$): most light refracts into the grease film and is returned by the foil as a pure *specular* reflection ($f_{ss} = 1$). It is retained as the lower-bound systematic. In both cases `REFLECTIVITY` is the normal-incidence reflectance of evaporated aluminium computed from the McPeak *et al.* n, k data [20] (0.904 at 650 nm rising to 0.925 at 360 nm).

Fidelity. Two idealisations. (i) The tabulated reflectance is that of a smooth evaporated film and is therefore an *upper bound*: household foil is rougher and carries a native oxide, both of which lower R and add a small diffuse component we do not model. (ii) The foil is assumed to cover the plate uniformly, whereas real wrapping leaves wrinkles and trapped air. The production campaign uses the `air` backing, which retains the full TIR cone and recovers the measured near-equality of aluminium and glossy paper, reaching $\approx 92\,\%$ of the glossy yield in the direct-coupled layouts. The `grease` backing is retained as the lower-bound systematic (Sec. 4.3). Note that the air backing is itself the *optimistic* end of the physical range: it is the limiting case of a perfectly trapped air layer, whereas the experiment describes grease applied *selectively*, i.e. a foil that contacts the plate in places and is decoupled in others. Neither extreme is the real object, and the scan of Sec. 4.3 brackets them.

3.3.4 High-gloss photo paper

Physical object. A commercial super-high-gloss photographic paper (Print4Life, $300\,\mathrm{g\,m^{-2}}$) wrapped around the plate with an air gap (no grease).

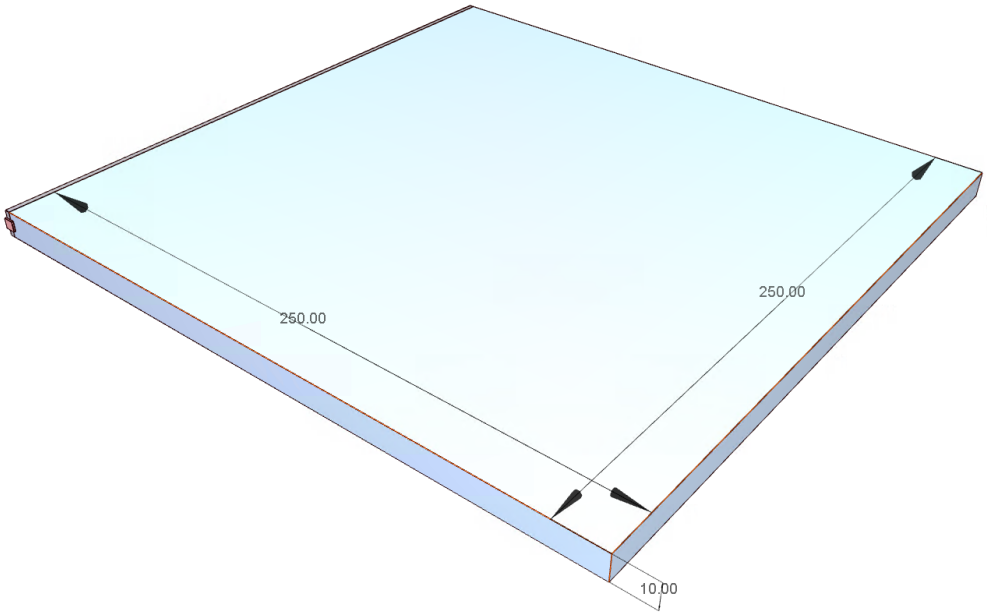
Model. `polishedbackpainted` with the backing index set to air ($n = 1.0003$), so the plate retains its full TIR cone and only the sub-critical fraction crosses the gap to the paper, which

returns it as a specular reflection ($f_{ss} = 1$) with a fitted $R(\lambda) \approx 0.900$–0.955—slightly lossier than the aluminium.

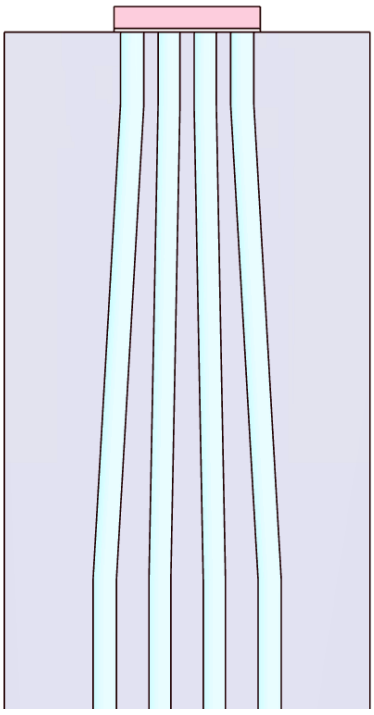
Fidelity. The reference reports glossy paper and aluminium foil as the two best wrappers (large, nearly equal signals), which motivates modelling glossy paper as a *slightly lossy mirror* rather than a diffuser. This is a model choice rather than a measurement: real photo paper is not a true mirror and has a diffuse component, and its $R(\lambda)$ is again fitted (no measured data), so this is the reflector whose optical model is least constrained by independent data.

3.4 Config 4: wavelength-shifting fibre clamp

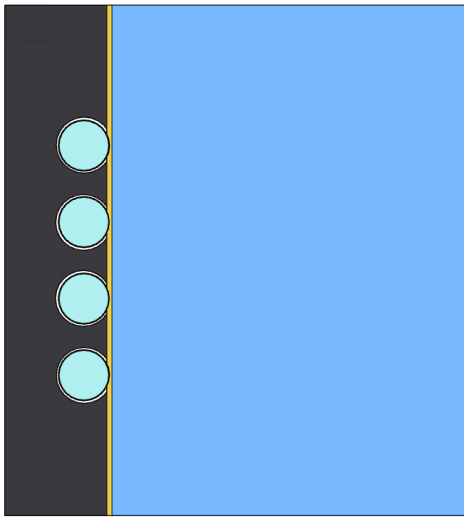
The fourth layout follows the single-SiPM, WLS-readout variant of Ref. [1]. Four 1.0 mm-diameter Kuraray Y-11(200) multi-clad wavelength-shifting fibres [16, 17] run the full 250 mm length of the $-x$ narrow face in a single row. Over most of the length they are parallel at a 2.5 mm centre-to-centre pitch—so a solid web of clamp material separates the milled 1.1 mm channels—and converge over a 41 mm length (39 mm ramp plus a 2 mm stub) to a touching 1.0 mm bundle that presents all four ends to a single 4 mm×4 mm SiPM at the $-y$ end; their far $+y$ ends are closed by individual aluminium mirror caps. Each fibre is modelled as a multi-clad guide: a Y-11 core ($n = 1.59$) inside a PMMA inner cladding ($n = 1.49$) inside a low-index fluoropolymer outer sheath ($n = 1.42$)—the low-index staircase that sustains core–cladding total internal reflection. In the production geometry the fibre solid is an analytic, mitered set of `G4CutTubs` legs (one per straight segment of the bent centreline), placed so adjoining legs meet at exact miter planes; this facet-free construction is what the runs use. (An earlier tessellated swept cylinder, which occasionally produced stuck optical tracks, is retained as an optional visualisation-only variant.) A thin 0.10 mm grease film couples the plate face to the clamp, and each fibre protrudes 0.05 mm through its channel mouth into that film, keeping the grease continuous against the scintillator. The deliberate 0.05 mm air annulus between the 1.0 mm fibre and its 1.1 mm channel is essential: the fibre must not touch the clamp ($n = 1.55$), and the outer sheath ($n = 1.42$) needs the low-index clearance to keep guiding the light. The clamp body is 3D-printed white plastic, modelled by default as a Lambertian reflector with 85 % reflectivity applied to *every* clamp face (channel walls and grease-facing front included), so light leaking out of the fibre is returned rather than lost. The far-end mirror is a full aluminium disc whose logical volume carries a reflecting skin as well as the fibre–cap border surfaces, so that light travelling in the *core*—not only the cladding—is also turned around at the far end. WLS re-emission is drawn from the digitized Y-11 emission spectrum with an exponential delay of 7 ns (Sec. 3.8), and the `G4OpWLS` process is enabled automatically when this geometry is built. In the report's top-down frame the clamp runs along the *top* edge with the single read-out SiPM at the *top-left* corner, as in Fig. 3(b) of Ref. [1]. Unlike the side-mounted sensors of configs 1–3, this read-out SiPM is coupled to the slab as well as to the fibres. Its 4 mm×4 mm grease pad is centred at $(x, y) = (-125.55, -125.05)$ mm and therefore overlaps the slab's $-y$ face by 1.45 mm×4 mm (5.8 mm^2); this is a real optical contact, not a gap. The contact is implemented with a pair of `G4LogicalBorderSurface` objects carrying the polished Fresnel coupling surface in both directions, so that it locally overrides the reflector skin and the SiPM sees direct blue scintillation light through the pad. The direct path is enabled by default (`setWLSPadSlabContact true`); setting it to `false` restores the legacy fibre-only model, in which the pad is separated from the slab and the SiPM receives only wavelength-shifted light from the fibre ends.



(a) Three-dimensional view of the complete config-4 assembly, with the principal dimensions annotated.

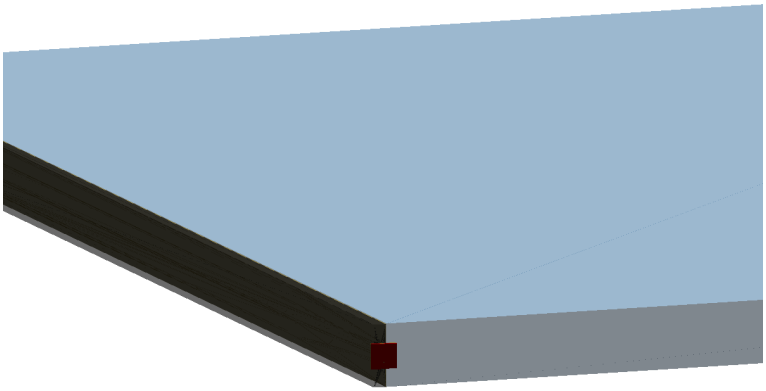


(b) Top view of the black fibre clamp: the four fibre channels and the read-out SiPM at the end of the converged bundle (report-frame top-left).

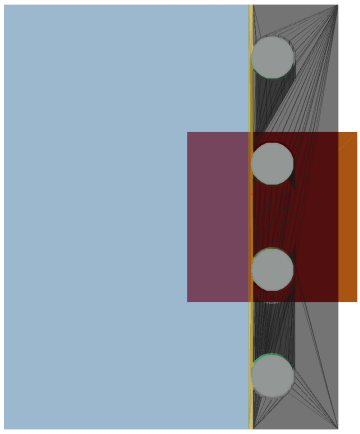


(c) Cross-section through the clamp: the scintillator, the thin grease film in the slab–clamp gap, the four fibre discs seated in their 1.1 mm channels with a 0.05 mm air annulus, and the outboard black clamp.

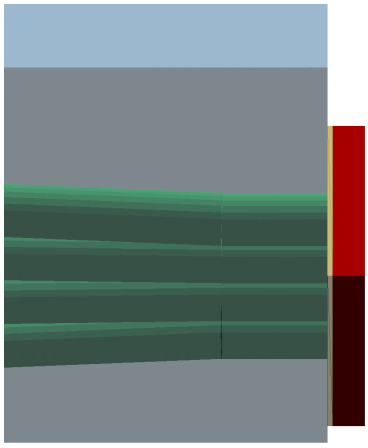
Figure 2: Geometry-only views of the config-4 WLS-fibre clamp, rendered from the Geant4 VRML geometry export (*no* beam-on). Following the event-display colour scheme, fibre cores are pale green, cladding dark green, the grease film amber, the black clamp and holder dark grey, and the read-out SiPM red. The four 1.0 mm fibres sit on the $-x$ narrow face in 1.1 mm channels separated by clamp webs, with only a thin grease film in the slab–clamp gap; they converge to a bundle read out at $-y$ by one SiPM and mirrored at $+y$. These screenshots show the clamp in its earlier black visualisation colour; the production clamp is white (Fig. 3).



(a) Oblique view of the whole plate with the clamp strip on the $-x$ narrow face and the single read-out SiPM at the $-y$ end; the fibres run the full plate length in y.



(b) Cross-section through the clamp: scintillator, grease film, and the four multi-clad fibre discs seated in their channels.



(c) Close-up of the four fibre ends converging onto the single read-out SiPM (here the Boolean clamp and grease film are hidden).

Figure 3: OpenGL (`TSG_OFFSCREEN`) geometry renders of the config-4 assembly (multi-clad $n = 1.59/1.49/1.42$ fibres, white clamp), generated head-lessly from Geant4 (no beam-on, drawn without wireframe edges). Panels (a) and (b) use the *tessellated* visualisation preset (`macros/muon_vis_wls_ogl.mac`): it is geometrically equivalent to the production analytic preset—identical round 1.0 mm fibres and clamp/grease volumes—but its clamp and grease film render as solids, whereas the analytic cut-tube cutters do not implement `CreatePolyhedron()` and would appear as point clouds. Panel (c) instead uses the *analytic production* geometry (`macros/muon_vis_wls_ogl_coupling.mac`) with only those two Boolean solids hidden, so that the plain `G4CutTubs` fibres, the read-out pad and the SiPM are seen coupling. These renders complement the hand-made VRML screenshots of Fig. 2.

3.5 Summary of detector parameters

The principal parameters of the three active components—the scintillator plate, the WLS fibre, and the SiPM—are collected in Table 3, alongside the corresponding values quoted for the reference experiment of Ref. [1]. Where a quantity is not reported in Ref. [1] or is not modelled here, a dash is shown; the simulation follows the experimental values wherever they are available, and the optical model of Sec. 3.7 is built entirely from the values in the last column.

Table 3: Principal parameters of the scintillator, the WLS fibre, and the SiPM, comparing the reference experiment of Ref. [1] with the values adopted in this simulation.

Parameter	Ref. [1]	This simulation
Scintillator (UPS-92S / PVT)		
Material	UPS-92S (AdvTECH UK)	PVT (`G4_PLASTIC_SC_VINYLTOLUENE`)
Dimensions	$250 \times 250 \times 10$ mm	$250 \times 250 \times 10$ mm
Refractive index	1.58	1.58
Density	1.05 g cm^{-3}	1.032 g cm^{-3}
Light output	50–60% of anthracene (≈ 8 to 10×10^3 ph/MeV)	1.0×10^4 MeV^{-1}
Emission peak	425 nm	425 nm (2.917 eV)
Bulk attenuation length	2000 mm	3.8 m (peak, falling in the UV)
Scintillation decay time	2.4 ns	2.1 ns
Rise time	—	0.9 ns
Birks constant	—	0.126 mm MeV^{-1}
Surface finish	all sides polished	all sides polished
WLS fibre (Kuraray Y-11(200))		
Fibre diameter	1.0 mm	1.0 mm (core+clad)
Cladding	multi-cladding	multi-clad: PMMA ($n = 1.49$) + fluoropolymer sheath ($n = 1.42$)
Core refractive index	—	1.59
Absorption peak	430 nm	430 nm (digitized Kuraray Y-11)
Emission peak	476 nm	476 nm
Attenuation length	> 3500 mm	$\gtrsim 5$ m (transport)
WLS absorption length at 430 nm	—	1.0 mm (systematic 0.3 mm to 3 mm, not scanned)
WLS decay time	—	7 ns (exponential)
Read-out arrangement	single row, 250 mm long	4 fibres, 2.5 mm pitch $\rightarrow$ 1.0 mm bundle (41 mm convergence)
Channel diameter	—	1.1 mm (0.05 mm air annulus)
SiPM (CEFLA S.C. ASD-NUV4S-P-40)		
Active area	4×4 mm^2	4×4 mm^2
Number of cells	9340	—
Peak sensitivity	420 nm	410 nm
Photon detection efficiency	43%	0.431 (peak, digitized curve)
Epoxy refractive index	1.5115	1.55
Optical coupling	BC-630 grease	0.1 mm grease film ($n = 1.46$)
Gain / dark-count rate / capacitance	10^6 / 50–100 kHz/mm^2 / 0.84 nF	— (light transport only)

3.6 Optical property tables

Figure 4 plots the spectral input tables actually fed to Geant4, over the 310 nm to 700 nm range used in the transport. The left panel compares the normalised spectral shapes: the scintillator emission peaks at 425 nm, the Y-11 absorption band at 430 nm, and the Y-11 re-emission lies in the green at 476 nm—the shift that makes the config-4 read-out spectrum green. Drawing the WLS absorption and emission on one axis makes their overlap explicit, while the absolute WLS absorption strength is carried by the middle panel. That panel gives the attenuation lengths on a logarithmic scale: the scintillator is transparent across the visible but its bulk attenuation

collapses in the UV, whereas the digitized Y-11 WLS absorption is effectively transparent redward of 500 nm and rises to its 430 nm peak (absorption length 1 mm); the WLS absorption onset is marked by the dashed line near 486 nm. The right panel is the SiPM photocathode efficiency = PDE(λ), the digitized ASD-NUV4S-P-40 curve peaking at 0.431 near 410 nm. These curves are exactly the tabulated points of `src/GeometryMuonScint.cc`; Geant4 interpolates them linearly, so the drawn segments are what the simulation uses. The left and middle panels together therefore carry the complete WLS spectral input (emission and absorption), and no separate WLS-spectrum figure is needed.

The remaining input data are shown in Figs. 5–7, all plotted directly from the same source arrays: the time structure of the two optical sources—the scintillation pulse on linear and logarithmic scales, and the exponential Y-11 re-emission delay (Fig. 5); the tabulated reflectivity $R(\lambda)$ of the three reflective wrappers, also read out at the scintillation and WLS emission peaks (Fig. 6); and the refractive-index ladder, the multi-clad fibre's radial index staircase, and the critical angles implied by it (Fig. 7).

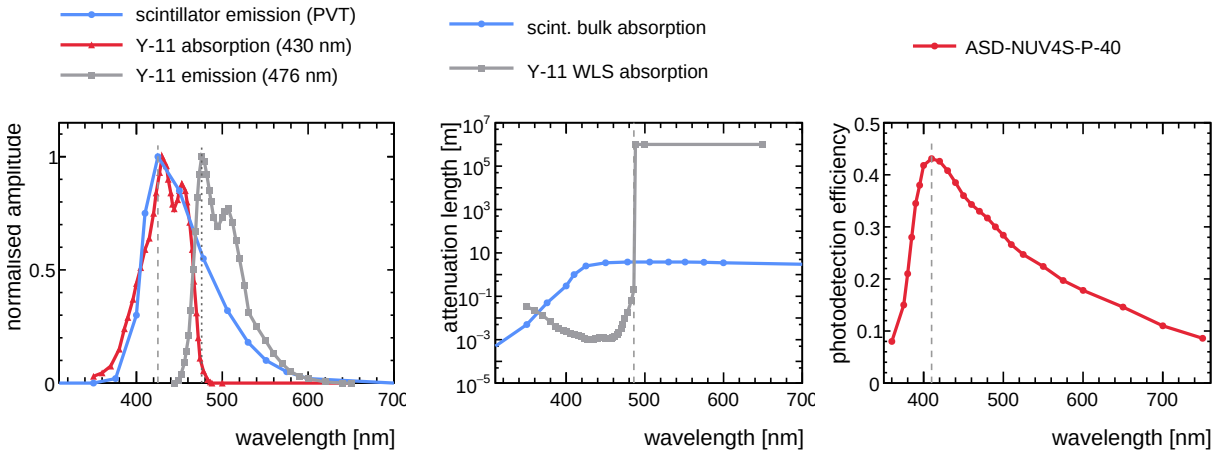


Figure 4: Optical property tables used by the simulation. *Left:* normalised spectral shapes of the PVT scintillator emission (peaking at 425 nm) and of the digitized Y-11 WLS fibre absorption (430 nm) and re-emission (476 nm); the absorption/emission overlap is the input to the `G4OpWLS` energy constraint $E'_\gamma \leq E_\gamma$. *Middle:* attenuation lengths on a logarithmic scale—the scintillator bulk absorption and the fibre WLS absorption (absolute scale set to 1 mm at the 430 nm peak). *Right:* SiPM photocathode PDE(λ), the digitized curve peaking at 410 nm. The vertical dashed line in each panel marks the characteristic wavelength of that panel's process—the scintillator emission peak at 425 nm and the Y-11 emission peak at 476 nm (left), the WLS absorption onset at 486 nm (middle), and the SiPM PDE peak at 410 nm (right); the scintillator emission peak and the SiPM PDE peak therefore nearly coincide, whereas the WLS absorption onset and the Y-11 emission peak are red-shifted into the WLS band. Markers are the tabulated points; segments are the linear interpolation Geant4 applies.

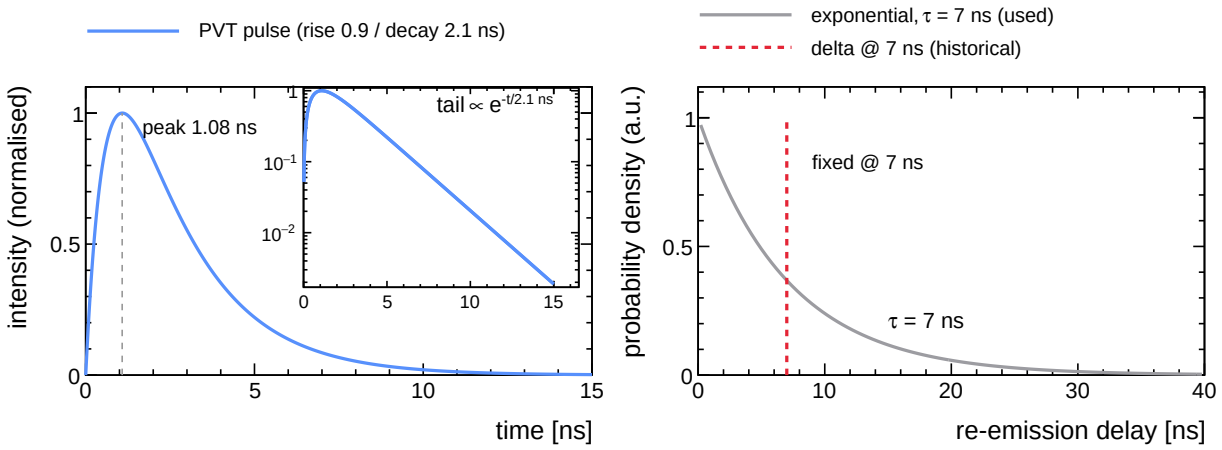


Figure 5: Time structure of the two optical sources. *Left:* the PVT scintillation pulse $f(t) \propto \left(1 - e^{-t/\tau_r}\right) e^{-t/\tau_d}$ with rise $\tau_r = 0.9\,\text{ns}$ and decay $\tau_d = 2.1\,\text{ns}$, normalised to its peak and shown on a linear scale; the dashed line marks the maximum at 1.08 ns. The inset (upper-right) replays the same pulse on a logarithmic ordinate, exposing the pure-exponential tail e^{-t/τ_d}. *Right:* the Y-11 WLS re-emission probability density: exponential with $\tau = 7\,\text{ns}$ used in production (solid), compared with the historical fixed-delay profile at 7 ns (dashed). The corresponding spectral tables are in Fig. 4.

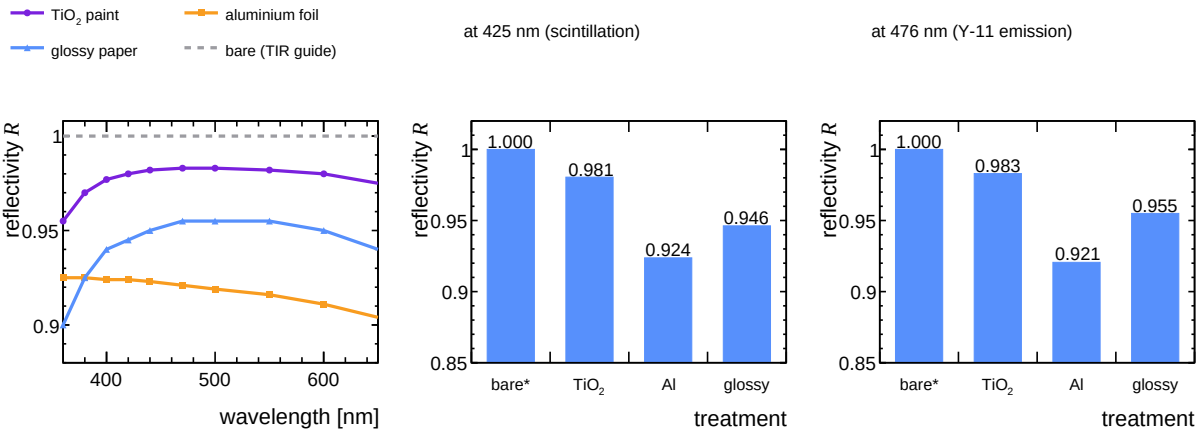


Figure 6: Reflector treatments. *Left:* tabulated reflectivity $R(\lambda)$ of the three reflective wrappers (TiO_2 paint, aluminium foil, high-gloss paper) on the shared 10-point grid, with the bare plate's lossless total-internal-reflection guide line. *Middle:* R at the scintillation emission peak (425 nm). *Right:* R at the Y-11 emission peak (476 nm), the wavelength of the config-4 re-emitted light. In both bar panels the bare plate is shown at unity as the TIR ceiling, not as a reflector.

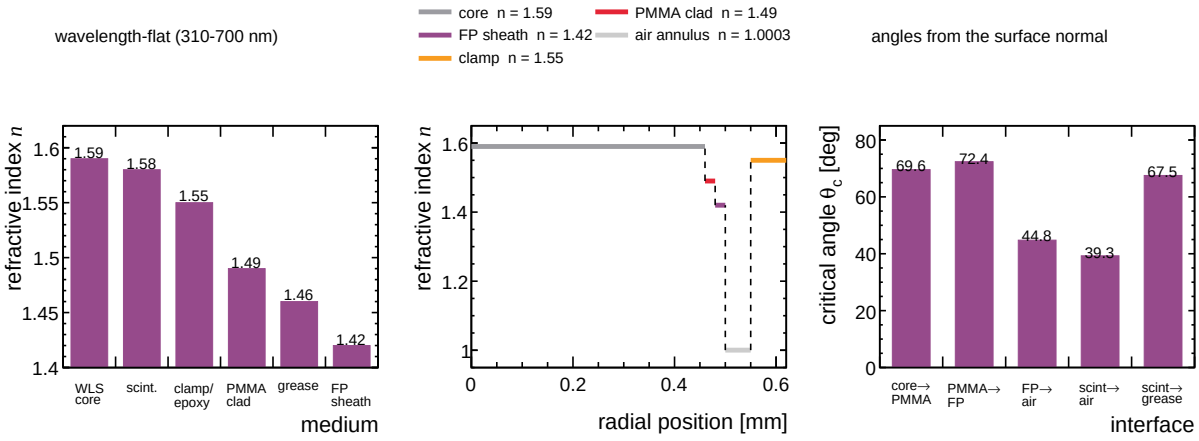


Figure 7: Refractive-index structure of the optical model. *Left:* the index ladder of the bulk media, all wavelength-flat across the visible—fibre core, scintillator, clamp/holder/epoxy, PMMA inner cladding, optical grease and fluoropolymer sheath. *Middle:* the radial index profile of the multi-clad Y-11 fibre from the core axis outward: core ($n = 1.59$), PMMA cladding ($n = 1.49$), fluoropolymer sheath ($n = 1.42$), the deliberate 0.05 mm air annulus, and the surrounding clamp ($n = 1.55$). *Right:* the critical angles $\theta_c = \sin^{-1}(n_2/n_1)$ of the successive layer pairs, measured from the interface normal. The large thresholds at the core–clad and clad–sheath boundaries are what sustain core–cladding total internal reflection, while the plate's air interface sets the usual escape cone.

3.7 Optical physics implementation

Optical photons are transported by the standard Geant4 optical physics module [4], with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes all active. Multiple scattering (Mie) is disabled; wavelength shifting (the `G4OpWLS` process) is enabled only for the config-4 fibre layout (Sec. 3.4) and inert for all others. The boundary physics of the Geant4 unified surface model, the `RINDEX` requirements it imposes, and the three code paths used by the four finishes are described in Appendix A, where they are verified line-by-line against the Geant4 source. This subsection states the two process-specific pieces of the model: the wavelength shifter and the photocathode.

3.8 Wavelength shifting

In the config-4 layout, photons that reach the fibre cores are absorbed and re-emitted isotropically at longer wavelength by the `G4OpWLS` process, which is configured through the fibre material's `WLSABSLENGTH`, `WLSCOMPONENT` and `WLSTIMECONSTANT` properties. Both spectra are the digitized Kuraray Y-11(200) curves: the absorption band peaks at 430 nm (onset $\approx$ 486 nm) and the re-emission peak is 476 nm (Fig. 4). The absolute absorption strength is set by the 430 nm absorption length, 1.0 mm in production (with a systematic band of 0.3 mm to 3 mm available through the `setWLSAbsLength` command, but not scanned here; see Sec. 10). The re-emission is sampled with an exponential time profile, $\tau = 7$ ns, rather than a fixed delay. `G4OpWLS` samples the secondary from `WLSCOMPONENT` subject to $E'_\gamma \leq E_\gamma$, so the emission table is cut at the absorption onset; a small overlap of the emission tail with the absorption band gives mild self-absorption. The process is switched on by the physics list when the WLS geometry is detected (the Geant4 optical-physics constructor reads the process activation at build time, so the flag is set before the base construction). Because most of the shifted emission lies red-ward of the absorption band, the re-emitted light is transported along the fibre by TIR to the read-out SiPM—which is why the detected config-4 spectrum is green-shifted (mean $\sim$ 510 nm) relative to the direct-coupled configurations.

3.9 Photodetection: the SiPM photocathode

Photodetection is delegated entirely to Geant4, rather than being drawn manually in the sensitive detector. The grease/SiPM boundary is a `dielectric_metal` surface whose MPT carries the SiPM `EFFICIENCY` as a function of photon energy (equivalently, of the wavelength λ), i.e. the photon detection efficiency (PDE) of an ASD-NUV4S-P-40 device [10], peaking at 0.431 near 410 nm (the digitized 25-point curve of Fig. 4; values fall to ≈ 0.30 at 490 nm), with `REFLECTIVITY` $= 0$.

The mechanism is worth stating precisely because it is the reason the sensitive detector needs no random draw of its own. In `DielectricMetal()`, when the Fresnel–PDE roll places a photon in the "absorbed" channel, `DoAbsorption()` is called. Inside it:

$$f_{\text{status}} = \begin{cases} \text{Absorption}, & \xi \geq \text{EFFICIENCY}(\lambda), \\ \text{Detection}, & \xi < \text{EFFICIENCY}(\lambda), \end{cases} \qquad \xi \sim U(0,1), \tag{2}$$

where EFFICIENCY(λ) is the tabulated PDE. The absorbed photon is assigned a non-zero `LocalEnergyDeposit` *only* when the PDE draw succeeds. Back in `PostStepDoIt`, if the status is `Detection` *and* the invoke-SD flag is set, the sensitive detector is invoked:

$$f_{\text{status}} = \text{Detection} \ \wedge \ f_{\text{InvokeSD}} \implies \text{InvokeSD(step)}. \tag{3}$$

The flag is enabled in the physics list via `G4OpticalParameters::SetBoundaryInvokeSD(true)`. Consequently the sensitive-detector `ProcessHits` fires *only* for genuine photoelectrons (it tests that the step energy deposit is positive), counts one photoelectron per call, and kills the photon (the SiPM is opaque). This is a clean separation of concerns: the physics of the photocathode lives in the optical surface, and the sensitive detector only books hits.

3.10 Per-event optical accounting

Two per-event counts recur throughout this report. Write $N_{\text{pe}}^{(e,s)}$ for the number of photoelectrons detected in SiPM s during event e, and $N_{\gamma}^{(e)}$ for the number of optical photons generated in the slab during event e (scintillation plus Čerenkov), irrespective of their later fate.

A thread-local diagnostic record (`OpticalDiag`) accumulates, per event: scintillation photons, Čerenkov photons, terminated photons, detected photoelectrons, and photons that escaped to the world. The primary muon's first-entry position in the slab is captured on the first step inside the scintillator logical volume and stored in slab-local coordinates. At end of event the sensitive detector writes one row per (event, SiPM), *including zero-detection rows*, so the sample is unbiased in the muon interaction position. Each row carries the entry position, the event's produced-photon count (the efficiency denominator), the detected count in that SiPM, the first arrival time, the mean and RMS detected wavelength, and the track weight. Three variable-length branches additionally list the individual photoelectrons: their global arrival times (`tDetected`), wavelengths (`wlDetected_nm`) and creating processes (`creatorDetected`, with 1 = `OpWLS`, 2 = scintillation, 3 = Čerenkov). The per-row count, timing and spectral data therefore support per-SiPM, combined and spatially-binned collection-efficiency analyses together with offline pulse-shape and process-mix studies.

Figure 8 shows a head-less event display of a single vertical muon in the glossy-wrapped plate (config 1). For legibility the display uses a 30 MeV muon—just below the Čerenkov threshold of the scintillator—with the scintillation yield rescaled, so that the individual optical tracks can be followed: photons are generated along the muon path, propagate by total internal reflection and specular reflection off the wrapper, and terminate on the SiPM patches at the four edges.

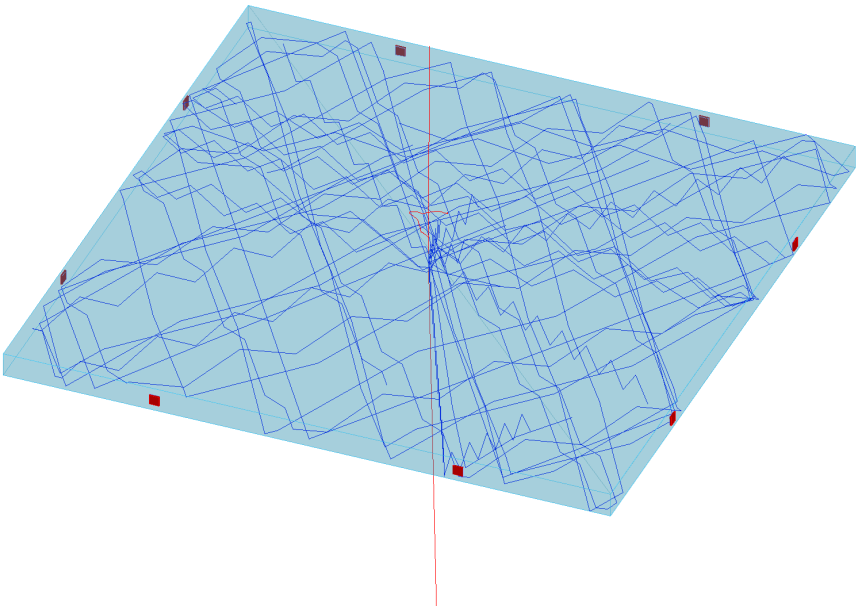


Figure 8: Head-less (`TSG_OFFSCREEN`) event display of one vertical muon in the glossy-wrapped plate with the config-1 layout, on a white background. The red line is the muon and the blue lines are optical photons (here purely scintillation, since the muon is below the Čerenkov threshold); the plate and its reflective wrapper are drawn as translucent surfaces with their edges outlined, and the SiPMs are opaque red. Both the muon energy (30 MeV, below the Čerenkov threshold) and the photon yield are chosen for visualisation only, so that individual tracks are resolved rather than merging into a saturated blob. The display is generated reproducibly by `macros/muon_vis.mac`.

4 Simulation campaigns and surface calibration

4.1 Beam, irradiation patches, and efficiency definition

The primary source is a vertical muon pencil beam: a 50 mm×50 mm square beam of 4 GeV μ^- normally incident on the plate, with the beam centre placed at three positions, each irradiating a 50 mm×50 mm patch lying *entirely on* the polished plate. With the plate spanning $[-125, 125]$ mm in both x and y, the patch centres are

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

so the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$ respectively (report frame: $+y$ is the top, $-x$ the left). This places the edge and corner regions on the bottom ($-y$) edge, opposite the single instrumented face of config 3 (top, $+y$), and it matches the 50 mm×50 mm active area of the muon coincidence detector (MCD) that defines the regions of interest in Ref. [1]. All three patches have the same area, so the denominator of Eq. 4 is the same at all three positions and the three points of a curve are directly comparable. Each (config, reflector, position) runs 2500 events after a single `/run/initialize`. The matrix therefore comprises

$$4\ \text{configs} \times 4\ \text{reflectors} \times 3\ \text{positions} = 48\ \text{files}, \qquad 48 \times 2500 = 120\,000\ \text{events}.$$

Two focused calibration campaigns bracket the remaining surface inputs and are presented first. A bare-contact scan runs configs 1–3 at eleven contact fractions $f = 0, 0.005, \ldots, 0.050$ at the three beam positions, $3 \times 11 \times 3 = 99$ files with 1000 events each (99 000 events); an aluminium-backing scan runs configs 1–3 with `grease` and `air` backing at the three positions, $3 \times 2 \times 3 = 18$ files

with 1000 events each (18 000 events). The Al scan is taken from the earlier production campaign because the Al model is unchanged and that scan is complete; the contact scan is the new extended campaign. Together with the whole-plate run of Sec. 5 ($16 \times 25\,000 = 400\,000$ events), the campaign totals 637 000 events, of which 237 000 are matrix-style (matrix + contact + Al).

Production runs were executed on the HPC cluster. The optional per-slab optical track-length flux map is disabled for the matrix (it is retained as a separate diagnostic); the collection efficiency is reconstructed entirely offline from the `sipmHits` tree at any desired binning, so the map is not needed.

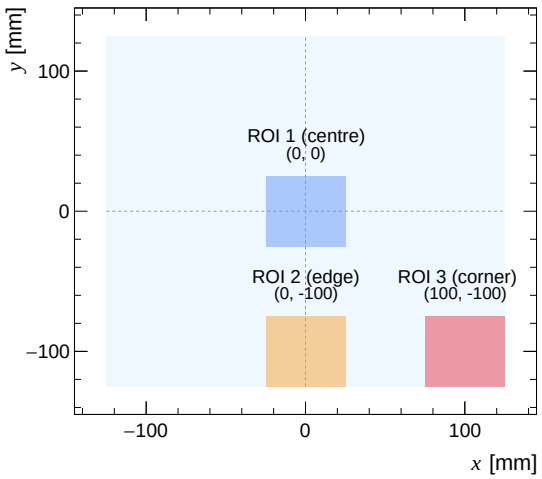


Figure 9: Plan view of the three irradiation sites (regions of interest) on the 250 mm×250 mm plate, in the report frame ($+y$ top, $-x$ left). Each coloured square is the 50 mm×50 mm footprint of the vertical muon beam: ROI 1 (centre, blue) at $(0, 0)$, ROI 2 (edge, orange) at $(0, -100)$, and ROI 3 (corner, red) at $(100, -100)$. The edge and corner sites lie on the bottom ($-y$) edge, opposite the single instrumented face of config 3.

The collection efficiency is defined, for a set $\mathcal{S}$ of SiPMs and an event e, as

$$\varepsilon = \frac{\sum_e \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_e N_\gamma^{(e)}}, \tag{4}$$

where the outer sum runs over all simulated events and the inner sum over the SiPM set $\mathcal{S}$. The all-SiPM sum over the three beam positions is the per-position figure of merit. A *plate-averaged* efficiency is the appropriate global figure of merit, but the three patches sample only part of the plate; it is therefore obtained from the uniform whole-plate irradiation of Sec. 5 and reported there.

For presentation, all position-resolved results (the maps of Fig. 9 and the whole-slab maps of Sec. 6) are shown in the report frame with the top edge at $+y$, the bottom at $-y$, the left at $-x$ and the right at $+x$. The internal simulation frame is this view rotated by 90° clockwise; the primary muon entry points and the SiPM markers are mapped into the report frame by $(x, y) \to (y, -x)$ before plotting. This rotation is a rigid change of view and leaves every scalar result (yields, spectra, efficiencies) unchanged; it only fixes the orientation of config 3 (top edge) and of the edge/corner irradiation sites (bottom edge).

The optical measurements do not fix two remaining surface inputs: the effective bare-holder contact fraction f and the aluminium backing. We determine both before presenting any production result, so that every matrix and whole-slab yield below uses one consistent nominal surface model. Both calibrations use the same beam patches and efficiency definition just described.

4.2 Bare-plate holder contact

The clean bare model assumes a 1.2 mm air gap between the polished plate and the black holder. A real clamped plate touches the holder at scattered high spots, and each contact frustrates total internal reflection and removes trapped light. To bracket this unmeasured loss channel we repeat the three-patch scan for a mean-field contact fraction f: every hit on a bare face is absorbed with probability f and otherwise follows the normal Fresnel/TIR path, implemented as a polished `dielectric_dielectric` skin with `REFLECTIVITY` $= 1 - f$ and `TRANSMITTANCE` $= 0$ (Sec. 3.3.1). The extended scan covers configs 1–3, the three beam positions and eleven fractions $f = 0$, 0.005, ..., 0.050, with 1000 events per case (99 files, 99 000 events). It is an effective area fraction, not a measurement of the holder gap.

Figure 10 shows the absolute yield versus f, with the matrix TiO_2 reference drawn as dashed lines, and the normalised centre yield. At $f = 0$ the config-1 bare plate collects 304.95, 295.37 and 288.55 N_{pe}/event at the centre, edge and corner; the corresponding config-2 values are 160.21, 153.68 and 152.25, and the config-3 values 81.27, 79.20 and 80.23. A 1% contact fraction already removes about half the light: at the config-1 centre the yield falls to 155.01 N_{pe}/event, and at the nominal $f = 0.045$ to 74.37. The loss is therefore strongly non-linear in the first few percent of contact area, as expected when a small fraction of the surface provides a fast escape path for the trapped mode volume.

The nominal value is chosen from the crossings themselves, listed in Table 5. At the centre the crossing is remarkably stable across layouts, $f = 0.0167$–0.0172 for configs 1–3. At the edge and corner the required fraction drops to 0.0034–0.0088 in the four-sided layouts, where the diffuse TiO_2 reference is bright, while in the single-sided config 3 it rises to 0.0387 (edge) and 0.0408 (corner), because the TiO_2 edge and corner patches there are the dimmest of the three positions. No single f matches every cell exactly; the production nominal $f = 0.045$ is the smallest sampled fraction that places all nine cells below TiO_2, at the cost of over-suppressing the four-sided centre. At that value the nine bare yields are 74.37/64.42/63.25 (cfg 1), 37.60/32.35/32.04 (cfg 2) and 19.22/16.32/16.04 (cfg 3) for centre/edge/corner. The same global f thus matches the ordering in every layout and position; we retain $f = 0$ as the clean-air reference and lower bound on the loss.

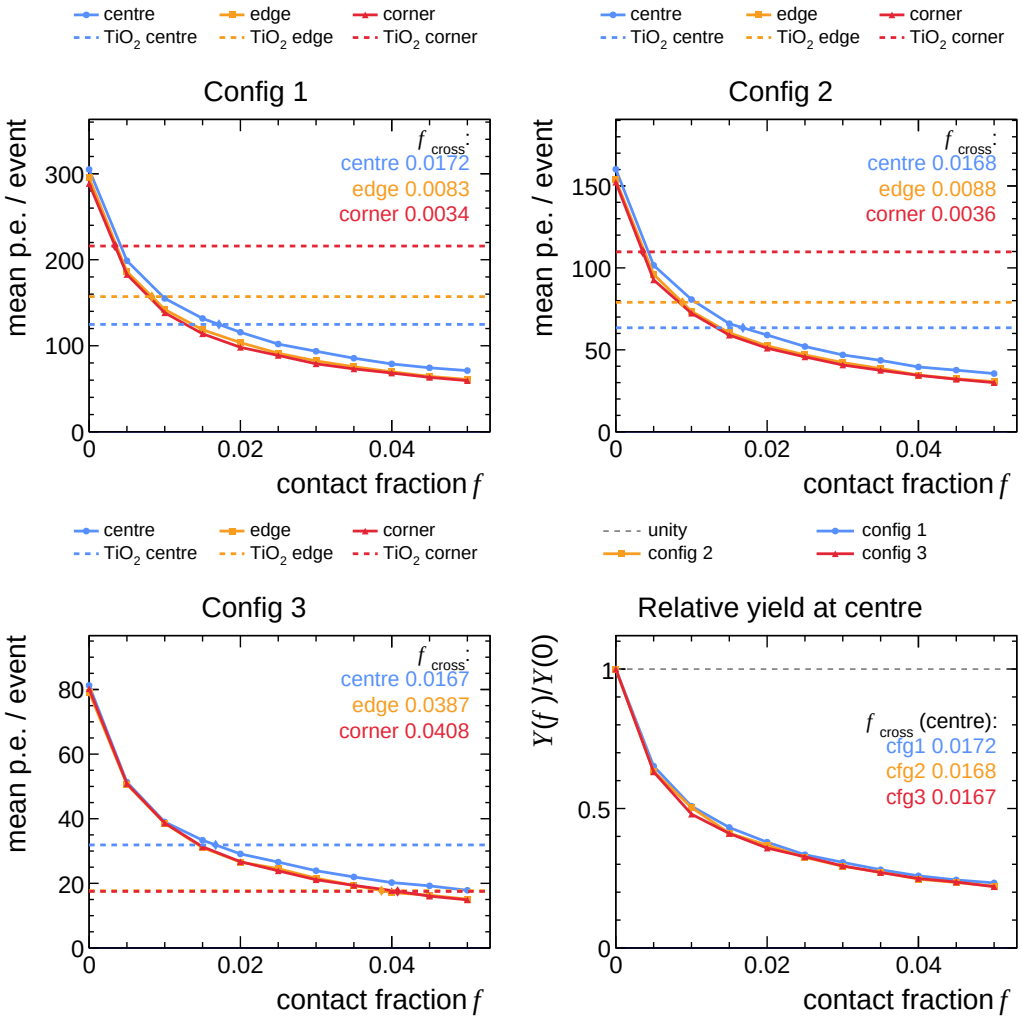


Figure 10: Bare-plate contact-fraction scan. The first three panels show the mean photoelectron yield versus contact fraction f for configs 1–3, with the centre, edge and corner patches as separate curves and the corresponding matrix TiO_2 reference as a dashed line in the same colour. A full diamond marks the interpolated crossing where the bare yield falls below TiO_2; the crossing fractions are listed, colour-matched to the curves, in the empty upper-right corner of each of the first three panels. The bottom-right panel shows the centre yield normalised to $f = 0$, with the centre crossing fractions of configs 1–3 listed in the region between the unity line and the curves. The contact model is a mean-field per-hit loss ($R = 1 - f$, $T = 0$), so f is an effective area fraction, not a measured holder contact area.

Table 4 lists the config-1 centre scan in detail, with the signed difference from the matrix TiO_2 reference of 124.79 N_{pe}/event.

Table 4: Config-1 centre contact-fraction scan. N_{pe} is the mean photoelectron yield per event with its statistical uncertainty, $Y(f)/Y(0)$ is normalised to the air-gap value, and Δ is the signed difference from the matrix TiO_2 reference (124.79 pe/event). The production nominal is $f = 0.045$.

f	N_{pe}	$Y(f)/Y(0)$	Δ (pe)
0.000	304.95 ± 2.75	1.000	+180.16
0.005	198.78 ± 1.74	0.652	+73.99
0.010	155.01 ± 1.47	0.508	+30.21
0.015	131.75 ± 1.18	0.432	+6.95
0.020	115.67 ± 1.09	0.379	−9.13
0.025	101.94 ± 0.83	0.334	−22.85
0.030	93.61 ± 0.80	0.307	−31.18
0.035	85.51 ± 0.77	0.280	−39.28
0.040	78.87 ± 0.71	0.259	−45.92
0.045	74.37 ± 0.61	0.244	−50.43
0.050	71.12 ± 0.68	0.233	−53.67

Table 5: Interpolated contact fraction f_{cross} at which the bare yield falls below the matrix TiO_2 reference, for each layout and beam position, and the bare yield at the production nominal $f = 0.045$. The TiO_2 reference is the v6p5 matrix value (Sec. 5).

Config	Position	Bare at $f = 0$	TiO_2	f_{cross}	Bare at 0.045
1	centre	304.95	124.79	0.0172	74.37
1	edge	295.37	157.14	0.0083	64.42
1	corner	288.55	215.91	0.0034	63.25
2	centre	160.21	63.46	0.0168	37.60
2	edge	153.68	78.99	0.0088	32.35
2	corner	152.25	109.78	0.0036	32.04
3	centre	81.27	31.89	0.0167	19.22
3	edge	79.20	17.74	0.0387	16.32
3	corner	80.23	17.50	0.0408	16.04

4.3 Aluminium backing: grease versus air

The aluminium foil is the one wrapper whose optical contact is explicitly ambiguous in the reference construction: grease is applied in places, but wrinkles and trapped air leave other regions decoupled. The production model uses the `air` backing ($n = 1.0003$, critical angle 39.3° at the scintillator), which leaves the full TIR cone intact, so only the sub-critical fraction reaches the foil. The alternative `grease` backing ($n = 1.46$, critical angle 67.5°) lets most trapped light refract into the grease and return specularly from the foil; it is retained as the lower-bound systematic. Both use the same tabulated aluminium reflectance ($R = 0.904$–0.925). The backing scan covers configs 1–3 at the three positions, with 1000 events per case, using `/dcs-monitor/det/muonScint/setAlBacking air|grease`. It is taken from the earlier complete production campaign because the Al model is unchanged; the v6p5 matrix air-backed centre value (482.23 pe/event) agrees with the v5 scan value (480.05) to better than 0.5 %.

Figure 11 and Table 6 show the result. Air backing raises the aluminium yield by a factor 2.38–2.59 depending on layout and position. At the config-1 centre the foil goes from 201.92 to 480.05 pe/event, at the config-2 centre from 102.51 to 254.57, and at the config-3 centre from 51.98 to 127.90. The air-backed foil reaches 92.2 %, 92.5 % and 90.9 % of the glossy yield from the production matrix (Sec. 5) in those three cases, i.e. it recovers the measured near-equality of aluminium and glossy paper. The grease-backed branch is roughly a factor 2.4 lower—about 37 %–39 % of the glossy yield—and would put the foil below the clean-air ($f = 0$) bare plate, so the backing choice is the dominant Al systematic. The enhancement is slightly larger at the

edge and corner than at the centre for config 3, but the position dependence within each backing remains weak.

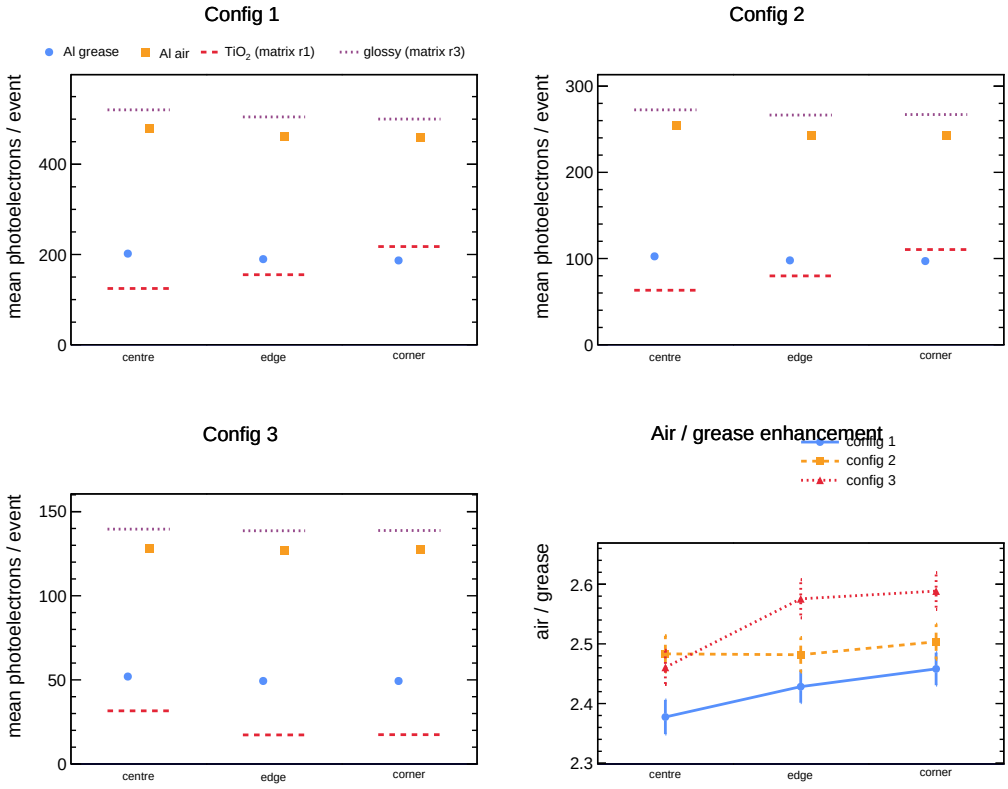


Figure 11: Aluminium backing scan. The first three panels compare the grease and air backings at the three beam positions for configs 1–3, with the matrix TiO_2 and glossy references shown for orientation; the bottom-right panel shows the air/grease enhancement ratio. The backing index changes the critical angle, not the tabulated aluminium reflectance, so the difference is entirely due to whether the plate retains its full TIR cone.

Table 6: Aluminium yield for the grease and air backings, and the air/grease ratio, for configs 1–3 and the three beam positions (1000 events per case). The production campaign uses the air backing; the grease column is the lower-bound systematic. The scan is from the earlier complete Al campaign (the Al model is unchanged); the v6p5 air matrix centre agrees with it to better than 0.5 %.

Config	Position	Grease (pe)	Air (pe)	Air/grease
1	centre	201.92 ± 1.79	480.05 ± 3.94	2.377 ± 0.029
1	edge	189.72 ± 1.58	460.71 ± 3.39	2.428 ± 0.027
1	corner	186.80 ± 1.45	459.18 ± 3.67	2.458 ± 0.027
2	centre	102.51 ± 0.90	254.57 ± 2.16	2.483 ± 0.030
2	edge	97.85 ± 0.86	242.85 ± 1.86	2.482 ± 0.029
2	corner	97.06 ± 0.86	243.00 ± 1.87	2.504 ± 0.029
3	centre	51.98 ± 0.45	127.90 ± 1.00	2.461 ± 0.029
3	edge	49.35 ± 0.46	127.10 ± 1.04	2.575 ± 0.032
3	corner	49.34 ± 0.44	127.71 ± 0.99	2.588 ± 0.031

The two backings bracket the as-built foil: uniform grease is the lower bound and a perfectly trapped air layer the upper bound. The air-backed result is consistent with the reference measurement's near-equality of aluminium and glossy paper. Together with the contact scan, the calibration fixes the production surface model: $f = 0.045$ for the bare plate and the air backing

for aluminium. Every matrix and whole-slab result in Secs. 5 and 6 uses these nominal values; $f = 0$ and grease are retained only as systematic bounds.

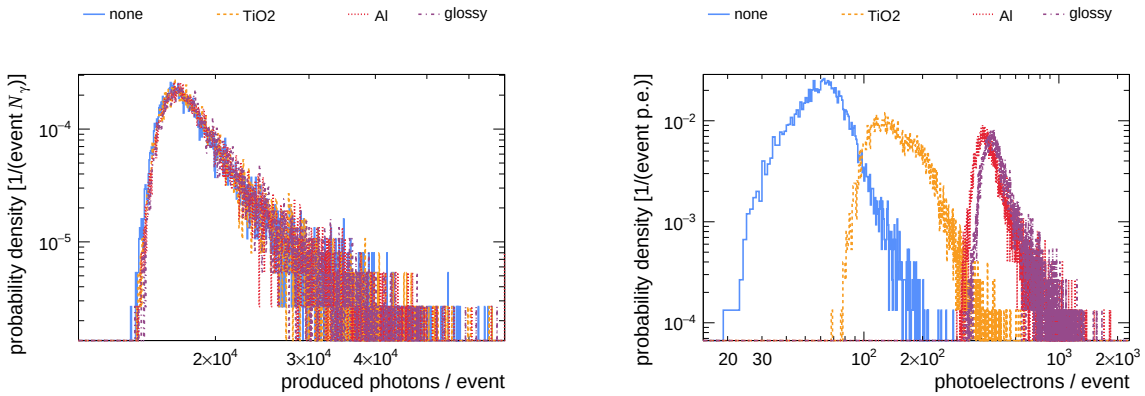
5 Light collection: yield and the wrapper ordering

With the surface model fixed at $f = 0.045$ and air-backed aluminium, we now present the production results.

5.1 The three-patch matrix scan

Figure 12a shows the per-event optical-photon yield (the denominator of Eq. 4) for the four reflectors. The mean yield is essentially independent of the reflector—about 19 600 photons per event—which is the correct physical behaviour: the reflector changes how photons are transported, not how many are produced. The small spread reflects event-to-event variation in the muon path length.

Figure 12b shows the corresponding photoelectron spectra. Here the reflector has a clear effect: at the config-1 centre the calibrated model gives 74.3 N_{pe}/event for the contact-limited bare plate, 124.8 N_{pe}/event for TiO_2, 482.2 N_{pe}/event for air-backed aluminium and 520.7 N_{pe}/event for glossy paper. The ordering is bare $<$ TiO_2 $<$ Al $<$ glossy, with the air-backed foil close to glossy paper. The bare curve is therefore a broad, low-occupancy distribution rather than the full-cone TIR distribution obtained at $f = 0$, which still loses the whole sub-critical fraction at the black wall; the difference is the contact-loss systematic of Sec. 4.2.



(a) Yield (produced photons/event). **(b)** Photoelectrons/event.

Figure 12: Per-event distributions for config 1, plotted as a probability density (unit area, 1 event per unit of the abscissa) with logarithmic x- and y-axes. The production yield (left) is reflector-independent, as it must be; the detected photoelectron count (right) depends strongly on the wrapper. The log y-axis is needed because the four curves have the same event count but very different widths and heights: the narrow, low-occupancy TiO_2 distribution and the broad, high-occupancy bare and glossy distributions cannot both be resolved on a linear scale.

The efficiency scale is set by the SiPM PDE and the coupling: of order 19 600 photons are produced and the best wrapper delivers $\approx$ 521 photoelectrons, i.e. $\approx$ 2.65%. Given a PDE of $\sim$0.4 and the geometric acceptance of a side-mounted sensor, this is consistent.

The three irradiation patches separate the response by position, and the tempting single-coordinate summary is the distance r of the patch from the plate centre, but r is blind to azimuth, and the shortcoming is decisive here: configs 3 and 4 instrument a *single* face, so the collection depends on the *direction* from which the plate is irradiated, not merely on the distance. All three patches also lie on the half of the plate opposite the instrumented face of the single-sided layouts, so their distances are not even monotonic in the proximity to a sensor. We therefore keep the three patches separate and compare them directly.

Figure 13 groups the three patches by layout (one panel per config, four reflector curves) and Fig. 14 groups them by reflector (one panel per wrapper, four layout curves). The conspicuous feature is that the specular wrappers are nearly *flat* across the three patches: across all four layouts the maximum/minimum patch ratio spans 1.0 % to 4.2 % for aluminium and 0.8 % to 3.6 % for glossy paper. The diffuse TiO_2 paint is the opposite extreme, varying by 74.9 % to 88.9 % (a factor 1.75–1.89). The contact-limited bare plate lies between them: 8.3 % to 17.5 %, with the four-sided and config-3 layouts all near 17 % and config 4 at 8.3 %.

The *sign* of the position trend is wrapper- and layout-dependent, and this is the first place where the simulation and the measurement part company. For the bare plate, aluminium and glossy paper the three-patch ordering is centre > edge > corner in every layout, i.e. the corner is the dimmest patch, exactly as reported in Ref. [1]. For the diffuse TiO_2 paint the trend is *reversed* in the four-sided layouts 1 and 2 (corner > edge > centre, a factor $\approx$1.75), while in the single-sided layouts 3 and 4 the centre is again the brightest patch. The mechanism behind this reversal is discussed in Sec. 5.3; it is the same mechanism that sets the uniformity of the full-plate maps and the ability of the SiPM pattern to localise the muon.

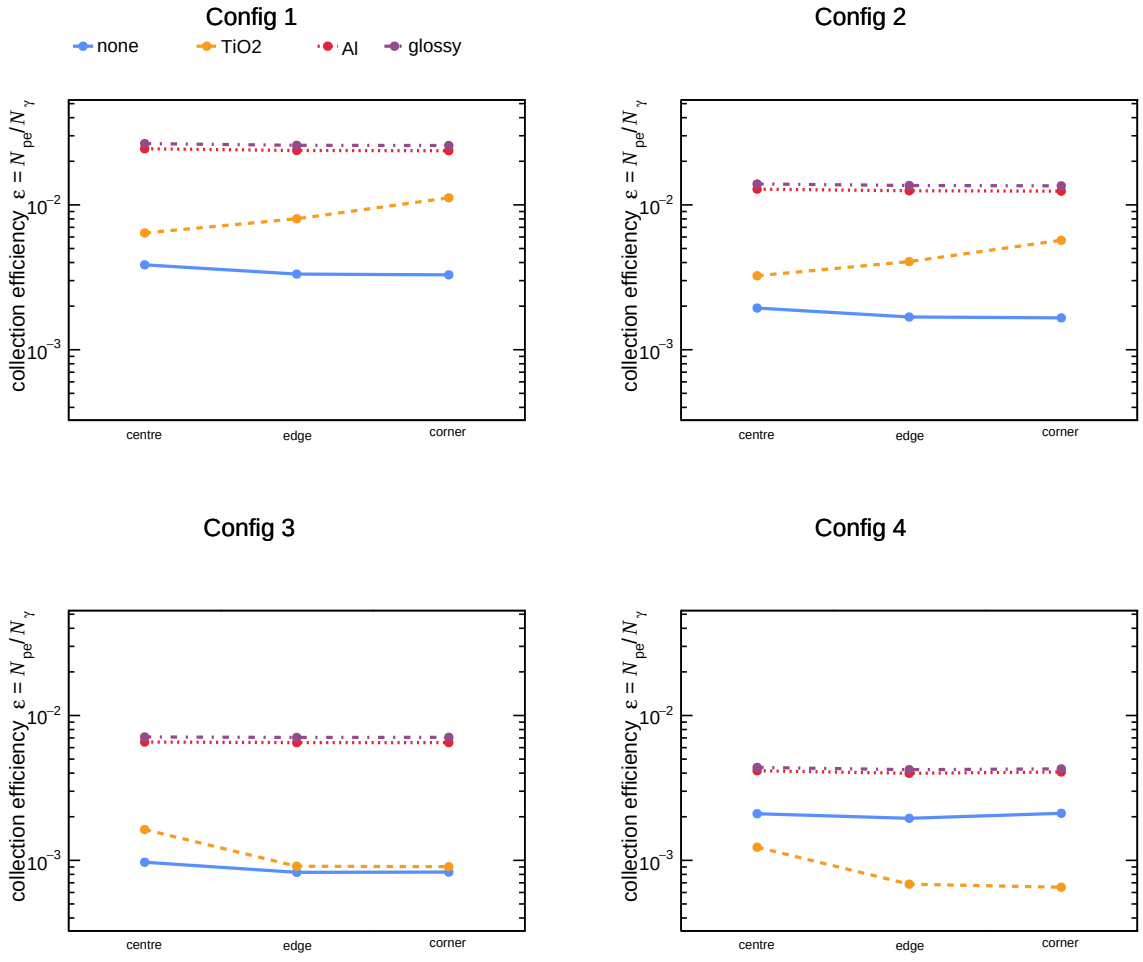


Figure 13: Collection efficiency at the three irradiation patches (centre, edge, corner), one panel per detector layout, with the four reflector curves (logarithmic y-axis). All three patches have equal area and lie fully on the plate, so the points are directly comparable. Aluminium and glossy paper are nearly flat across the three patches in every layout; the contact-limited bare plate varies by up to 18 %, and the diffuse TiO_2 paint shows by far the strongest position dependence.

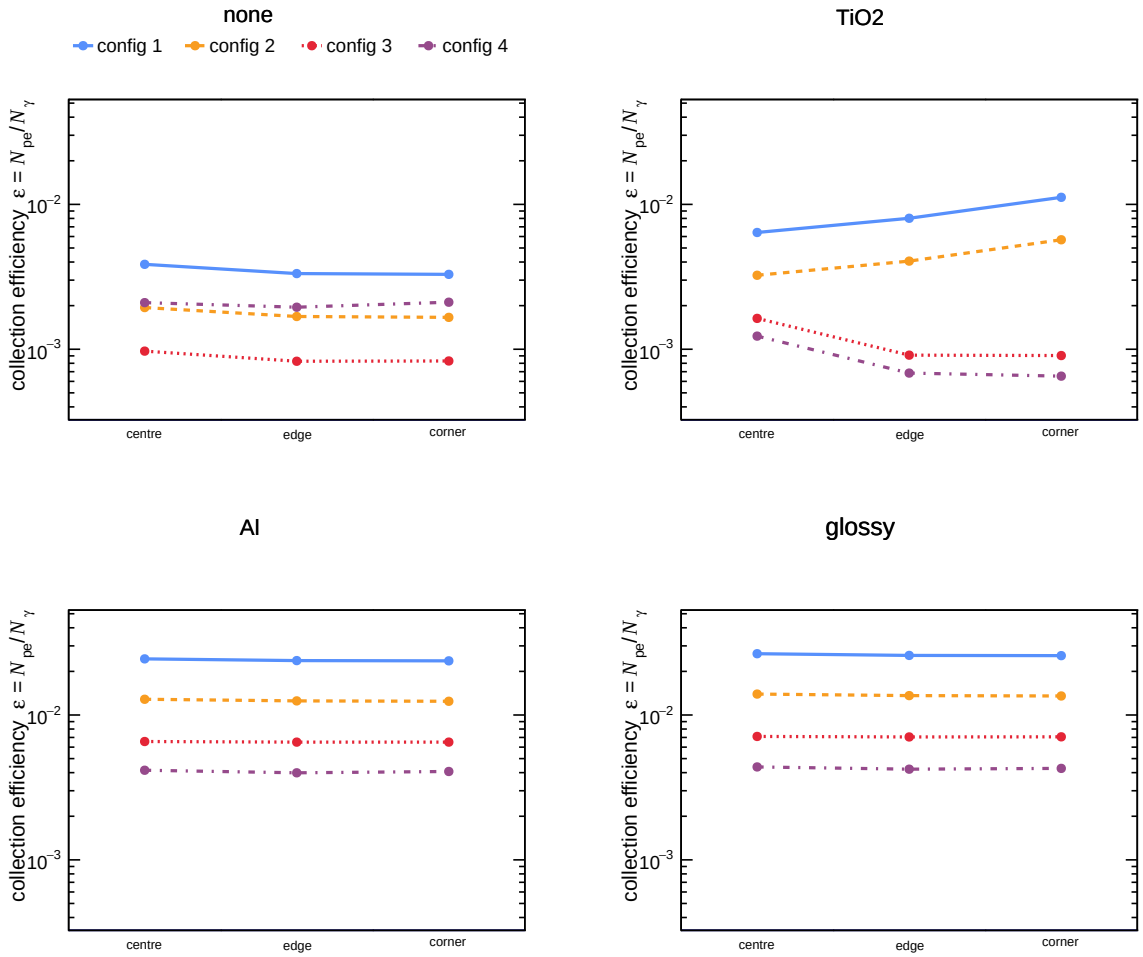


Figure 14: The same data grouped by reflector (one panel per wrapper), with the four layouts as curves. Aluminium and glossy paper are nearly flat in every layout, while the contact-limited bare plate varies by up to 18 %. The diffuse TiO_2 paint varies by a factor $\approx$1.75 in the four-sided layouts (corner highest); in the single-sided layouts (3 and 4) it drops from centre toward the edge and corner patches, which sit on the far side of the plate from its sensors.

Figure 15 resolves the directional response most directly. For config 1 (the layout with the most sensors) it shows the *share* of the collected light seen by each of the eight sensors, in one panel per reflector, with the three irradiation patches as grouped bars (dashed line: the uniform expectation of 1/8). The difference between the wrappers is instructive. Glossy paper and aluminium spread the light almost uniformly: at the corner patch the largest and smallest single-sensor shares differ by only 1.23 and 1.23. The diffuse TiO_2 paint concentrates it strongly: the largest and smallest shares differ by a factor 11.20, and the two sensors on the faces bordering the illuminated corner together collect 73.6 % of the photoelectrons (about three times the uniform share each), while the remaining six share the rest. The contact-limited bare plate is intermediate (5.35), a direct consequence of the position-dependent holder-loss model. The total efficiency still varies only modestly across the three patches for Al and glossy (Table 7), so the cavity keeps the *amount* of collected light nearly constant while rearranging its spatial division. A radius-only summary averages these distinct per-sensor patterns into a single number and hides precisely this information.

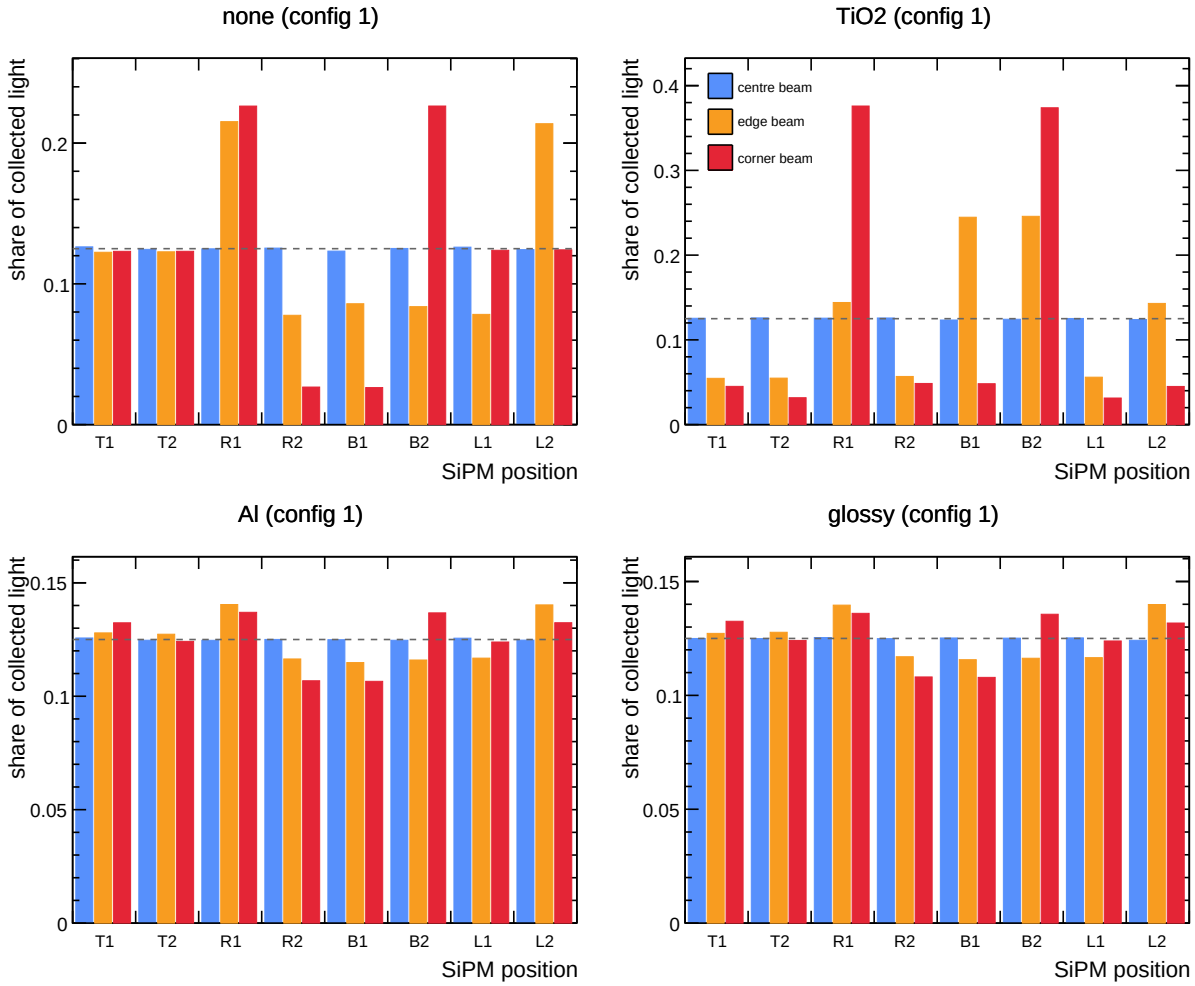


Figure 15: Direction-aware light sharing, config 1, one panel per reflector. Each group of bars is one sensor (x axis) and the three colours are the irradiation patches (blue: centre, orange: edge, red: corner); the height is that sensor's share of the light collected at that patch, and the dashed line marks the uniform expectation 1/8. Glossy paper and aluminium spread the light almost uniformly over all eight sensors (corner max/min 1.23 and 1.23), the contact-limited bare plate is intermediate (5.35), whereas the diffuse TiO_2 paint concentrates it on the two sensors nearest the beam (corner max/min 11.20).

Table 7 lists the absolute centre/edge/corner efficiencies for config 1; the fully position- and direction-resolved maps over the *entire* plate are given in Sec. 6.

Table 7: Position dependence of the collection efficiency, config 1 (all SiPMs). Each beam patch is 50 mm×50 mm and lies fully on the plate, so the produced-photon denominators are equal.

Reflector	Centre	Edge	Corner
Bare	3.857×10^{-3}	3.329×10^{-3}	3.288×10^{-3}
TiO_2	6.398×10^{-3}	8.019×10^{-3}	1.119×10^{-2}
Al	2.440×10^{-2}	2.374×10^{-2}	2.365×10^{-2}
Glossy	2.649×10^{-2}	2.578×10^{-2}	2.569×10^{-2}

Figure 16 collects the per-event photoelectron spectra over the whole matrix scan in the layout of Fig. 10 of Rao et al. [1]: the three irradiation regions (centre, edge, corner) are the rows and the detector configurations are the columns, with the four reflector treatments overlaid in every cell. Two adaptations are made. First, the voltage pulse-height spectra of the measurement are replaced by the simulation's number of photoelectrons per event, the quantity that is directly comparable between our model and the experiment. Second, the fourth configuration (single

SiPM + WLS fibre clamp) is added as a fourth column, so that all four layouts appear in one figure on a common logarithmic axis (panel (b)) rather than in a separate figure.

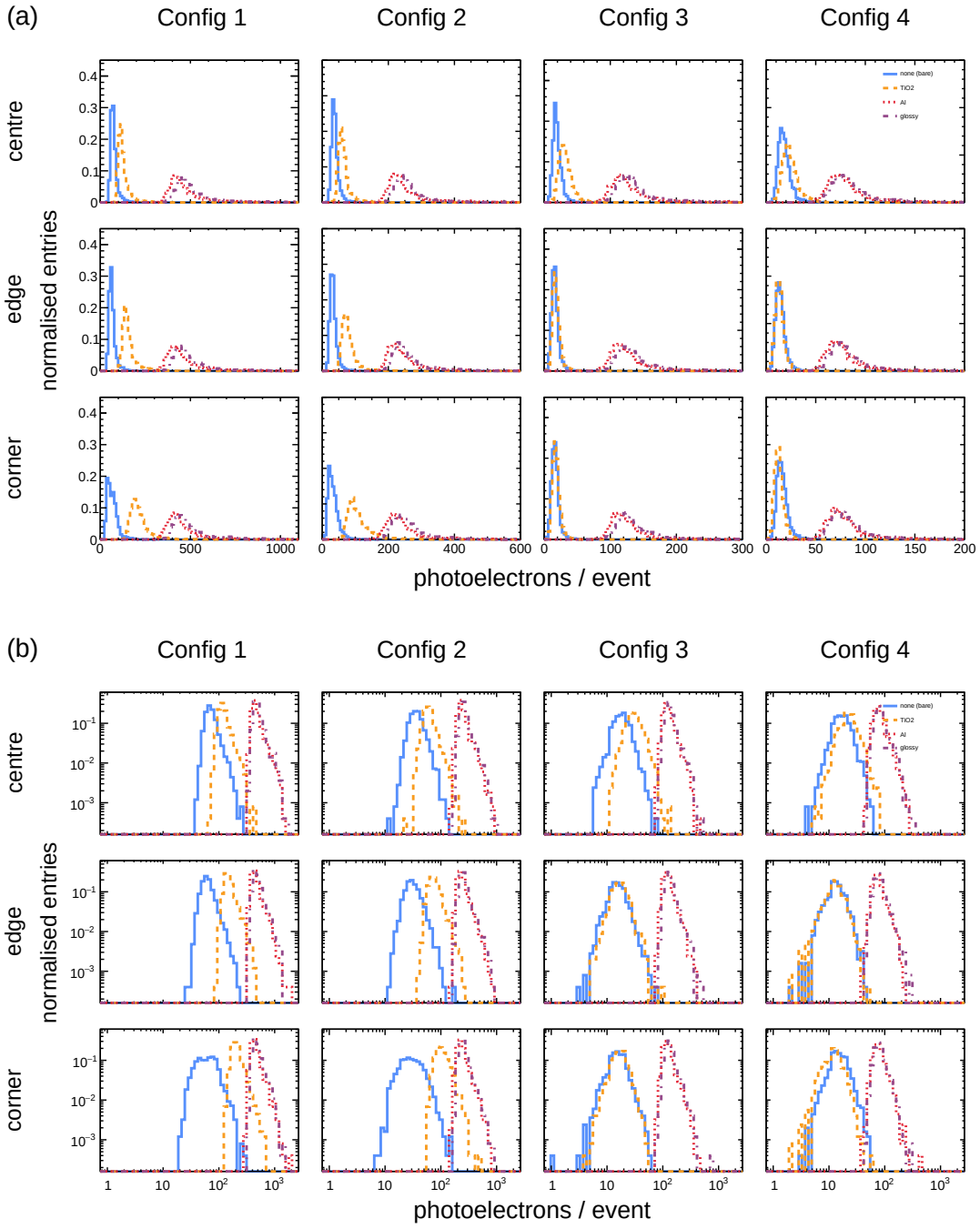


Figure 16: Photoelectron spectra in the layout of Fig. 10 of Rao et al. [1]: rows are the irradiation regions (centre, edge, corner) and columns the detector configurations (1–4, with the WLS-clamp layout as the fourth column). Each cell overlays the four reflector treatments. The histograms are normalised entry distributions (each cell is scaled so its entries sum to unity). The *upper* grid (a) uses linear axes and the *lower* grid (b) logarithmic axes; both share the same reflector colours and styles and the same units. Panel (a) gives each column its own linear abscissa, truncated at that configuration's 99.5th percentile (rounded up) and binned linearly, so that the narrow high-density core of every layout is resolved. A single shared linear abscissa would instead collapse the WLS-clamp layout (config 4), which collects roughly an order of magnitude fewer photoelectrons than the direct-coupled layouts, into the first few percent of the axis. Panel (b) shows every cell over its full distribution on a single shared logarithmic abscissa with logarithmic bins, so the four layouts remain directly comparable there. Zero-photoelectron events form a pulse below threshold and lie off the abscissa. The wrappers shift and broaden the spectrum from the low, contact-limited bare distribution through TiO_2 and air-backed Al to the wide, bright glossy one; apart from TiO_2 and the bare plate the spectra vary by at most 3.4 % across positions.

The figure condenses the trends established above. At the config-1 centre the mean yields are 74.3, 124.8, 482.2 and 520.7 pe/event for bare, TiO_2, Al and glossy, so the calibrated ordering bare $<$ TiO_2 $<$ Al $<$ glossy holds in every direct-coupled cell, with the contact-limited bare plate the lowest and glossy the highest. The position dependence is wrapper-specific: for the diffuse TiO_2 paint the corner spectrum is shifted highest in the four-sided layouts (a factor $\approx$1.75 in the mean, configurations 1 and 2) but the centre is highest in the single-sided layouts 3 and 4. Aluminium and glossy paper vary by at most 4.2 % across positions, whereas the contact-limited bare plate varies by up to 17.5 %. This is the same reflector-versus-position interplay summarised in Table 7 and the efficiency maps, presented here as the pulse-height distributions that a threshold-based read-out would actually see.

5.2 The whole-plate average

The matrix scan pools only three 50 mm×50 mm patches, all on the $-y$ half of the plate, so the efficiency it yields depends on the choice of patches and is *not* a plate average. The whole-slab run removes that bias: the GPS plane is expanded to 250 × 250 mm (`halfx` = `halfy` = 125 mm) and 25 000 events are generated per (config, reflector), so the primary muon's first-entry position is sampled *uniformly* over the entire plate and the pooled ratio $\sum N_{\mathrm{pe}} / \sum N_{\gamma}$ is the *plate-averaged* collection efficiency—the appropriate global figure of merit. Figure 17 shows it for every (config, reflector) pair, and Table 8 lists the numbers.

With the calibrated surfaces every layout reproduces the measured ordering,

$$\text{bare} < \mathrm{TiO_2} < \mathrm{Al} < \text{glossy} \quad (\text{configs 1–4}),$$

with glossy paper the brightest wrapper in every layout and air-backed aluminium within 92 % of its yield in configs 1–3 (95 % in config 4). The contact-limited bare plate is now the weakest wrapper in every layout, as the experiment finds. The spread from weakest to brightest is a factor 6.5–6.9 in configs 1–3, set mainly by the bare contact loss, and 4.5 in config 4. The efficiency scales roughly with the photosensitive area: the eight-SiPM layout (config 1, $\varepsilon = 4.08 \times 10^{-3}$) is highest, four SiPMs on four faces (config 2, 2.06×10^{-3}) roughly halve it, and concentrating the two sensors of config 3 on a single face (1.05×10^{-3}) halves it again. The single-SiPM WLS-fibre layout (config 4, 1.01×10^{-3}) is comparable to config 3: it is lower for TiO_2, Al and glossy by 20 %–40 %, and now slightly lower for the bare plate as well (1.007 versus 1.048×10^{-3}).

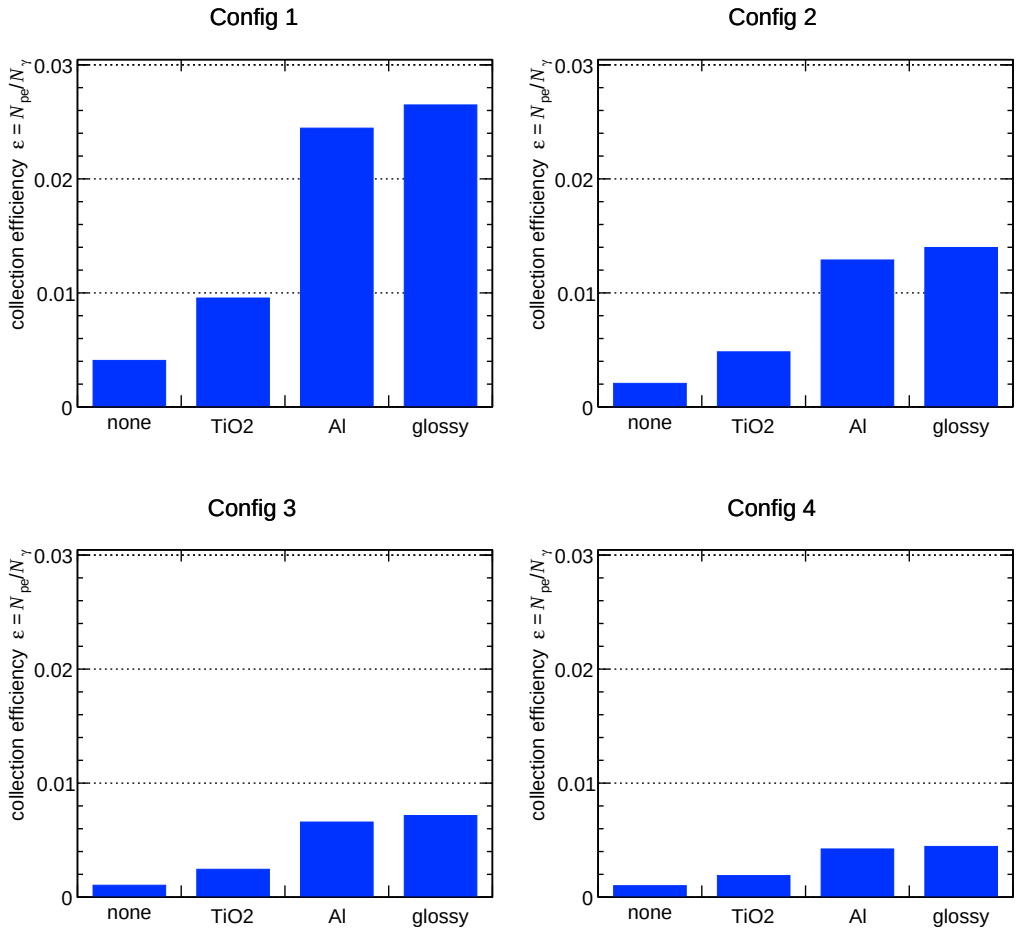


Figure 17: Plate-averaged collection efficiency $\varepsilon = \sum N_{\mathrm{pe}} / \sum N_{\gamma}$ for the four reflector treatments and four detector layouts, from the uniform whole-plate irradiation (25 000 events per panel). Glossy paper is brightest in every layout and air-backed aluminium is second; the contact-limited bare plate is the weakest wrapper in every layout.

Table 8: Plate-averaged collection efficiency (all SiPMs, uniform whole-plate irradiation; the pooled ratio of Eq. 4 over all events). The statistical uncertainties are below the quoted digits.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	4.081×10^{-3}	9.551×10^{-3}	2.445×10^{-2}	2.649×10^{-2}
Config 2 (4 SiPMs)	2.068×10^{-3}	4.834×10^{-3}	1.289×10^{-2}	1.398×10^{-2}
Config 3 (2 SiPMs)	1.048×10^{-3}	2.447×10^{-3}	6.588×10^{-3}	7.158×10^{-3}
Config 4 (1 SiPM, WLS)	1.007×10^{-3}	1.891×10^{-3}	4.228×10^{-3}	4.448×10^{-3}

A complementary, absolute figure of merit is the mean photoelectron yield per event, and it is the quantity the experiment actually measures, because the experiment reports a pulse height (or a threshold crossing) rather than a dimensionless efficiency. It is also the quantity that does *not* carry the model dependence of the PDE and the coupling, which is why both are quoted. Because the whole-plate beam covers the entire scintillator, almost every primary enters the slab, and 399 986 of the 400 000 events register at least one photoelectron (detection probability 0.999965). All 14 events without a detected photoelectron are in config 3 bare. The mean per event and the mean per *detected* event therefore differ negligibly. Table 9 lists the mean per event for every (config, wrapper) pair. Since the optical-photon production is set by the muon energy deposit and is independent of the wrapping, the yield is simply the plate-averaged efficiency of

Table 8 scaled by the (layout-independent) mean number of photons produced per event, so it reproduces exactly the same wrapper ordering.

Table 9: Mean photoelectron yield per event for the full-slab irradiation (whole-plate beam, 25 000 events per cell), for each layout and reflector treatment. Of the 400 000 whole-plate events, 399 986 detect at least one photoelectron (detection probability 0.999965; all 14 events without a detected photoelectron are in config 3 bare), so the mean per detected event differs negligibly from the mean per event; the statistical uncertainties are below the quoted digits.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	79.86	186.13	480.75	520.70
Config 2 (4 SiPMs)	40.38	94.12	253.79	275.13
Config 3 (2 SiPMs)	20.45	47.86	129.22	140.73
Config 4 (1 SiPM, WLS)	19.52	36.88	82.81	87.00

5.3 Survival versus mixing: the two axes that organise every result

The ordering of Table 8 is not a single effect, and it is worth separating it into two, because they pull in opposite directions and because almost every other result in this report follows from the same pair.

Survival is the fraction of the trapped light that survives the transport. It is governed by the critical angle at the plate's non-instrumented faces and by what happens to the sub-critical fraction that leaves the guided mode:

- the bare plate keeps the full TIR cone, but the sub-critical 39.3° cone that crosses the air gap reaches the black holder and is absorbed—so the bare plate keeps the cone but loses everything outside it;
- the diffuse paint destroys the cone altogether: there is no TIR at a `groundfrontpainted` surface, and the light is re-emitted isotropically, so most of it is re-absorbed on the next encounter;
- the two mirrors keep the cone and return the sub-critical fraction with a reflectivity of 0.90–0.96, which is why they are the brightest.

The bare plate is thus *not* the brightest, even though it keeps the full TIR cone, because the region outside the cone is exactly where a black holder catches everything. Adding the contact loss of Sec. 4.2—a loss on *every* bounce, not only on the sub-critical cone—takes it from third-brightest to weakest, which is what the measurement requires.

Mixing is the degree to which the light that survives has forgotten where it was produced. A cavity bounded by a specular mirror and a TIR guide is a good mixer; a cavity whose walls are Lambertian is a poor one, because each re-emission randomises the direction and the light leaves towards whichever wall is nearest, so the probability of reaching a sensor depends on the geometry of the plate and on the sensor positions. The three-patch data of Sec. 5 already shows the two extremes: the specular wrappers are flat to within 0.8 % to 3.4 % across the three patches, while the diffuse paint varies by a factor 1.75–1.87.

The useful statement, and the one that organises the rest of the report, is that these two axes are *not* independent. The wrappers that survive well (glossy, Al) mix well, because both properties come from the same specular+TIR boundary; the wrapper that mixes badly (the paint) survives badly, for the same reason. The bare plate is the interesting case: it mixes *excellently* (it is a perfect TIR guide, like the two mirrors) but survives badly (absorbing wall, plus the contact loss), and it is the only wrapper for which the two axes disagree. This is visible in the data as a bare plate that is dim, yet whose response still varies with position by a few to 19 %—the residual structure of a cavity that mixes well but not perfectly.

The consequence for the position reconstruction of Sec. 8 is direct: light sharing carries position information exactly when the mixing is poor, so the diffuse paint—the dimmest of the

reflective wrappers and the least uniform—is the only one whose *integrated charges* localise the muon, while the bright, perfectly mixing specular wrappers have to be read out in time instead.

6 The uniformity of the response over the plate

6.1 Full-plate maps

The same whole-slab events that give the plate average of Sec. 5 also resolve the response by entry position. Reconstructing the efficiency from the `sipmHits` tree in 5×5 mm cells gives the field

$$\varepsilon(x, y) = \frac{\sum_{e\,:\,(x_e,y_e)\in\text{cell}} \sum_{s\in\mathcal{S}} N_{\text{pe}}^{(e,s)}}{\sum_{e\,:\,(x_e,y_e)\in\text{cell}} N_{\gamma}^{(e)}}, \tag{5}$$

where the sums run over the whole-slab events whose primary-muon first-entry position (x_e, y_e) falls inside the cell, i.e. identically to Eq. 4 but binned in the entry position.

Figure 18 shows the complete set of full-plate maps for config 1, one panel per reflector. The plate-averaged efficiencies are 4.081×10^{-3} (bare), 9.551×10^{-3} (TiO_2), 2.445×10^{-2} (Al) and 2.649×10^{-2} (glossy). The contact-limited bare plate has a modest inner deficit—3.841×10^{-3} within $r < 60$ mm versus 4.135×10^{-3} outside, about 7.1 %—reflecting the holder-loss model. Aluminium and glossy paper are essentially flat (2.431×10^{-2} versus 2.448×10^{-2} and 2.636×10^{-2} versus 2.651×10^{-2}, i.e. 0.7 % and 0.6 %). TiO_2 is the only wrapper with a strong gradient, and it runs the other way: the inner disc ($r < 60$ mm, 6.729×10^{-3}) is *dimmer* than the outer annulus (1.018×10^{-2}), a factor of 1.51. The map itself is not a radial gradient at all: it is a *bright frame around a dark central cross*, and the brightest cells are the ones immediately adjacent to a SiPM (the cells reaching 1.215×10^{-2} against 6.40×10^{-3} in the central 50×50 mm cell). The specular transport erases the entry position almost completely.

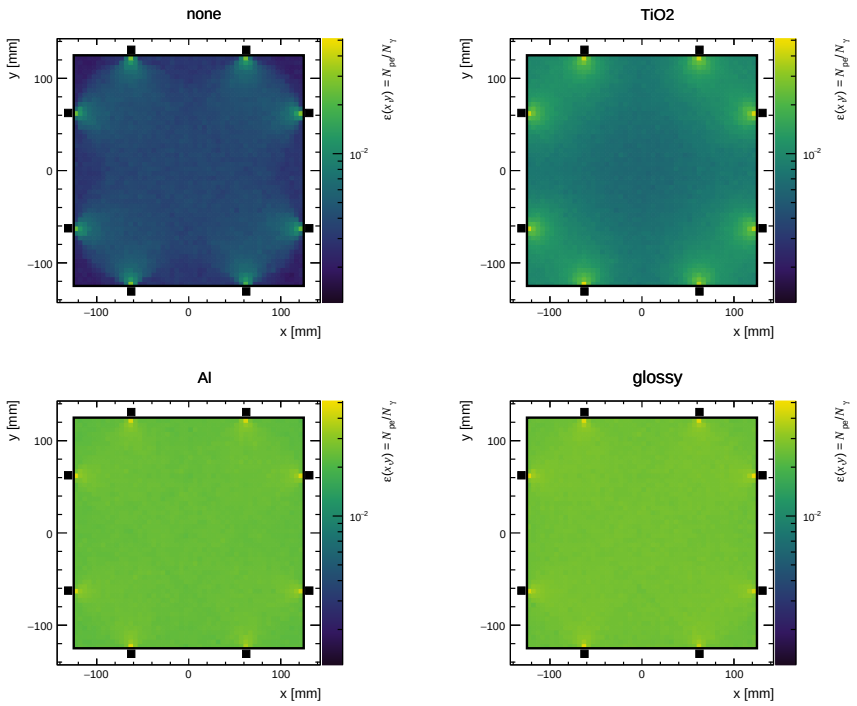


Figure 18: Full-plate collection-efficiency maps for config 1, from 25 000 whole-slab events per panel (5×5 mm cells, on a logarithmic colour scale shared by the panels). Aluminium and glossy paper are almost flat; the contact-limited bare plate (top left) is the weakest and has a modest inner deficit, while TiO_2 (top right) is the only one with a strong gradient, dimmest in a central cross and brightest towards the edges and corners. The bold rectangle is the 250 mm×250 mm slab outline and the black squares mark the eight SiPM positions.

Figure 19 repeats the whole-slab scan for config 2 (four sensors, one on each lateral face at the config-1 offsets). Every configuration is weaker than its config-1 counterpart—roughly halved, as expected from the reduced sensor area—while the ordering is unchanged. The bare plate gives $\varepsilon = 2.068 \times 10^{-3}$ (1.954×10^{-3} inside $r = 60$ mm versus 2.094×10^{-3} outside); TiO_2 reaches 4.834×10^{-3} with the same centre deficit (3.407×10^{-3} versus 5.153×10^{-3}), and Al and glossy reach 1.289×10^{-2} and 1.398×10^{-2}, both nearly flat. The bare $<$ TiO_2 $<$ Al $<$ glossy sequence is therefore robust against the sensor count: fewer sensors lower the overall light collection but do not change the physics of the wrapper.

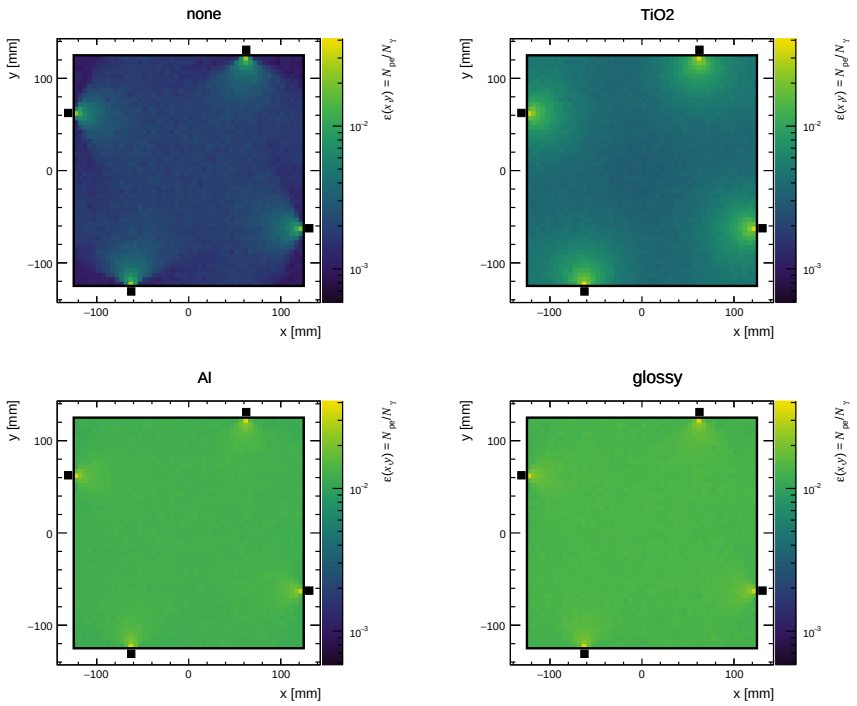


Figure 19: Full-plate collection-efficiency maps for config 2 (four SiPMs, one per lateral face at the config-1 offsets), otherwise identical to Fig. 18. The reflector ordering is the same, Al and glossy are almost flat, and each efficiency is about half the config-1 value.

Figure 20 shows config 3, in which both sensors sit on the same (top, $+y$) face. The integrated efficiencies are bare 1.048×10^{-3}, TiO_2 2.447×10^{-3}, Al 6.588×10^{-3} and glossy 7.158×10^{-3}, roughly half their config-2 values because concentrating both sensors on one face halves the solid angle available to photons born on the opposite side. Aluminium and glossy paper remain nearly flat (2.2 % and 2.0 % between the inner and outer regions). The TiO_2 panel is brightest near the instrumented top ($+y$) edge and dimmest on the far ($-y$) side: in the report frame the top/bottom efficiencies are 3.839×10^{-3} and 1.073×10^{-3}, a factor 3.58. The contact-limited bare plate shows the same sign but a much weaker asymmetry (1.217×10^{-3} versus 0.881×10^{-3}).

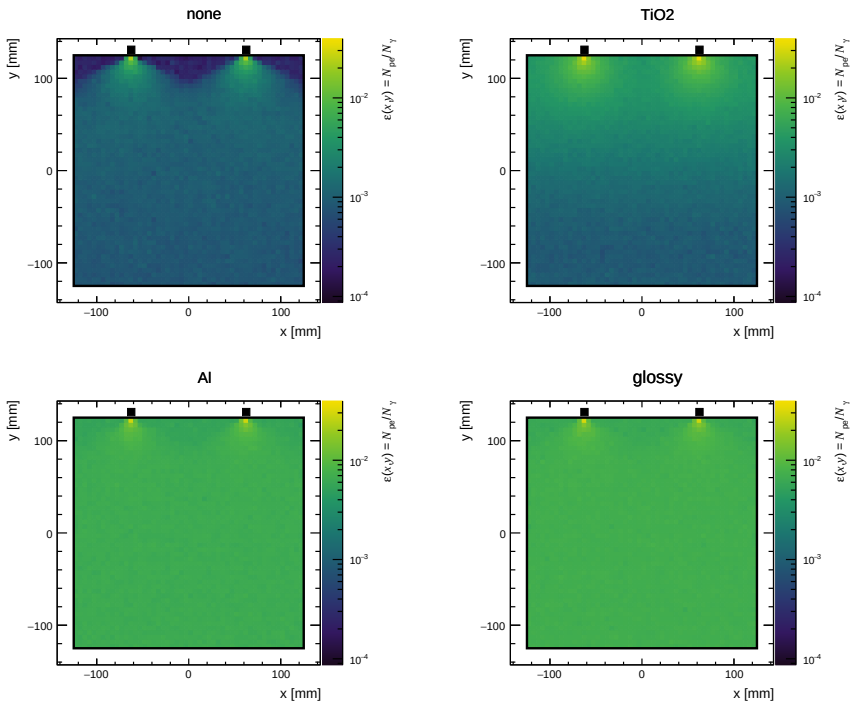


Figure 20: Full-plate collection-efficiency maps for config 3 (two SiPMs, both on the top $+y$ face). The TiO_2 panel shows the clearest top–bottom asymmetry, the contact-limited bare plate shows a weaker one, and the Al and glossy panels remain nearly flat.

Figure 21 completes the set with the single-SiPM WLS-fibre layout (config 4). The efficiencies are bare 1.007×10^{-3}, TiO_2 1.891×10^{-3}, Al 4.228×10^{-3} and glossy 4.448×10^{-3}, comparable to config 3 and with the same ordering bare $<$ TiO_2 $<$ Al $<$ glossy. Glossy paper is again the most efficient wrapper and air-backed Al is second. TiO_2 shows the clearest gradient: in the report frame the top/bottom efficiencies are 2.995×10^{-3} and 0.808×10^{-3}, a factor 3.71. The contact-limited bare plate shows a weaker but visible gradient (1.225×10^{-3} versus 0.783×10^{-3}, a factor 1.56), and Al and glossy agree to within about 9 % between the top and the bottom half, so even with a single readout face the specular transport largely erases the entry-point dependence.

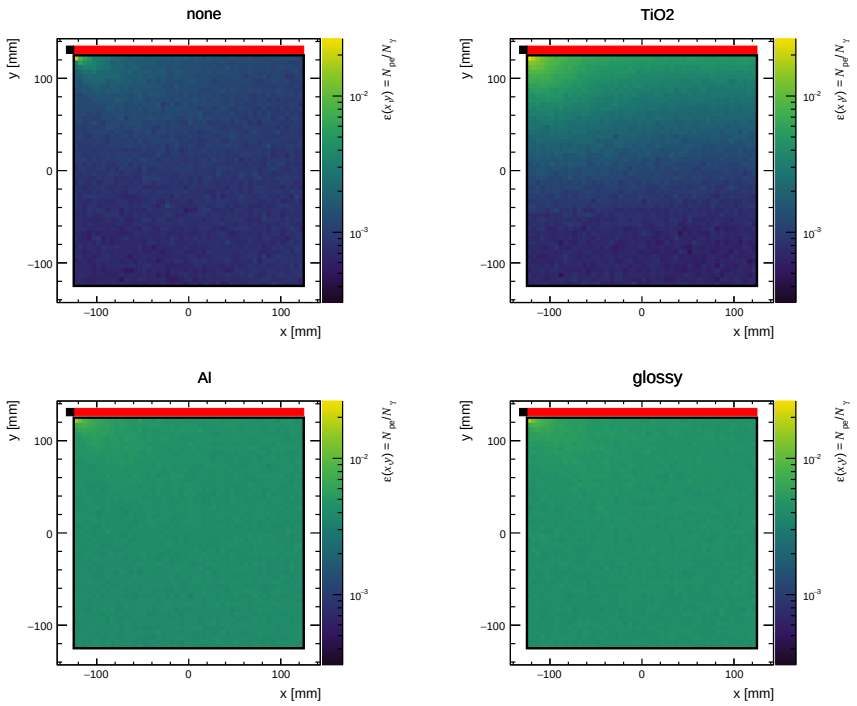


Figure 21: Full-plate collection-efficiency maps for config 4 (single SiPM + WLS fibre clamp). The efficiencies are comparable to config 3 and follow the same ordering, with the contact-limited bare plate now the weakest. The TiO_2 panel retains a strong top–bottom gradient, the bare plate a weaker one, while the Al and glossy panels are almost flat. The red bar along the top ($+y$) edge marks the WLS fibre clamp (the fibres run the full plate length along it) and the black square at its left end is the single read-out SiPM, so the fibres can be located relative to the read-out. Same report frame and binning as Figs. 18–20, each panel on its own logarithmic colour scale.

Across all four layouts glossy paper is the brightest wrapper and air-backed aluminium the second, while the weakest is the contact-limited bare plate in every layout. The efficiency scales roughly with the collected photosensitive area, confirming that the readout layout sets the overall scale while the wrapper sets the ordering.

6.2 A single number for the spread: $\kappa_{2\mathrm{D}}$

A purely radial contrast is deliberately *not* used. The read-out layouts have no continuous rotational symmetry—configs 1 and 2 are four-fold symmetric, configs 3 and 4 single-sided—and the plate is square, so the collection efficiency is not a function of radius alone. Beyond the inscribed circle ($r > 125\,\mathrm{mm}$) only the four corner wedges of the plate are populated, so a fixed-width radial bin would mix a shrinking range of directions and its radial profile $\varepsilon(r)$ would reflect the geometry of the square rather than the detector. The uniformity is therefore quantified directly in the plate plane (x, y).

The *two-dimensional contrast*

$$\kappa_{2\mathrm{D}} \;=\; \frac{\max_{\mathrm{cell}}\,\varepsilon}{\min_{\mathrm{cell}}\,\varepsilon} \tag{6}$$

collapses the field to a single number, the ratio of the brightest to the darkest region; it answers "how large is the spread" and is the quantity that ranks the layouts and reflectors. $\kappa_{2\mathrm{D}}$ is evaluated on a coarser 5×5 grid of 50 mm cells, because on the 5 mm cells of the maps the per-cell statistics are too thin for a stable max/min—Poisson noise in the extreme cells would inflate it. The measure is therefore deliberately the *robust* dynamic range, and is smaller than the raw peak-to-trough range of the fine maps. No symmetry is assumed, so $\kappa_{2\mathrm{D}}$ is meaningful for the single-sided layouts 3 and 4 as well as for the four-sided ones.

The direction-sensitive contrast κ_{2D} ranks the reflectors consistently in *every* layout,

$$\mathrm{TiO_2} \;>\; \text{bare} \;>\; \mathrm{Al} \;\gtrsim\; \text{glossy},$$

i.e. the diffuse TiO_2 paint is the *least* uniform wrapper and glossy paper the most uniform. The diffuse paint is also the dimmest direct-coupled wrapper, while the bright specular wrappers are the most uniform: the specular transport mixes the light so thoroughly that the entry position is largely erased ($\kappa_{2D} \approx 1.1$–1.4), whereas the diffuse paint retains a pronounced positional response. The contact-limited bare plate sits between them because the holder loss is itself position dependent. Numerically the TiO_2 contrast grows from $\kappa_{2D} = 1.911$ and 2.579 in the four-sided layouts 1 and 2 to 8.413 and 10.897 in the single-sided layouts 3 and 4; the bare plate spans 1.56–4.14, aluminium 1.07–1.37 and glossy 1.07–1.36. A muon crossing anywhere on the plate is therefore collected within about 40 % of the most favourable point for Al/glossy, within a factor of 4.1 for the contact-limited bare plate, and within a factor of 10.9 for TiO_2 in the single-sided layouts. Figure 22 collects all sixteen values as bar charts, one panel per layout; Table 10 lists the same values numerically, resolving the near-unity entries that the linear bar scale cannot.

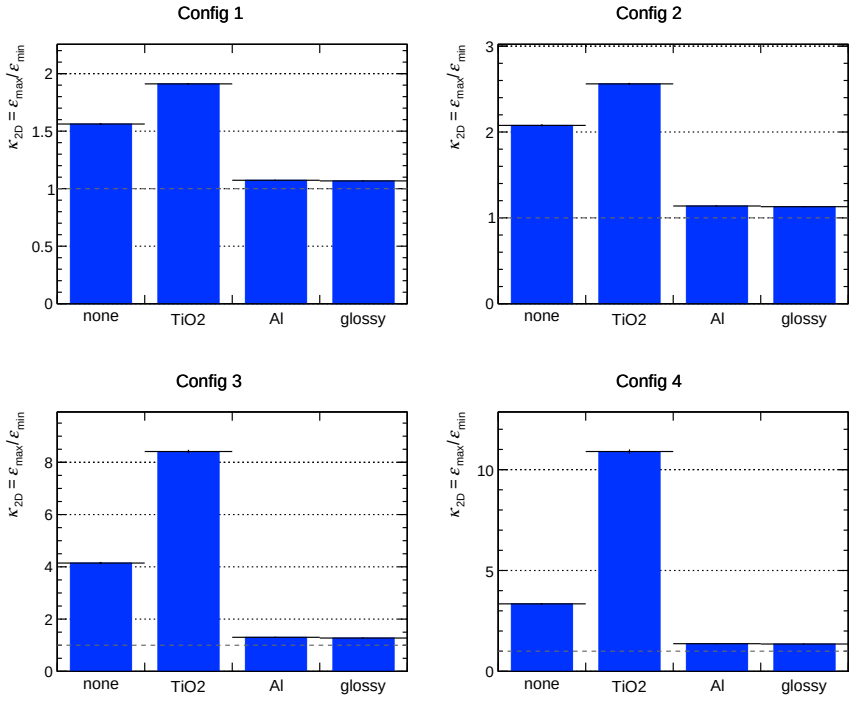


Figure 22: Full-plate position contrast $\kappa_{2D} = \varepsilon_{\max}/\varepsilon_{\min}$ over the 50 mm cells of the whole-slab map, one panel per layout and one bar per reflector (each panel on its own linear scale). This is the same kind of comparison as the three near-edge patches, but evaluated over the entire plate; the dashed line marks unity (a perfectly uniform response). TiO_2 shows by far the largest contrast in every layout, the contact-limited bare plate is second, and Al and glossy paper stay closest to unity.

Table 10: Full-plate position contrast κ_{2D} over the 50 mm cells of the whole-slab map, per layout and reflector, with the propagated Poisson uncertainty. These are the values plotted in Fig. 22; the table resolves the near-unity entries that the linear bar scale cannot, and shows the TiO_2 > bare > Al $\gtrsim$ glossy ordering in every layout. The number in parentheses is the uncertainty in the last digits of the preceding value; because the statistic is an extreme value, the propagated Poisson uncertainty is a lower bound on the true spread.

Reflector	Config 1	Config 2	Config 3	Config 4
Bare	1.562(8)	2.077(14)	4.145(43)	3.346(33)
TiO_2	1.911(7)	2.561(12)	8.413(67)	10.897(104)
Al	1.073(2)	1.138(3)	1.303(5)	1.371(6)
Glossy	1.068(2)	1.130(3)	1.275(5)	1.356(6)

The single-sided layouts are the most demanding. The largest TiO_2 two-dimensional contrast of the entire study appears in config 4 ($\kappa_{2D} = 10.897$), just above config 3 (8.413): the front–back asymmetry of the diffusely wrapped plate is far stronger than any radial average reveals. The Al and glossy wrappers in the same layouts stay below 1.4, whereas the contact-limited bare plate reaches 4.14 in config 3 and 3.35 in config 4. A single-sided layout is simply not described by a radius, so only the full (x, y) map—not a radial profile—carries the information.

6.3 The shape of the field: distance to the nearest sensor

The contrast number of the previous subsection says how large the spread is, but not what shape it has, and the maps show that the shape is not radial at all. For config 1 the bare plate maps to a *checkerboard*: the 50×50 mm cell efficiencies alternate between 3.3×10^{-3} and 5.2×10^{-3} from cell to cell, with the bright cells being exactly those that contain a SiPM. For config 2 the bare plate shows the same thing more sharply: the four cells that contain the four sensors read 3.21–3.42 $\times 10^{-3}$ while the remaining twenty-one read 1.65–2.07 $\times 10^{-3}$. For config 3 the bare plate has two bright cells (2.20×10^{-3} and 2.45×10^{-3}) at the two sensor positions, but the cell *between* them—on the instrumented face, but not on a sensor— is the darkest of the whole plate (0.59×10^{-3}): light produced there leaves through the uninstrumented part of that face before it can reach a sensor. The diffuse TiO_2 paint shows the complementary structure: a bright frame, with the cells adjacent to the sensors reaching 1.215×10^{-2} and the central cross falling to 6.4–8.0 $\times 10^{-3}$.

These patterns are all produced by the same rule, and the rule has a single coordinate: *the response tracks, on average, the distance from the interaction point to the nearest collection site.* Where the cavity mixes the light well, the response is independent of that distance; where it mixes poorly, the response falls off as the interaction point moves away from the sensors, because the light then has a longer and more lossy path to reach one. The rule is a statement about the *envelope* of the map and not a point-by-point description of it: the cell between the two sensors of config 3 quoted above is a counterexample to a strict monotonicity, and the reason is visible—that cell lies on the instrumented face, so its light leaves the plate immediately through the part of that face that carries no sensor. The distance statistic below integrates over such configurations, so what it measures is the trend that survives the averaging.

Figure 23 tests this directly, by binning the whole-slab events in the distance d to the nearest collection site and plotting the efficiency normalised to the plate average against it. For configs 1–3 the collection sites are the SiPM tiles; for config 4 the collection aperture is the fibre clamp, a line along the top edge, so d is the distance to that line. The four curves of each panel share one shape but not one level:

- the two specular wrappers are flat to within 5 % over the whole range, in every layout—they are near-perfect light mixers;
- the contact-limited bare plate shows a clear excess in the first 25 mm (1.44×, 2.12× and 3.36× the plate average for configs 1–3) and then relaxes to slightly *below* unity, which is the

signature of a cavity that mixes well but loses a little light on every extra bounce;

- the diffuse paint shows the largest excess (1.63×, 2.40× and 4.52×) and the steepest decay, reaching 0.36–0.37 of the average at the far side of the plate in the single-sided layouts.

This single figure therefore contains the explanation of the whole uniformity result: the specular wrappers are uniform because they mix, the paint is non-uniform because it does not, and the bare plate sits in between because the contact loss destroys the perfect mixing of the air-gap guide without destroying its geometry. It also explains the sign flip of the position trend seen in the three-patch scan: in the four-sided layouts the corner patch is close to two sensors and is therefore the brightest, while in the single-sided layouts the sensors are on the far side of the plate and the centre of the plate—the point furthest from any sensor—is not the dimmest point, because what matters is the distance to the *nearest* sensor, and the single sensor still beats the far corners.

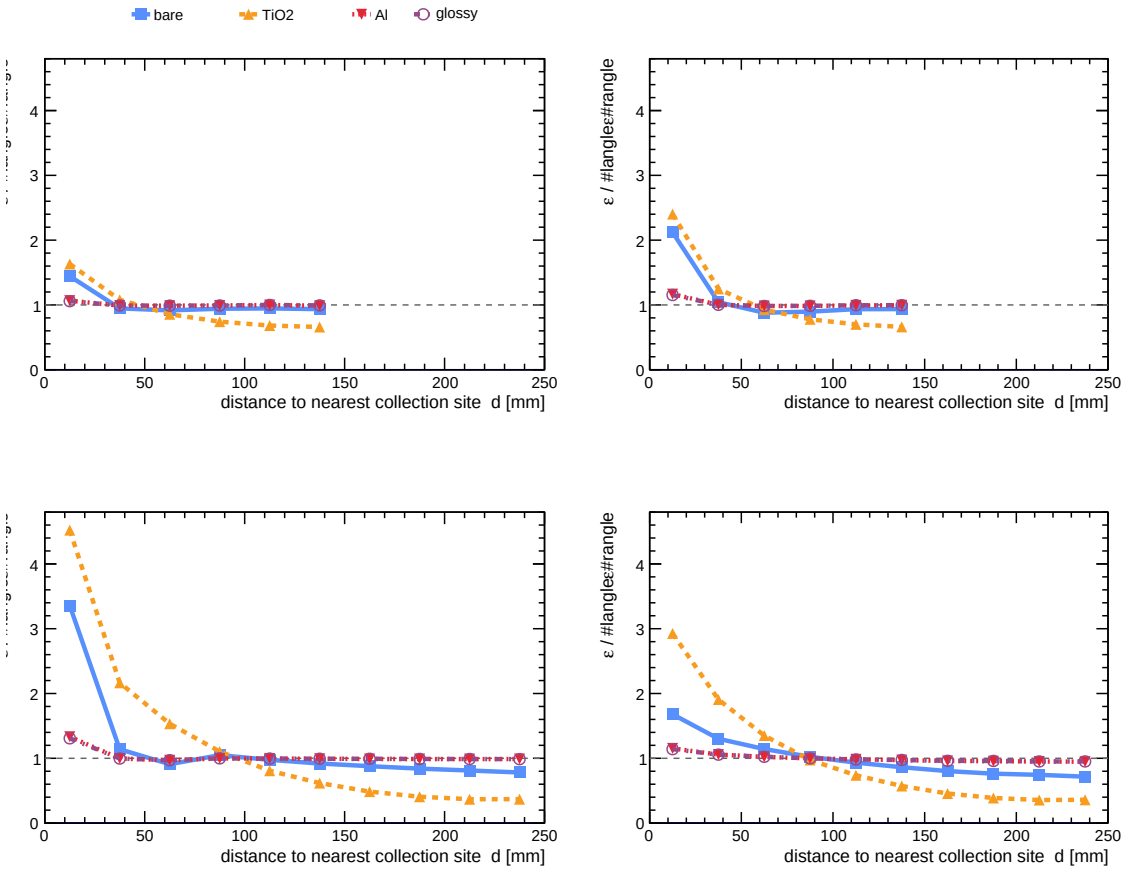


Figure 23: The response as a function of the distance to the nearest collection site. Each panel is one layout; the abscissa is the distance d from the muon's entry point to the nearest SiPM tile (configs 1–3) or to the WLS fibre line along the top edge (config 4), and the ordinate is the collection efficiency normalised to the plate average of the same layout and treatment. The dashed line is the plate average. The four treatments share one shape and differ only in level: the specular wrappers are flat because they mix the light thoroughly, while the diffuse paint and the contact-limited bare plate are brightest close to a sensor and fall away from it. Error bars are the propagated Poisson uncertainties.

7 The spectral and temporal signature of the detected light

7.1 Wavelength of detected photons

Figure 24 shows the wavelength distribution of detected photoelectrons, weighted by their count, for the four reflectors. The distribution peaks near 456 nm for the reflective wrappers (config-

1 centre means of 456.86 nm for glossy, 456.15 nm for Al and 454.45 nm for TiO_2), i.e. slightly red-shifted with respect to the 425 nm emission peak. This is the expected consequence of self-absorption: the shorter-wavelength part of the emission overlaps the absorption band and is attenuated over the multi-bounce path, so the surviving and detected spectrum is red-shifted. The contact-limited bare plate is bluer (449.46 nm) because the holder-contact loss suppresses the longest multi-bounce paths before they can redden the spectrum; its lower yield still leaves a well-defined distribution.

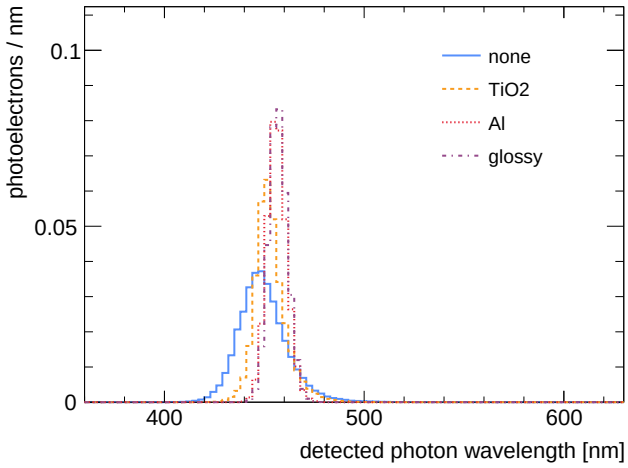


Figure 24: Wavelength distribution of detected photoelectrons (config 1), weighted by detected count. The reflective wrappers give a distribution peaking near 456 nm, red-shifted from the 425 nm emission peak by self-absorption over the multi-bounce path.

Figure 25 compares the detected-photon spectrum of the four configurations for each wrapper. Configurations 1–3, in which the SiPMs view the tile directly, all detect the same blue scintillation spectrum: the *mean* wavelength is essentially independent of the number of sensors, moving only from 457.01 nm (glossy, config 1) to 457.50 nm (config 3), the small increase reflecting the slightly longer collection path of the single-sided layouts. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, so the detected spectrum is shifted to 498.5 nm (glossy) — a green-shift of some 42 nm. The same shift appears for every wrapper, including the bare tile (near 499 nm), confirming that it originates in the fibre wavelength shifter rather than in the wrapper. The detected config-4 light is redder than the Y-11 emission peak (476 nm) because the blue side of the emission is preferentially re-absorbed along the fibre before reaching the read-out.

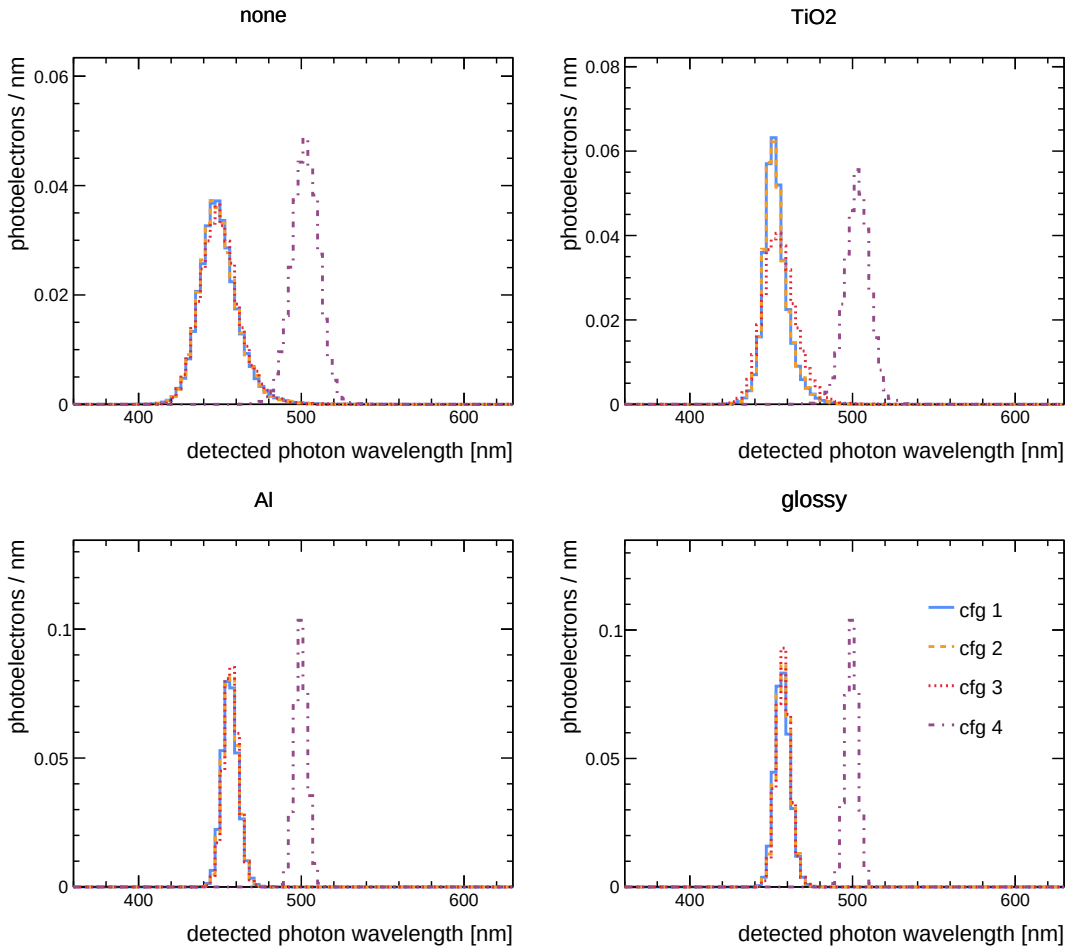


Figure 25: Detected-photon wavelength distribution, weighted by photoelectron count, for each wrapper (panel) and each configuration (curves), using the pooled data of the three irradiation positions. Configurations 1–3 follow the 457 nm blue scintillation spectrum regardless of sensor count, whereas the WLS-fibre readout of config 4 is green-shifted to ≈ 510 nm by the Y-11(200) emission band and the self-absorption along the fibre.

7.2 Arrival times and pulse persistence

The collection-efficiency maps above treat every detected photoelectron as a point; the per-photoelectron timing columns of the `sipmHits` tree let the *arrival-time structure* of the detected light be examined as well. Figure 26 shows the oscilloscope-style persistence map of the arrival-time profile for the four layouts (glossy paper). For each event the per-SiPM arrival times are summed over the layout's sensors and histogrammed in time, giving a trace of photoelectron count versus arrival time; stacking events and recording, for every 0.1 ns time bin, how many events showed a given count, yields a two-dimensional map whose abscissa is the arrival time, whose ordinate is the instantaneous photoelectron count, and whose colour is the number of events. The rising edge and the decaying tail therefore show how the detected light builds up and dies away, and the vertical spread at a given time is the event-to-event fluctuation of the instantaneous rate. For the three directly-read layouts the modal mean arrival times are 3.85 ns, 3.95 ns and 3.75 ns; the prompt peaks reach 54, 43 and 47 photoelectrons per 0.1 ns bin in configs 1–3, well above the 30 photoelectrons per bin that the shared ordinate of the figure can show, so the top of the prompt peak is clipped in the display; the single-fibre WLS read-out (config 4) peaks markedly later, at 7.95 ns and at only 18 photoelectrons per bin, which is therefore fully visible. The fractions arriving within 5 ns are 31.6, 30.4, 29.8 and 13.4 % for configs 1–4, within 10 ns 63.2, 61.3, 60.2 and 40.3 %, while the late ($t > 20$ ns) tail carries 10.7, 12.3, 13.1 and 21.5 % of the

photoelectrons.

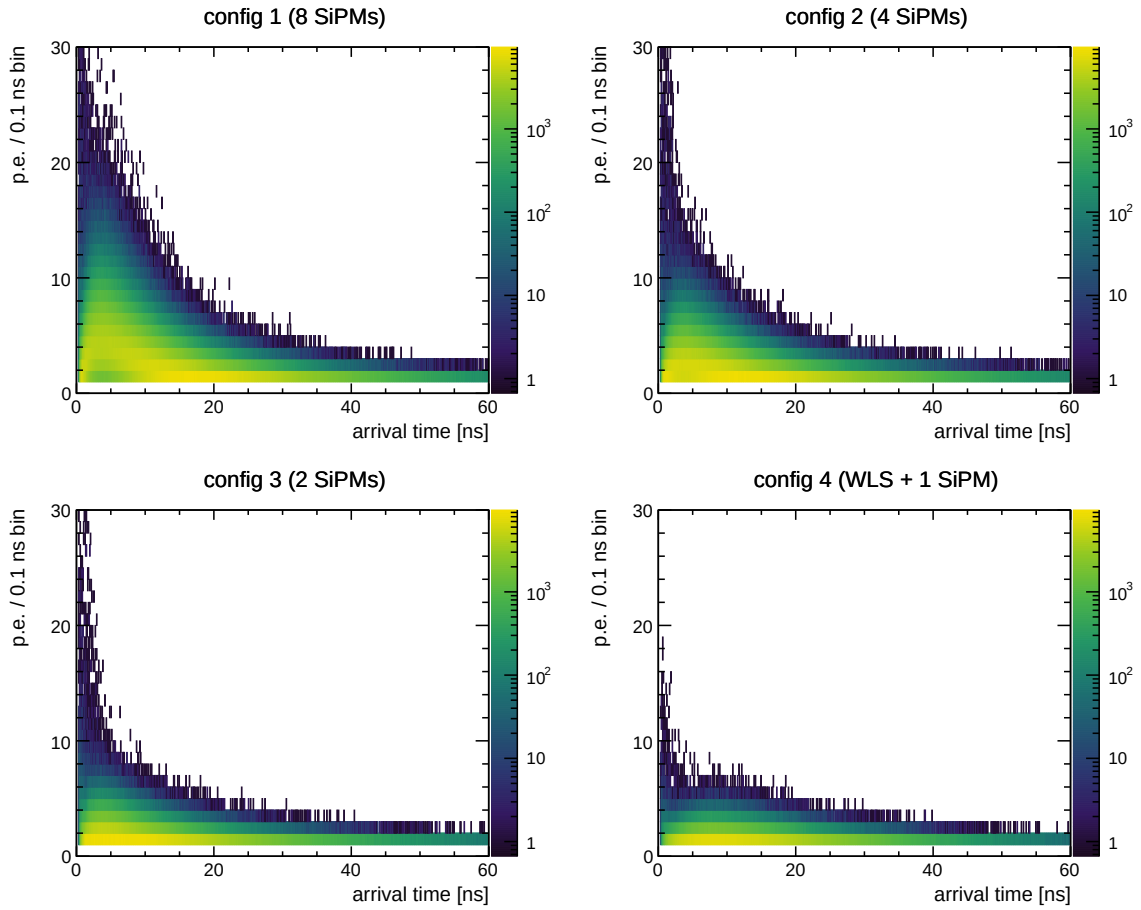


Figure 26: Arrival-time persistence map of the detected photoelectrons, one panel per layout (glossy paper, whole-plate irradiation). The abscissa is the global arrival time, the ordinate the number of photoelectrons in the corresponding 0.1 ns time bin (summed over the layout's SiPMs), and the colour the number of events reaching that point (shared logarithmic scale). Empty time bins are omitted; the ordinate is shared across the four panels and linear over 0–30 photoelectrons per bin (1 p.e. per bin), the arrival-time axis over 0–60 ns. The prompt rise reflects the direct scintillation and Čerenkov light, while the slower tail is the multi-bounce transport and delayed (self-absorbed and re-emitted) light. The prompt peak of configs 1–3 extends above the shared ordinate and is therefore clipped; the WLS layout (config 4) builds up roughly 4.1 ns later than the others, its prompt peak shifted by the fibre absorption and re-emission.

The same arrival-time data can be collapsed into one average waveform per layout by averaging the per-event traces of Fig. 26 over events; Fig. 27 overlays the four averages. This makes the amplitude hierarchy and the time shifts explicit. Integrating the mean rate over the full 60 ns window recovers 520.7, 275.1, 140.7 and 87.0 photoelectrons per event for configs 1–4 (the same ordering found in the collection-efficiency maps), and the rising edges of the three directly-read layouts coincide within a fraction of a nanosecond, whereas the WLS read-out of config 4 rises about 4 ns later and decays more slowly – its late tail stands out on the logarithmic ordinate as a straighter, shallower fall.

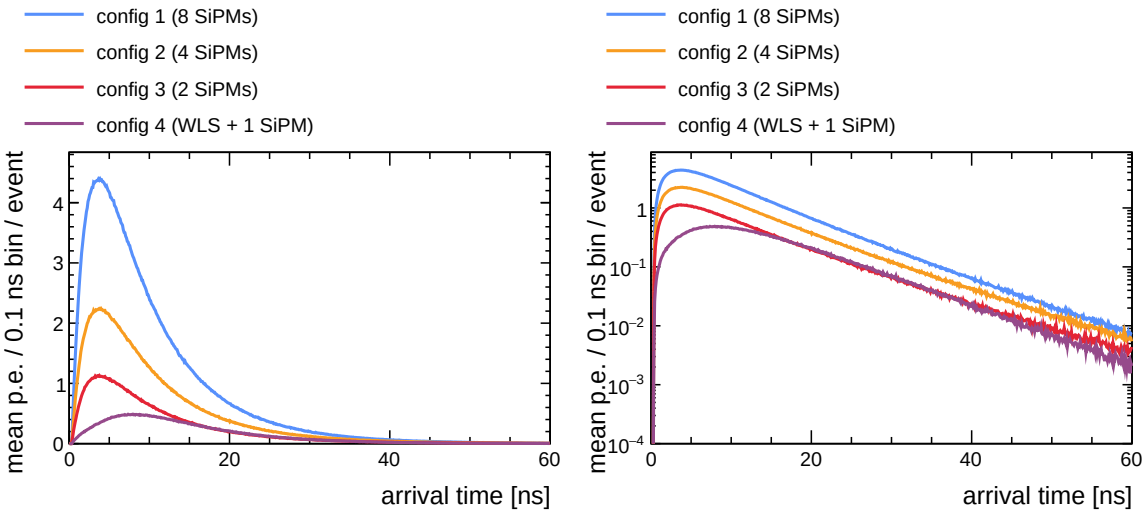


Figure 27: Average arrival-time waveform of the detected photoelectrons, one curve per layout (glossy paper, whole-plate irradiation), obtained by averaging the per-event traces of Fig. 26 over events. The abscissa is the global arrival time and the ordinate the mean number of photoelectrons in each 0.1 ns bin per event. The left panel uses a linear ordinate and the right a logarithmic one; all four configurations are drawn on each panel (the prompt peaks of configs 1–3 nearly coincide and extend above the linear range). The integrated mean yields are 520.7, 275.1, 140.7 and 87.0 photoelectrons per event for configs 1–4. The WLS layout (config 4) peaks roughly 4.1 ns later than the others and retains a long, slowly-decaying tail, the average-waveform counterpart of the persistence map.

Figure 28 combines the two offline observables: the arrival time and the wavelength of each detected photoelectron. Configurations 1–3 collect the blue scintillation spectrum directly, with a photoelectron-weighted mean of 456.7–457.2 nm, whereas the Y-11 WLS read-out of config 4 is green-shifted to 498.5 nm, consistent with the steady-state spectra of Fig. 25. In every layout the spectrum reddens with arrival time: the mean wavelength rises from 450.9 nm ($t < 5$ ns) to 467.7 nm ($t > 20$ ns) for configs 1–3, and from 471.8 nm to 507.7 nm for config 4, while the rms spread narrows (37.9 $\rightarrow$ 35.5 nm and 44.1 $\rightarrow$ 26.5 nm, respectively). The config-4 early light is the direct blue component transmitted through the slab–pad contact; the late light is dominated by the green WLS component. The depleted blue edge of the late light is the signature of self-absorption along the multi-bounce path and, in config 4, of the additional wavelength-dependent attenuation of the WLS fibres.

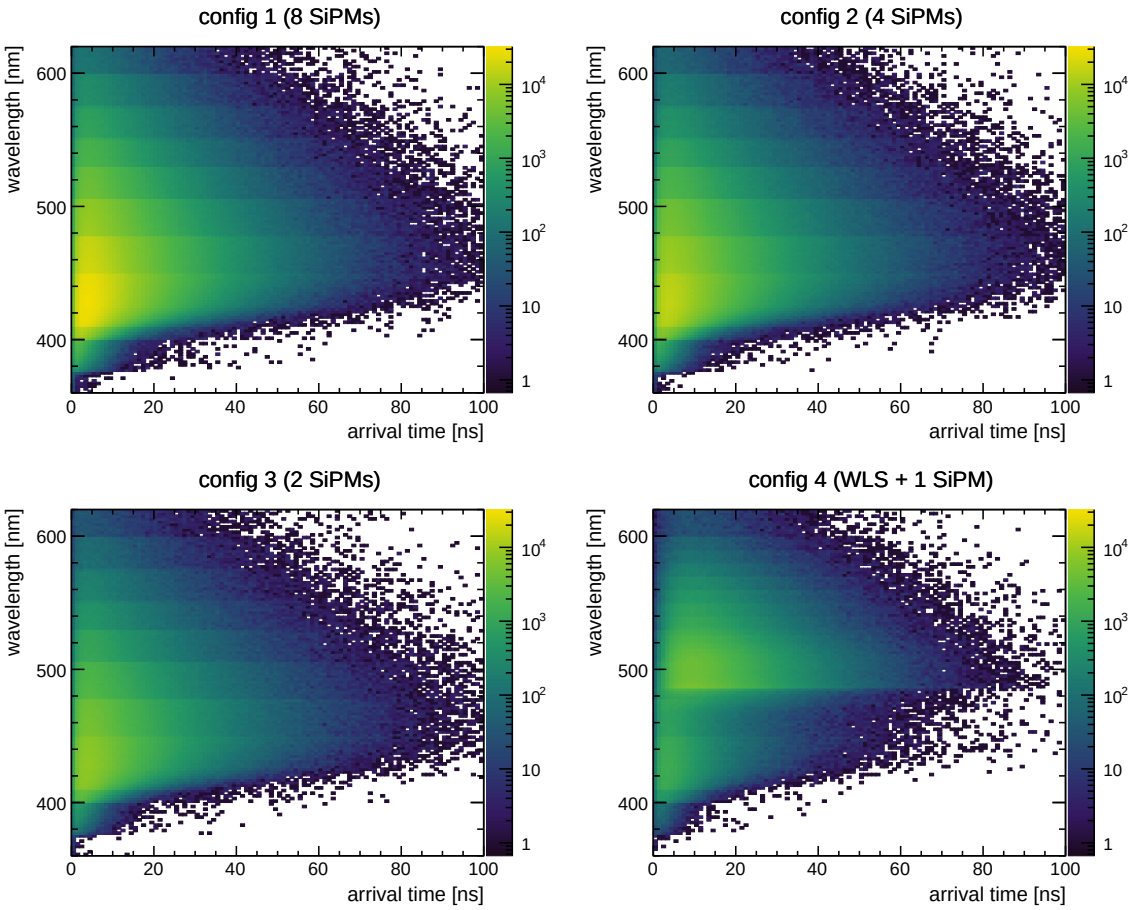


Figure 28: Detected wavelength versus arrival time, one panel per layout (glossy paper, whole-plate irradiation), on a shared logarithmic colour scale (number of photoelectrons). Configurations 1–3 populate a narrow band around the 457 nm blue scintillation spectrum; config 4 is green-shifted by the Y-11 WLS fibres. The time dependence of the mean wavelength traces the self-absorption of the shorter-wavelength light over the multi-bounce path (from 450.9 nm to 467.7 nm in configs 1–3 and from 471.8 nm to 507.7 nm in config 4, between the early and late arrival windows).

8 Position reconstruction from the SiPM pattern

The eight SiPMs of config 1 view the same scintillation event from eight directions. A natural question for a large-area muon trigger built this way is whether the *relative* pattern of the eight channels—their integrated charges, their arrival-time wavefronts, and the prompt fraction of their charge—carries enough information to localise the interaction point inside the plate. The whole-plate irradiation of Sec. 5 is the data set that answers it: it samples the interaction point uniformly over the 250 mm×250 mm footprint and records, for every event, both the ground truth (the muon's first-entry position) and the complete per-SiPM photoelectron timestamp list.

8.1 What the eight channels measure

For every event the `sipmHits` tree carries, per SiPM $s \in \{0, \ldots, 7\}$, the number of detected photoelectrons N_s, the ordered arrival times $t_{(1),s} \le t_{(2),s} \le \cdots \le t_{(N_s),s}$ of the individual photoelectrons in the variable-length branch `tDetected`, the number of photoelectrons that arrived within 3 ns of the interaction N_s^{prompt} (the prompt fraction), and the mean detected wavelength. These are the optical counterparts of the quantities a read-out chain provides: an integrated charge, a leading-edge or constant-fraction timestamp, and a pulse-shape parameter.

It is worth being explicit about what N_s is, because the distinction matters for the electronics.

N_s counts *every* photoelectron detected in the channel over the whole event, so it is the *integral* of the light pulse delivered to that SiPM (multiplied by the gain), not the amplitude of the pulse peak. This is the natural choice: the integral carries the full Poisson statistics of the collected charge, whereas a peak-amplitude measurement uses only the prompt subset of the light and is additionally biased low by the spread of the photoelectron arrival times. In the present data the prompt fraction (photoelectrons arriving before 3 ns) is 53.0 % of the light for the bare plate, 33.7 % for the TiO_2 paint and only 15 % to 16 % for the two specular wrappers, so a peak-amplitude read-out would work with a small minority of the collected photons. No peak-amplitude observable is used anywhere in this section; the position information is extracted from pulse integrals and from arrival times.

Only the transverse position can be reconstructed: the muon is vertical and traverses the whole 10 mm thickness, so the light is produced along a vertical track and the side pattern carries no usable z information (the corresponding path-length difference between the two faces of the plate is 10 mm, i.e. 0.05 ns, an order of magnitude below the timing signals below).

8.2 A better clock than the first photon

The simplest timing observable is the arrival time of the first photoelectron, $t_s^{\mathrm{first}} = t_{(1),s}$. It is convenient, but it is a poor clock, for two reasons that act in opposite regimes. A channel that collects more photoelectrons triggers earlier simply because it has more chances to catch an early photon: the shift of the trigger time scales as $N_s^{-1/2}$, the same amplitude walk that a fixed-threshold discriminator suffers in an electronic chain. In the diffuse TiO_2 wrapper, where a corner muon delivers ~ 68 photoelectrons to the two sensors at that corner and only ~ 6 to the opposite sensors, this walk alone shifts t^{first} at the bright sensors by several hundred picoseconds, with no relation whatsoever to the distance of the muon. In the opposite regime, a channel that collects only a handful of photoelectrons is timed by a single photon, and its trigger time fluctuates from event to event depending on whether one prompt photon happens to enter the sensor aperture on its first pass. The extreme-value statistics that quantify the first effect are collected in Appendix B.

The standard cure in an electronic read-out chain is a constant-fraction discriminator, which triggers not on the amplitude of the pulse but on the moment at which the pulse has reached a fixed fraction α of it [8]. The exact optical counterpart is the *wavefront time* of the channel: the time by which a fraction α of the light of that channel has arrived, i.e. the mean arrival time of the leading α fraction of its photoelectrons,

$$\bar{t}_{\le\alpha,s} = \frac{1}{q_s}\left(\sum_{j=1}^{m_s} t_{(j),s} + (q_s - m_s)\, t_{(\min(N_s,\, m_s+1)),s}\right), \qquad q_s = \max(1,\, \alpha N_s), \quad m_s = \lfloor q_s \rfloor. \tag{7}$$

Averaging over the leading photoelectrons removes the $N_s^{-1/2}$ amplitude walk, because every channel is then timed on the same fraction of its own pulse rather than on its first photon; and it suppresses the single-photon shot noise, because the timing information is spread over the $\lceil \alpha N_s \rceil \approx 8$ photoelectrons of the prompt component instead of resting on one. Below we call this the *optical constant-fraction (CFD) time* of the channel.

We use $\alpha = 12.5\,\%$. The choice is dictated by the cavity physics of Sec. 7.2: in the specular wrappers the prompt, ballistic component carries 15 % to 16 % of the total pulse ($\sim$ 8–10 photoelectrons per channel) while the remaining 84 % to 85 % forms the spatially mixed multi-bounce reverberation tail ($t > 3$ ns). A high constant fraction such as $\alpha = 50\,\%$ therefore times the reverberating cavity tail rather than the ballistic wavefront and progressively washes out the timing information, whereas $\alpha = 12.5\,\%$ integrates across the ballistic rising edge just below the reverberation transition.

That argument is checked directly in Fig. 29, which measures the reconstruction error as a function of α over $\alpha = 3\,\%$ to $50\,\%$ for all four wrappers. The result confirms the picture and

shows that the choice is not critical: over the whole range 3 % to 20 % the error of the learned estimator changes by less than 6 % (e.g. 16.9 mm at $\alpha = 3\,\%$ and 16.2 mm at $\alpha = 12.5\,\%$ for the glossy wrapper), and the nominal value is within 1 % of the best one over the scanned range. The position of that minimum is not resolved more precisely than about $\alpha = 12.5\,\% \pm 2.5\,\%$: the differences between neighbouring points near the bottom of the curve ($\lesssim 1\,\%$) are of the order of the statistical precision of the error itself (about 1 % with 5000 test events). The transparent linear read-out is more sensitive, because in the two specular wrappers it is the only channel that carries the position information: there the error has a clear minimum at $\alpha \simeq 10\,\%$ to 15 % and grows by a factor ~ 1.5 at $\alpha = 50\,\%$ (from 182 mm to 279 mm in glossy paper), which is exactly the loss of timing sensitivity when the constant fraction reaches the reverberation tail. The bare plate and the diffuse paint, whose positions are set by the light sharing and not by the timing, are flat in α throughout. At $\alpha = 50\,\%$ the learned estimator still keeps most of its resolution (from 16.2 mm to 16.4 mm in glossy paper), because the charges and the prompt share continue to carry the position information, whereas the transparent timing read-out loses half of its sensitivity.

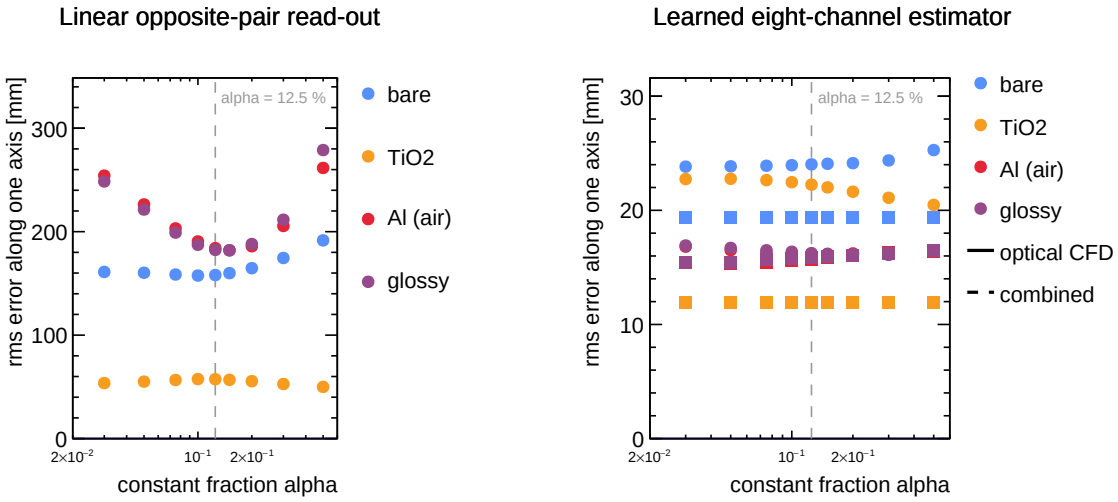


Figure 29: Position resolution as a function of the constant fraction α of the optical CFD time. *Left*: the transparent linear opposite-pair read-out; *right*: the learned eight-channel estimator, with the optical CFD channel (filled markers, solid lines) and the combined estimator (open markers, dashed lines). One curve per wrapper; the dashed vertical line marks the nominal $\alpha = 12.5\,\%$. The learned estimator is almost insensitive to the choice of α, while the linear timing read-out of the two specular wrappers has a clear minimum near $\alpha = 10\,\%$ to 15 % and degrades by a factor ~ 1.5 once the fraction reaches the multi-bounce reverberation tail.

Pairing the wavefront onset $t_{(1),s}$ with the leading-α wavefront mean $\bar{t}_{\leq\alpha,s}$ captures *two* physical characteristics of the pulse in each channel: the arrival time of the ballistic wavefront and its *optical rise time*

$$\Delta t_{\mathrm{rise},s} = \bar{t}_{\leq\alpha,s} - t_{(1),s}, \tag{8}$$

which measures the temporal dispersion of the rising edge (steep for a sensor receiving a direct burst from a nearby muon, and broad for a distant sensor accumulating multi-bounce light).

8.3 Two ways to localise the muon

Two families of estimator are evaluated on the whole-plate dataset. The first is transparent and purely analytic: the *opposite-pair* observables

$$A_x = \frac{N_{\mathrm{right}} - N_{\mathrm{left}}}{N_{\mathrm{right}} + N_{\mathrm{left}}}, \qquad T_x = \bar{t}_{\mathrm{right}} - \bar{t}_{\mathrm{left}}, \qquad T_{\mathrm{CFD},x} = \bar{t}_{\leq 12.5\%,\mathrm{right}} - \bar{t}_{\leq 12.5\%,\mathrm{left}}, \tag{9}$$

formed from the two SiPMs on the right and the two on the left edge (and the analogous A_y, T_y, $T_{\mathrm{CFD},y}$ for the bottom and top edges). N_{right} and N_{left} are the summed pulse integrals of the two channels on each edge, while $\bar{t}_{\mathrm{right}}$ and $\bar{t}_{\leq 12.5\%,\mathrm{right}}$ are the edge-averaged first-arrival and 12.5 % wavefront times. All three observables are odd functions of the coordinate x, so a linear fit against x measures the first-order response slope $\partial O_x/\partial x$ and rms residual σ_O, yielding an implied linear resolution $\sigma_x = \sigma_O/|\partial O_x/\partial x|$.

Because the eight-channel pattern in a rectangular light guide is strongly non-linear—particularly in the timing channel and near the plate corners—a linear combination of opposite edges extracts only a fraction of the available information. The second estimator therefore learns the inverse map $\mathbf{f} \mapsto \mathbf{y} = (x, y) \in [-L, L]^2$ ($L = 125\,\mathrm{mm}$) non-parametrically from a calibration training set of size $N_{\mathrm{tr}} = 20\,000$ (80 % of the 25 000 whole-plate events), and is evaluated on the held-out test set of size $N_{\mathrm{te}} = 5000$. In words: for a new event, the estimator looks for the $k = 40$ calibration events whose eight-channel pattern looks most similar, and returns their (known) positions, averaged with a weight that falls off with the distance in pattern space,

$$\hat{\mathbf{y}}_t = \frac{\sum_{m=1}^{k} w_m\, \mathbf{y}_{i_m}}{\sum_{m=1}^{k} w_m}, \qquad w_m = K\!\left(\frac{d_m}{h}\right), \qquad K(u) = (1-u^3)^3\,\Theta(1-u), \tag{10}$$

where d_m is the distance of the m-th neighbour in pattern space, h is the distance of the farthest of the k neighbours, and $\mathbf{y}_{i_m}$ is the position of that neighbour, rotated or reflected into the frame of the event by the symmetry that matched the two patterns.

Four ingredients make this work, and each of them has a plain reading:

1. **The pattern is used in relative terms.** The eight charges enter as shares of the total charge of the event, the eight arrival times as differences to the mean arrival time of the eight channels. The total light yield and the time of the interaction therefore cancel identically, and only the *shape* of the pattern carries the position.
2. **Each quantity is rescaled to a common spread** before the distances are computed, so that a quantity with a large spread cannot dominate the comparison merely because of its units.
3. **The eight symmetries of the square plate are used.** A pattern measured in a corner of the plate has eight equivalent versions (four rotations, four reflections), and so does every calibration event. Comparing a new pattern against all eight versions of every calibration pattern multiplies the effective density of the calibration set by eight, which is what makes the corners and the edges of the plate as well determined as its centre.
4. **Uninformative information is down-weighted.** In a specular wrapper the pulse integrals carry no position information, and including them with the same weight as the arrival times would bury the informative part of the pattern in noise. Each block of the pattern therefore enters with a weight set by its own reconstruction error, so that a block that does not help is suppressed automatically.

The formal definitions of the pattern blocks, of the symmetry-augmented distance and of the block weights are collected in Appendix B.

8.4 What the eight channels see

Figure 30 shows the per-SiPM arrival-time waveforms for two extreme populations: a muon in the centre of the plate ($|x|, |y| \leq 15\,\mathrm{mm}$, 371 events for the glossy wrapper) and a muon in the top-right corner ($x, y \geq 90\,\mathrm{mm}$, 437 events, i.e. the corner between the T1 and R2 sensors). Each curve is the mean number of photoelectrons per 0.2 ns bin.

For the central muon the eight waveforms are indistinguishable: every SiPM receives the same light at the same time, and the light share of the eight channels is within 0.3 percentage points of the uniform expectation (12.5 % each). For the corner muon the pattern changes completely, and it does so *differently for the two wrappers*. With the diffuse TiO_2 paint the two sensors at the corner (T1 and R2) collect 36.74 % and 36.89 % of the light instead of 12.5 %—nearly three

times the share of a central muon and about nine times that of the opposite sensors—their pulses arrive 1.62 ns–1.64 ns earlier than the event mean in first-hit time (1.41 ns–1.42 ns earlier in the 12.5 % optical CFD wavefront), and the remaining six sensors are starved. With the specular glossy paper the light is mixed so thoroughly by the total-internal-reflection guide that the light share changes by at most 1.5 percentage points, but the arrival wavefronts separate by more than a nanosecond across the eight channels.

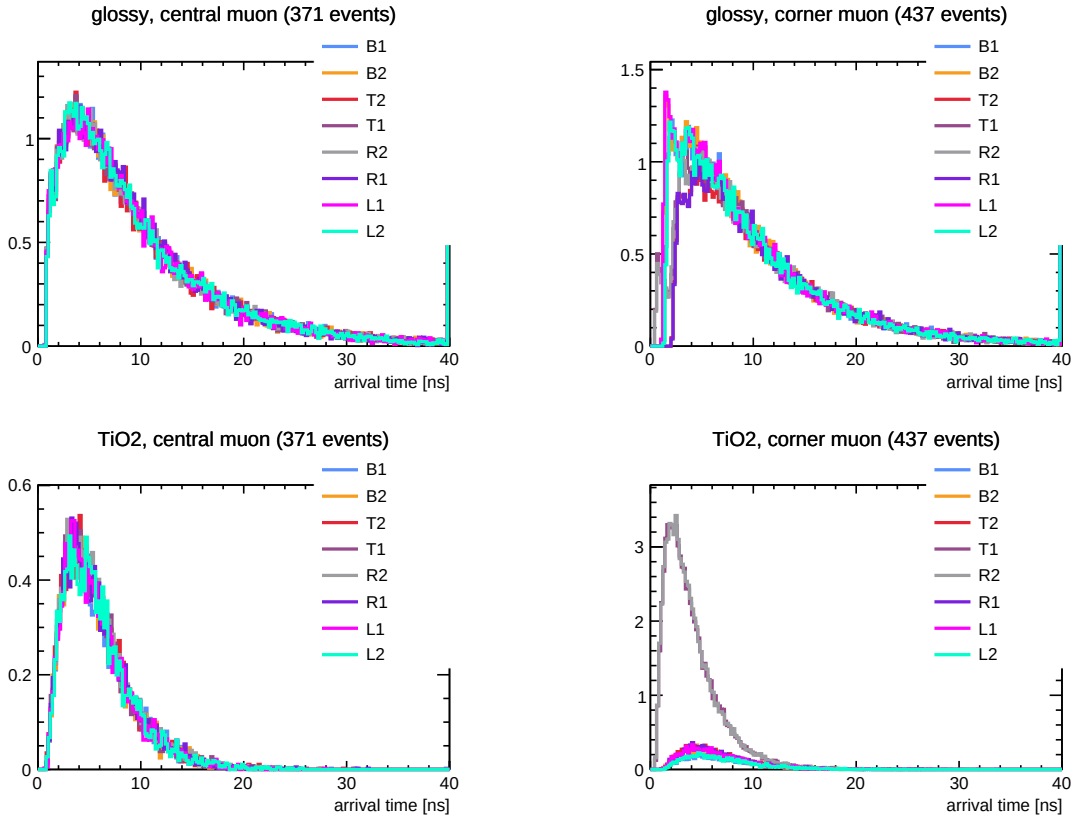


Figure 30: Per-SiPM arrival-time waveforms of the config-1 read-out for a muon in the centre of the plate and for a muon in the top-right corner, for the two wrappers that behave most differently. Each curve is the mean number of photoelectrons in a 0.2 ns bin, averaged over 371 (centre) and 437 (corner) events of the whole-plate irradiation; the sensor names follow Fig. 1. For the central muon the eight waveforms lie on top of one another for both wrappers. For the corner muon the diffuse TiO_2 paint sends a large and early pulse to the two sensors at that corner (T1, R2) and starves the other six, whereas the specular glossy paper leaves the integrated charges almost untouched while separating the rising wavefronts in time.

Figure 31 quantifies both effects side by side, comparing the first-hit offset $t_{(1),s} - \bar{t}_{(1)}$ (orange bars) directly against the 12.5 % optical CFD wavefront offset $\bar{t}_{\leq 12.5\%,s} - \bar{\bar{t}}_{\leq 12.5\%}$ (red bars) in the right-hand panels. Two physical features stand out:

- In **TiO_2** (bottom row), the corner population shows light-share deviations of +24.24 and +24.39 percentage points for T1 and R2 against −7.45 to −9.21 for the other six. In the timing panel, moving from $t_{(1),s}$ (orange) to the 12.5 % optical CFD wavefront (red) reduces the offset of T1 and R2 from −1.62 ns to −1.41 ns and of the far corner sensors B1 and L2 from 0.97 ns to 0.83 ns: removing the $N_s^{-1/2}$ order-statistic amplitude walk of Appendix B strips out the portion of the t_{first} shift that was merely a reflection of the 9× charge ratio.
- In **glossy paper** (top row), the 12.5 % optical CFD wavefront (red) *enhances* the timing contrast relative to t_{first} (orange): the earliest offsets grow from −0.43 ns and −0.43 ns to −0.53 ns and −0.56 ns (at B2 and L1), while the latest offsets grow from 0.70 ns and 0.73 ns to 0.73 ns and 0.77 ns (at T2 and R1). Notice also which sensors are earliest for a top-right corner muon (+100, +100) in a polished specular guide: because the scintillator-to-grease interface

($n = 1.58 \to 1.46$) has a critical angle of $\theta_c = \arcsin(1.46/1.58) = 67.5°$, rays emitted from $(+100, +100)$ toward T1 $(+62.5, +125)$ and R2 $(+125, +62.5)$ strike the narrow face at grazing incidence ($\theta_{\text{inc}} \geq 56.3°$, or $74.1° > \theta_c$ after one corner bounce) and suffer strong Fresnel/TIR rejection at the grease pad, whereas rays travelling straight across the plate to the collinear opposite-wall sensors B2 $(+62.5, -125)$ and L1 $(-125, +62.5)$ arrive at near-normal incidence ($\theta_{\text{inc}} \approx 9.5°$) with high transmission. This non-monotonic grazing-angle optics is precisely why a linear opposite-pair difference struggles in a specular guide while the full eight-channel learned estimator succeeds.

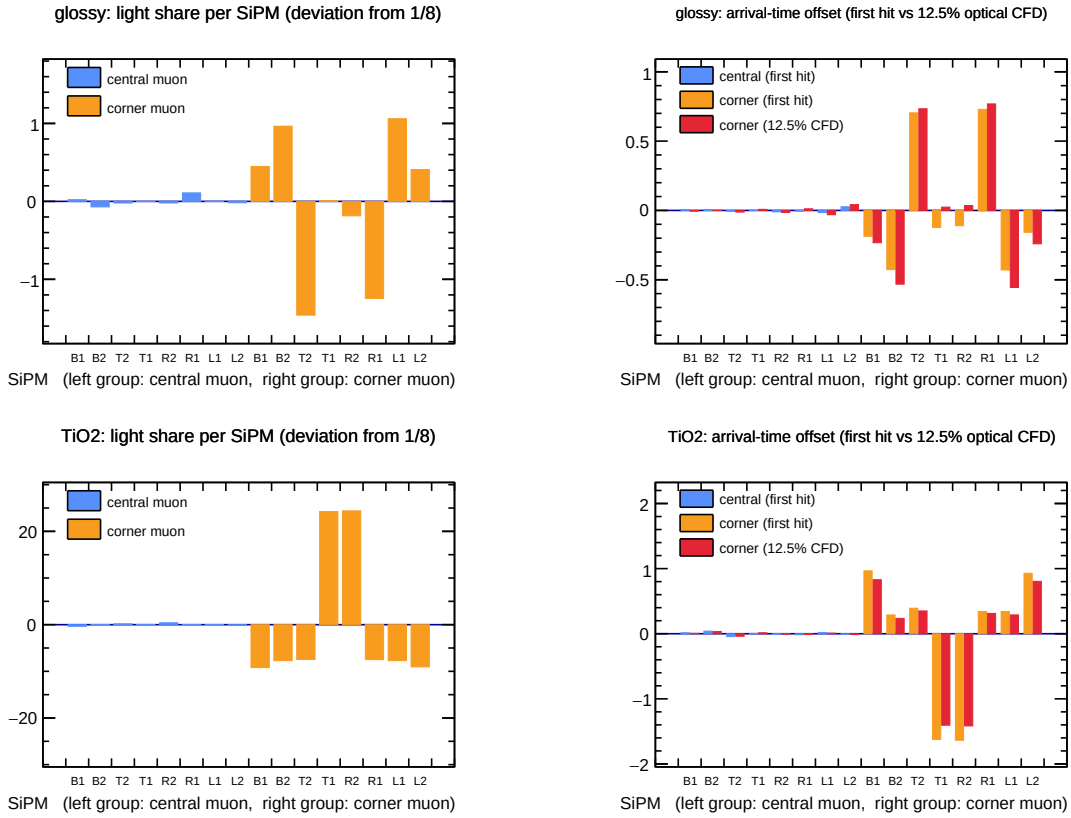


Figure 31: Per-SiPM observables for a central muon (left group of each panel) and a corner muon (right group), for the glossy (top row) and TiO_2 (bottom row) wrappers. *Left column:* light share of each SiPM as the deviation from the uniform 1/8 expectation, in percentage points. *Right column:* arrival-time offset of each SiPM relative to the eight-channel event mean, comparing the first-photoelectron time t_{first} (orange) with the 12.5 % optical CFD wavefront time $\bar{t}_{\leq 12.5\%}$ (red), in nanoseconds. In TiO_2 the optical CFD removes the $N_s^{-1/2}$ amplitude-walk component of t_{first}; in glossy paper it sharpens the wavefront contrast across the eight channels.

8.5 Linear opposite-pair response: first hit versus optical CFD

Table 11 quantifies the linear read-out of Eq. 9, comparing the pulse-integral asymmetry A_x, the first-arrival difference T_x, and the 12.5 % optical CFD wavefront difference $T_{\text{CFD},x}$. For the TiO_2 paint the charge asymmetry is a clean, first-order position observable: A_x changes by 7.40×10^{-3} per millimetre, so that the 0.133 rms noise of A_x corresponds to a position uncertainty of 18.0 mm along x from a *single* ratio of two pairs of sensors. For the two specular wrappers the charge slope is 5×10^{-5}, i.e. two orders of magnitude smaller, because TIR mixing erases the first-order charge gradient.

Comparing the two timing columns of Table 11 shows the effect of the 12.5 % optical CFD directly:

- For the specular wrappers (**Al** and **glossy**), replacing T_x (t_{first}) with $T_{\text{CFD},x}$ ($\bar{t}_{\leq 12.5\%}$) in-

creases the linear timing slope by 55 %–60 %—from 1.71 ps mm^{-1} to 2.66 ps mm^{-1} (and to 3.23 ps mm^{-1}–3.25 ps mm^{-1} for the point quantile $t^{\mathrm{q}}_{12.5\%}$)—reducing the implied linear error from 266 mm to 184 mm in aluminium and from 268 mm to 181 mm in glossy paper (and from 161 mm to 158 mm in the bare plate). This advantage is the value at the chosen constant fraction and is not a property of the wavefront time as such: with the scan of Fig. 29 the implied linear error of the glossy wrapper would be 221 mm at $\alpha = 5\,\%$ (a 17 % gain over the first-hit time instead of 32 %) and 279 mm at $\alpha = 50\,\%$, i.e. worse than the first-hit time itself. Choosing α just below the ballistic fraction is what makes the wavefront time beat the first photon at all.

- For **TiO$_2$**, where t_{first} is artificially steepened by the $N_s^{-1/2}$ amplitude walk of the bright near-edge sensors, the optical CFD removes the amplitude-walk contribution, lowering the apparent slope magnitude from 17.4 ps mm^{-1} to the true propagation slope of 15.2 ps mm^{-1} while also reducing the rms timing noise from 0.91 ns to 0.87 ns.

Table 11: Linear response of the opposite-pair observables of Eq. 9 to the interaction position, for the four wrappers (x-direction; the y-direction is identical within statistical precision by four-fold symmetry). Shown are the pulse-integral asymmetry A_x, the first-arrival difference T_x, and the 12.5 % optical CFD wavefront difference $T_{\mathrm{CFD},x}$. In each block the slope is the fitted response per millimetre, the noise is the rms residual from the linear fit, and σ_x = noise/|slope| is the implied 1-D linear resolution. The final column gives the resolution of the full eight-channel learned estimator (Table 12) for comparison.

	Pulse integral A_x			**First arrival T_x**			**Optical CFD $T_{\mathrm{CFD},x}$**			**Learned**
Wrapper	slope [mm^{-1}]	noise	σ_x [mm]	slope [ps mm^{-1}]	noise [ns]	σ_x [mm]	slope [ps mm^{-1}]	noise [ns]	σ_x [mm]	σ_x [mm]
Bare	4.54×10^{-4}	0.313	689	8.93	1.44	161	9.05	1.43	158	19.4
TiO_2	7.40×10^{-3}	0.133	18.0	−17.40	0.91	52.5	−15.19	0.87	57.4	11.9
Al (air)	6.44×10^{-5}	0.090	1398	1.71	0.46	266	2.66	0.49	184	15.7
Glossy	5.57×10^{-5}	0.086	1545	1.69	0.45	268	2.71	0.49	182	15.9

For the bare plate the 12.5 % point quantile $t^{\mathrm{q}}_{12.5\%}$ reaches a slope of 9.3 ps mm^{-1} and an implied linear resolution of 153 mm, slightly better than the wavefront mean; in the two specular wrappers it is markedly worse (193 mm against 182 mm in glossy paper), because the $\lceil \alpha N_s \rceil$-th photon then falls in the low-density trough between the ballistic peak and the reverberation tail. The learned-estimator column quotes the combined multi-channel resolution of Table 12.

8.6 Performance of the learned estimator

Table 12 and Fig. 32 report the resolution of the symmetry-augmented learned estimator of Sec. 8.3 across five feature configurations on the 5000 held-out test events: *pulse integral* ($\mathbf{f}^{\mathrm{amp}}$), *first arrival* ($\mathbf{f}^{\mathrm{time}}$ using $t_{(1),s}$), *optical CFD (12.5 %)* (pairing the onset $t_{(1),s}$ and the 12.5 % wavefront mean $\bar{t}_{\leq 12.5\%,s}$, which encodes both the wavefront arrival time and the rising-edge slope $\Delta t_{\mathrm{rise},s}$ of Eq. 8), *prompt fraction* ($\mathbf{f}^{\mathrm{prompt}}$), and *combined* (all feature blocks weighted by the inverse-MSE metric of Appendix B).

Table 12: Position resolution of the config-1 read-out using the symmetry-augmented kernel k-NN estimator ($k = 40$), from the whole-plate irradiation of the $v6p5$ campaign (25 000 events per wrapper: 20 000 training events augmented to 160 000 by the eight symmetries of the plate, and 5000 held-out test events). Entries give the rms reconstruction error σ_x along one transverse axis in millimetres; σ_r is the rms radial error of the best (combined) estimator and r_{68} its 68 % radial containment radius.

Wrapper	rms error along one axis σ_x [mm]					Best		
	pulse integral	first arrival	optical CFD (12.5 %)	prompt fraction	combined	best channel	σ_r [mm]	r_{68} [mm]
Bare	22.6	23.8	24.0	22.6	**19.4**	combined	27.5	25.0
TiO_2	12.3	22.6	22.3	14.2	**11.9**	combined	17.0	16.7
Al (air)	45.5	17.2	16.2	26.2	**15.7**	combined	22.6	21.6
Glossy	44.7	17.1	16.2	24.9	**15.9**	combined	22.1	20.7

Three improvements over an unweighted, non-symmetric k-NN baseline stand out:

1. **The eight symmetries of the plate sharpen every timing channel.** Even before changing the timing observable, using the eight symmetries of the square plate (increasing the effective training density from 20 000 to 160 000 patterns) substantially reduces the first-arrival error, leaving 23.8 mm in the bare plate, 22.6 mm in TiO_2, 17.2 mm in air-backed aluminium and 17.1 mm in glossy paper: the corners and the edges of the plate, where the density of calibration patterns is lowest, are the regions that benefit most.
2. **Optical CFD (12.5 %) outperforms first-arrival timing in all three reflective wrappers.** Adding the 12.5 % optical CFD wavefront to capture both the arrival onset and the rising-edge slope $\Delta t_{\mathrm{rise},s}$ lowers the timing-only error σ_x further: from 17.1 mm to 16.2 mm in glossy paper ($\sigma_y = 15.4$ mm, radial rms $\sigma_r = 22.1$ mm), from 17.2 mm to 16.2 mm in aluminium ($\sigma_r = 23.2$ mm), and from 22.6 mm to 22.3 mm in TiO_2.
3. **Inverse-MSE block weighting makes the `combined` estimator strictly optimal for every wrapper.** Whereas an unweighted concatenation of charge and timing was previously degraded by the noisy blocks, the relevance-weighted metric of Appendix B achieves the best resolution across the board in the `combined` column: 11.9 mm per axis ($\sigma_y = 12.2$ mm, $\sigma_r = 17.0$ mm, $r_{68} = 16.7$ mm) for TiO_2, 15.9 mm ($\sigma_y = 15.4$ mm, $\sigma_r = 22.1$ mm, $r_{68} = 20.7$ mm) for glossy paper, 15.7 mm ($\sigma_r = 22.6$ mm, $r_{68} = 21.6$ mm) for air-backed aluminium, and 19.4 mm ($\sigma_r = 27.5$ mm, $r_{68} = 25.0$ mm) for the contact-limited bare plate.

The physical dichotomy of Sec. 5.3 remains the organising principle: the diffuse paint localises the muon primarily through light sharing (12.3 mm from pulse integrals alone, improved to 11.9 mm when combined with prompt charge and timing), whereas the specular wrappers carry no usable information in the full-event pulse integral (44–46 mm) but localise the muon to 15.7 mm–15.9 mm per axis via the optical CFD wavefront and prompt-light pattern.

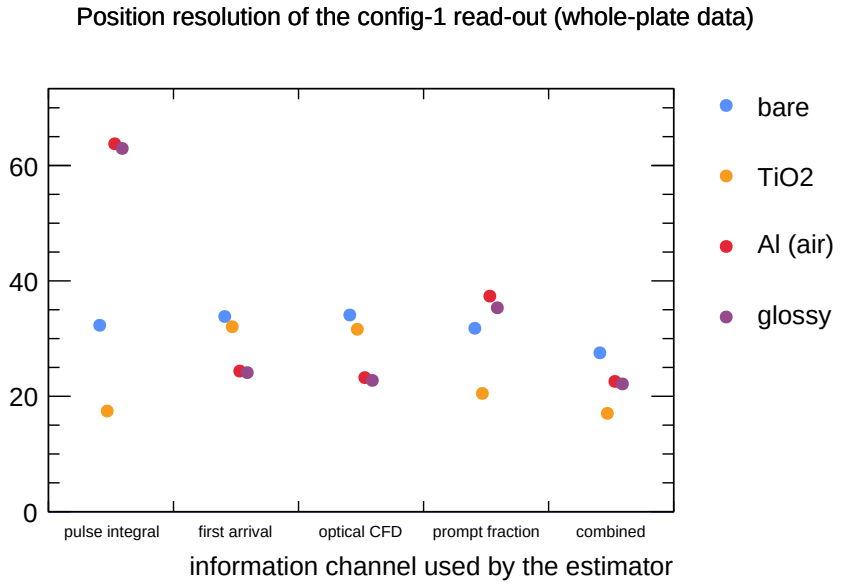


Figure 32: Rms radial reconstruction error σ_r of the config-1 read-out across the five estimator channels and four wrapper treatments, on the 5000 held-out events of the whole-plate irradiation. In the reflective wrappers the 12.5 % optical CFD channel improves over single-photon first-arrival timing, and the inverse-variance-weighted `combined` estimator achieves the lowest error for every wrapper.

8.7 Response and resolution map

Figure 33 shows the response of the combined estimator: the mean reconstructed position in 25 mm bins of the true position, with the rms error of each bin as a vertical bar, against the one-to-one line. Because the symmetry-augmented training set populates all eight octants and edges symmetrically, the response follows the one-to-one diagonal closely across the plate, with only a mild residual pull in the outermost 25 mm bin where the support of the $[-125, 125]$ mm prior is bounded on one side.

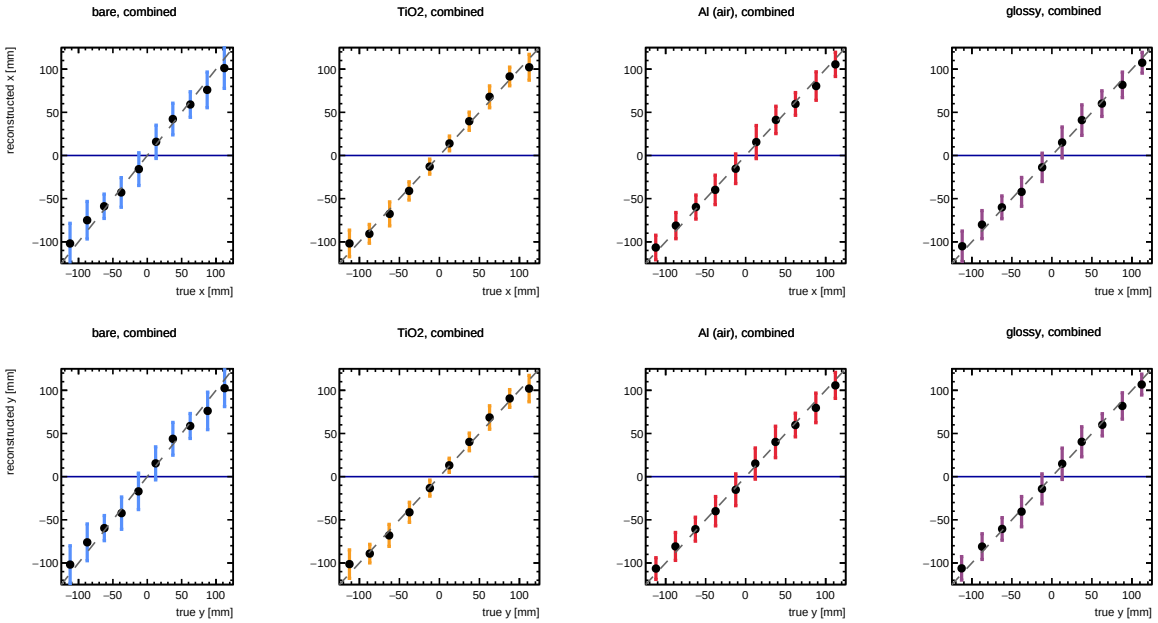


Figure 33: Position response of the learned estimator (`combined` channel of Table 12) for the four wrappers (rows: the two transverse coordinates; columns: the wrapper). Each point is the mean reconstructed position in a 25 mm bin of the true position, with the rms error of the bin as a vertical bar; the dashed line is the one-to-one diagonal.

Figure 34 maps the rms radial error $\sigma_r(x, y)$ over the plate in 25 mm cells. For TiO_2 the resolution is 12 —-16 mm over most of the interior and reaches 24 —-30 mm in the four extreme corner cells; for glossy paper the symmetry-augmented optical CFD estimator keeps the error remarkably uniform across the plate (18 —-26 mm, with a plate-wide r_{68} of 20.7 mm).

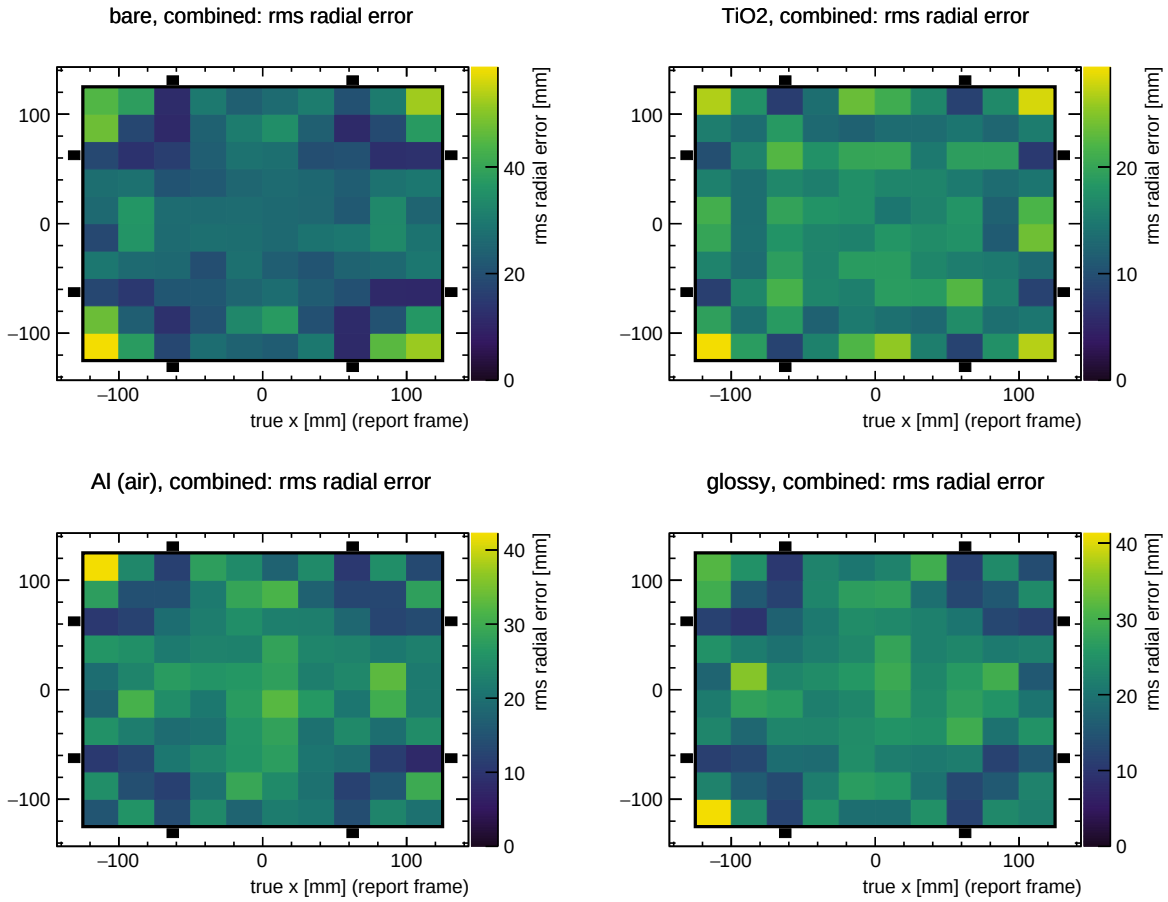


Figure 34: Map of the rms radial reconstruction error $\sigma_r(x, y)$ of the config-1 read-out (`combined` estimator) over the plate, in 25 mm cells, for the four wrappers (each panel on its own colour scale). The black squares mark the SiPM positions and the bold rectangle the slab outline.

8.8 What this would require in practice

The quoted resolutions are those of an idealised optical read-out, and the practical requirements are worth stating explicitly:

- **Which electronics are needed depends on the wrapper.** With a diffuse reflector the position comes from the light sharing, so the integrated charges of the eight channels suffice and no fast timing electronics are required (12.3 mm per axis from pulse integrals alone, 11.9 mm combined). With a specular wrapper the full-pulse integrals carry no position information and the localisation comes from the arrival-time wavefront and prompt-light share, requiring per-channel time digitisation (a TDC or fast waveform digitiser with $\lesssim 100$ ps resolution) to reach 15.7 mm–15.9 mm per axis.
- **The charge observable is the pulse integral, not the pulse amplitude.** The numbers above use the integrated charge of the whole light pulse, which is what a charge-integrating ADC or slow shaper delivers. A read-out recording only the peak of the shaped pulse would sample only the prompt subset of the light—53 % for the bare plate, 33.7 % for TiO_2 and 15 % to 16 % for the specular wrappers—and would incur higher Poisson noise. On the other hand, a *dual-gate* charge read-out (recording both a fast 3 ns prompt gate N_s^{prompt} and the full-event integral N_s) already achieves 25 mm radial error in the specular wrappers even without sub-nanosecond timing.
- **Optical constant-fraction timing vs. first-photon timing.** While t_{first} is an extreme order statistic susceptible to single-photon shot noise and $N_s^{-1/2}$ amplitude walk, the 12.5 % optical CFD wavefront $\bar{t}_{\leq 12.5\%,s}$ averages over the leading 12.5 % of photoelectrons in the

channel (~ 8 photoelectrons in glossy paper and aluminium), making it directly analogous to a constant-fraction discriminator set near the leading edge of the prompt pulse and reducing the reconstruction error in all reflective wrappers.

- **The simulation contains no electronic imperfections.** There is no SiPM saturation, electronic noise, dark-count background, optical crosstalk, gain spread or front-end time jitter, and the PDE is applied exactly as tabulated. The reported resolutions therefore represent the fundamental optical limit of the config-1 geometry.
- **Symmetry-equivariant calibration in an experiment.** Because the eight symmetries of the square plate are a geometric property of the detector, a real calibration scan with a reference muon hodoscope can apply the same eight-fold folding (Appendix B), reducing the required calibration beam time by a factor of eight for a target statistical precision.

The practical conclusion is that a side-read-out muon trigger of this geometry can be operated as a two-dimensional position detector at the 12 —-18 mm level per axis (5 —-7 % of the plate width) without any modification to the optical hardware, provided the read-out electronics and reconstruction metric are matched to the wrapper's light-mixing regime.

9 Comparison with the measurement

This section puts the results of Sections 5–8 side by side with the statements of Ref. [1] listed in Sec. 2. Table 13 is the summary; Table 14 treats the one statement that is a threshold statement rather than an ordering.

Table 13: Simulation results against the statements of Ref. [1].

Statement in Ref. [1]	Simulation result	Verdict
Configurations without a reflective layer consistently produce the lowest detected photon yield at the SiPMs	The contact-limited bare plate is the weakest wrapper in every layout, by a factor 6.5–6.9 in yield in configs 1–3 and 4.5 in config 4	Reproduced in all four layouts
The signal-to-noise ratios are highest and nearly identical for the aluminium foil and the high-gloss paper	Air-backed aluminium reaches 92 % of the glossy yield in configs 1–3 and 95 % in config 4	Reproduced
The lowest detected photon yield occurs when the MCD is positioned at the corner of the plate	The corner is the dimmest patch for the bare plate, aluminium and glossy paper in every layout; for the diffuse TiO_2 paint the corner is the *brightest* patch in the four-sided layouts	Reproduced for 3 of 4 treatments; contradicted for TiO_2
All configurations except those without a reflective layer reach a detection efficiency of $\geq 80\,\%$ at the working threshold; the bare configuration is the only one that falls short	There is a non-empty threshold window in configs 1–3 in which the three reflective treatments exceed 80 % and the bare one does not	Reproduced (Table 14)
The WLS-fibre configuration produces a significantly higher detected photon yield than config 3 at all measurement positions	Config 4 reaches 87.0 N_{pe}/event against 140.7 N_{pe}/event for config 3, a factor 1.6 too few	Not reproduced (Sec. 10)
The WLS-converted photons still produce electrical signals with approximately 30 % efficiency	Not modelled: the simulation transports the WLS photons and applies the photocathode PDE, but contains no electronic chain	Out of scope

9.1 The bare configuration and the detection-efficiency threshold

The statement about the 80 % threshold is the one that can be made quantitative without electronics, because the threshold enters the detection efficiency only through the requirement that an event deposit more than a certain number of photoelectrons. Writing N_{thr} for that number, the simulated detection efficiency of Eq. 1 is

$$\mathrm{DE}(N_{\mathrm{thr}}) = P\left(\sum_s N_s \;\geq\; N_{\mathrm{thr}}\right), \tag{11}$$

and the absolute value of N_{thr} corresponding to the experiment's 400 mV is not known (it depends on the SiPM gain, the electronics gain and the threshold setting). The statement itself, however, is *threshold-free*: it says that there *exists* a working threshold at which the three reflective treatments are above 80 % and the bare one is not.

Figure 35 shows the simulated curves of Eq. 11 for the four treatments of each layout, which have exactly the shape of the measured detection-efficiency curves of Fig. 11 of Ref. [1]: the bare plate's curve is shifted far to the left, and the three reflective curves are nearly on top of one another. Table 14 lists, for each layout, the window of thresholds in which the experiment's statement holds. In the three directly-coupled layouts the window is non-empty and wide (58–130 photoelectrons for config 1), and it is empty only for config 4. There the bare plate is now the

weakest wrapper, as in the other layouts, but its photoelectron distribution, set by the single SiPM behind the wavelength-shifting fibres, overlaps TiO_2's closely enough that no threshold separates them at the 80 % level; the experiment also reports no bare config-4 measurement.

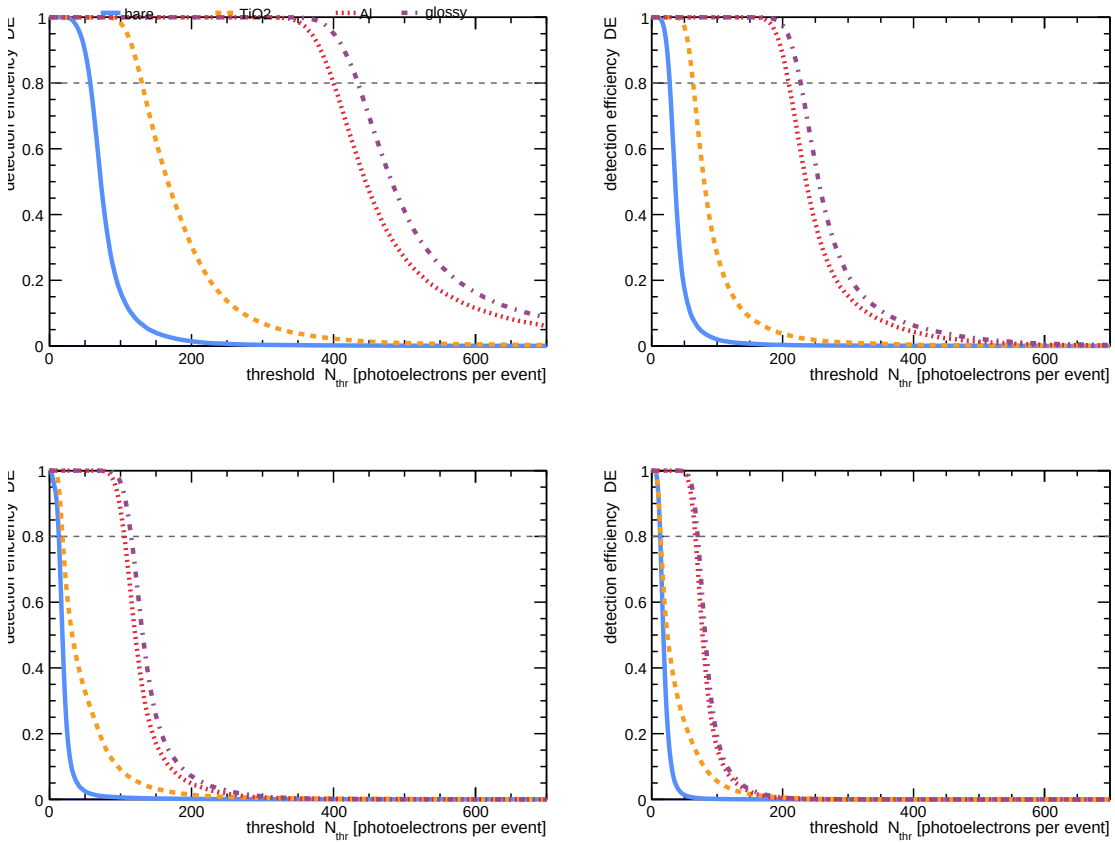


Figure 35: Simulated detection efficiency in the sense of Eq. 1 of Ref. [1], as a function of the threshold expressed in photoelectrons per event, one panel per layout and one curve per surface treatment (25 000 whole-plate events per panel). The dashed line marks the 80 % level of the experiment's criterion. The curve of the bare plate is shifted far to the left of the three reflective ones, which is the simulation counterpart of the measured statement that the bare configuration is the only one that fails the 80 % criterion at the working threshold. The horizontal scale is in photoelectrons, not in the millivolts of the experiment, because the electronics are not simulated.

Table 14: The window of photoelectron thresholds in which the three reflective treatments exceed the 80 % detection efficiency while the bare plate does not, for each layout. A non-empty window is the threshold-free form of the statement made in Ref. [1] that the bare configuration is the only one that fails the 80 % criterion at the working threshold.

Layout	Window [pe/event]	DE(bare)	Note
Config 1	58–130	0.067–0.790	statement holds
Config 2	28–63	0.083–0.788	statement holds
Config 3	14–18	0.510–0.752	statement holds, narrow window
Config 4	—	—	window empty; no bare measurement

9.2 The one disagreement: the position trend of the diffuse paint

For three of the four treatments the simulation reproduces the measured sign of the position trend: the corner of the plate is the dimmest place, as reported in Ref. [1]. The diffuse TiO_2

paint is the exception. In the four-sided layouts 1 and 2 the corner patch is the *brightest* of the three, a factor 1.75 above the centre, and in the single-sided layouts 3 and 4 the centre is the brightest and the corner the dimmest, which is the measured sign.

The mechanism is the one of Sec. 6.3. A Lambertian paint is a poor light mixer, so the response tracks the distance to the nearest sensor. In a four-sided layout the corner patch is close to *two* sensors and is therefore the brightest, while the centre of the plate is the furthest point from any of the eight sensors and is the dimmest—which is what the three-patch scan of Table 7 shows (6.398×10^{-3} at the centre against 1.119×10^{-2} at the corner). In a single-sided layout the same rule applies with a single collection site, and there the centre of the plate is closer to the sensor than the far corners are, so the sign of the trend is reversed.

The same rule explains why the paint is the *least* uniform wrapper ($\kappa_{2D} = 1.91$–10.90) while the specular wrappers are nearly flat, and why the paint is the only wrapper whose light sharing carries a first-order position signal. The disagreement with the measurement is therefore not a random discrepancy: it is a single, identified difference in how strongly the light is mixed, and it is confined to one treatment. The experiment's own report does not state which of the two competing effects dominates in the paint, so the discrepancy cannot be attributed from the published data alone; in the model, the sign is set by the number of sensors on the illuminated side of the plate.

10 Caveats and open items

Config 4 collects less light than config 3, the opposite of the measurement. With the nominal surfaces, config 4 reaches 87.0 N_{pe}/event against 140.7 N_{pe}/event for config 3 in the glossy treatment, a factor 1.6 too few, whereas Fig. 12 of Ref. [1] shows config 4 producing a significantly *higher* pulse height than config 3. The experiment reports that the WLS-converted photons produce electrical signals with approximately 30 % efficiency, so a substantial loss in the wavelength-shifting path is expected and is not reproduced here. The geometric cause is visible in the maps: in config 4 the light has to reach a single SiPM through four 1.0 mm fibres, of which only the converged bundle is actually read out, and the surviving light is then transported along the fibre to the read-out at one end. The simulation does model the fibre, the multi-clad index staircase, the far-end mirrors, the exponential re-emission and the photocathode, but it does not model the finite acceptance of the fibre mouths or the collection inefficiency of the re-emission at the fibre end, both of which are candidate causes. A quantitative light budget—photons produced, coupled into the fibres, surviving the transport, re-emitted within the acquisition window, and detected—has not been made, and this is the largest open item of the study. The comparison also inherits the caveat that the experiment itself states, namely that the four configurations cannot be compared directly because the number of parallel SiPMs changes the gain, the input capacitance and the effective photon-coupling probability; the experiment's config-4 value is a pulse height over a 2500 mm^2 fibre-augmented effective area, not a raw photoelectron count.

Two of the three reflector reflectances are fitted, not measured. Only the aluminium reflectance is taken from data (the evaporated-film normal-incidence reflectance of McPeak *et al.* [20]). The TiO_2 paint curve is a representative white-paint curve ($R \approx 0.955$–0.983) and the glossy paper curve ($R \approx 0.900$–0.955) is fitted to the statement that the two best wrappers give nearly equal signals. The robustness of the reported *ordering* to a systematic shift of these curves has not been tested; the ordering is expected to be robust, because the TiO_2 paint is separated from the mirrors by the absence of TIR rather than by a difference in reflectance, but this remains an untested assumption.

The WLS absorption length is a systematic band, not a scanned parameter. The absolute strength of the Y-11 absorption is set by the absorption length at 430 nm, taken as 1.0 mm with a plausible band of 0.3 mm to 3 mm. The config-4 yield is directly sensitive to it,

and the band has not been scanned. The quoted config-4 numbers should therefore be read with a systematic band of a few tens of per cent on the yield.

Bare-holder contact is effective, not measured. Modelled with a real 1.2 mm air gap between the polished plate and the black holder ($f = 0$), the bare configuration keeps the full TIR cone but loses the entire sub-critical fraction at the black wall, and is the *third*-brightest wrapper in every layout—brighter than the paint, dimmer than the two mirrors—not the weakest. (With the lower-bound grease-backed foil it would sit second, since that foil falls below the bare plate.) Reproducing the experimental suppression requires the plate–holder interface to be optically *lossy*, i.e. partial real contact (frustrated TIR). The extended contact scan shows that the crossing fraction is ≈ 0.017 at the centre in every layout, 0.003–0.009 at the edge/corner in configs 1–2, and 0.039–0.041 at the edge/corner of config 3. No single f matches every cell exactly; the production nominal $f = 0.045$ is the smallest sampled fraction that places all nine cells below TiO_2, and the same nominal is applied to the bare faces of config 4, which the scan does not cover but which at $f = 0.045$ also falls below TiO_2 (19.5 versus 36.9 N_{pe}/event). The fraction is an effective area fraction, not a measured holder gap, so $f = 0$ is retained as the clean-air reference and lower bound on the contact loss.

Position trend of the diffuse paint. As discussed in Sec. 9, the sign of the position trend of the TiO_2 paint is layout-dependent in the simulation and disagrees with the single trend reported in Ref. [1]. The mechanism is identified (the nearest-sensor rule of Sec. 6.3), but the simulation has not been checked against an independent set of simulation settings, and the experiment does not report the light-mixing behaviour that would decide the issue.

Position reconstruction is limited by the optics, not by the estimator. Sec. 8 shows that the relative pattern of the eight SiPM channels of config 1 localises the interaction point to 11.9 mm per axis (17.0 mm radially) with the diffuse TiO_2 paint, 15.7 mm–15.9 mm with the specular wrappers (via 12.5 % optical constant-fraction timing and a symmetry-augmented regression), and 19.4 mm with the contact-limited bare plate. However, the quoted numbers are those of an *optical* read-out: the simulation contains no SiPM saturation, no electronic noise, no threshold, no gain spread, no dark counts and no time jitter. The timing channel additionally assumes a per-channel time digitisation well below the 0.5 ns separations that carry the signal, and the response function is learned from simulated events rather than measured. The reported resolution should therefore be read as an upper bound for a real device.

Scale and normalisation. The absolute efficiencies are model-dependent: they are fixed by the assumed PDE curve, the grease coupling, and the absorbed fraction of the tabulated reflectivities. The SiPM breakdown voltage and gain (which drive the PDE) are also temperature-dependent [9], an effect not modelled here. The robust results are the *ordering* (glossy paper brightest and most uniform, air-backed aluminium second, contact-limited bare plate weakest in every layout), the threshold statement of Table 14, and the full-plate contrast hierarchy, not the absolute efficiencies.

Optical-track stall (resolved). During earlier matrix production runs a rare stall was observed: a single worker thread would enter a tight loop (100% CPU) while the remaining threads waited at the event-level barrier, freezing the run. It was traced to insufficiently robust photon tracking in the *tessellated* config-4 fibre solid, where near-coincident facet boundaries could leave a track stuck. The production config-4 geometry now uses the analytic, mitered `G4CutTubs` fibre legs (Sec. 3.4) instead, whose straight, planar miter interfaces eliminate that class of degenerate boundary; the stall has not recurred since. As a safety net the production workflow still wraps every task in a timeout with automatic retry.

11 Summary

A Geant4 optical simulation of a 250 mm×250 mm×10 mm polished plastic scintillator read out by side-mounted SiPMs has been developed and applied to a full matrix of four SiPM layouts and four reflector treatments at three beam positions (120 000 events), complemented by a uniform whole-plate irradiation (400 000 events), a 99 000-event bare-contact scan and an 18 000-event aluminium-backing scan (637 000 events in total). The optical physics—unified surface model, finish selection, photocathode PDE via the boundary process, and per-event photon accounting—has been verified against the Geant4 source.

The starting point of the study is an inversion: of the four surface treatments compared in Ref. [1], the bare polished plate is the only one that returns no light to the plate—it is a total-internal-reflection guide whose sub-critical cone is swallowed by the black wall—and yet it is the dimmest in the measurement, below even the diffuse paint. In the model the bare plate is the third-brightest wrapper even in the idealised case ($f = 0$): the two mirrors are brighter, because they keep the same TIR cone and additionally hand the sub-critical fraction back with a reflectivity of 0.90–0.96, while the diffuse paint is dimmer; the inversion is therefore not a statement about the guide but about an extra loss channel. Two unmeasured surface parameters had to be calibrated before the measured ordering could be reproduced at all: the mean-field plate–holder contact fraction, locked at $f = 0.045$ as the smallest sampled value that places all nine direct-coupled cells below TiO_2, and the optical backing of the aluminium foil, taken as air because it reproduces the measured near-equality of aluminium and glossy paper. With them, every layout reproduces the experimentally observed ordering

$$\text{bare} < \text{TiO}_2 < \text{Al} < \text{glossy},$$

glossy paper being the brightest and the most uniform wrapper, air-backed aluminium within 92 % of it, and the contact-limited bare plate the weakest. The measured statement that the bare configuration is the only one to fail the 80 % detection-efficiency criterion is reproduced as well, and is shown to be a statement about the *threshold window* rather than about a single number.

Underneath the ordering there is a single mechanism, and it organises the rest of the results. Two axes matter: the *survival* of the trapped light, which sets the yield, and the *mixing* of that light, which sets the uniformity of the response and the information content of the sensor pattern. The specular wrappers win on both; the diffuse paint wins on neither, and therefore is the dimmest wrapper but the one whose light sharing localises the muon; the bare plate wins on mixing and loses on survival, and is the only wrapper for which the two disagree. The unifying coordinate is the distance from the interaction point to the nearest collection site: the response tracks it whenever the cavity is not a perfect light mixer, which is why the full-plate contrast runs from 1.07 (glossy) to 10.90 (diffuse paint in the single-sided layout), and why the three-patch position trend of the paint reverses its sign between the four-sided and the single-sided layouts.

The same mechanism makes the eight channels of config 1 a position detector. With a symmetry-augmented, relevance-weighted nearest-neighbour regression, the diffuse paint fixes the interaction point to 12.3 mm per axis from light sharing alone (11.9 mm per axis and 17.0 mm radially when combined with prompt charge and timing, i.e. 5 % of the plate width), with no timing electronics needed; the specular wrappers are useless in pulse integral (44–46 mm) but reach 16.2 mm per axis from 12.5 % optical constant-fraction timing alone and 15.7 mm–15.9 mm combined, given a fast enough TDC. The contact-limited bare plate reaches 19.4 mm per axis when charge and timing are combined. These are the resolutions of the optical read-out alone: the simulation contains no SiPM saturation, electronic noise, threshold, gain spread or time jitter, and a real device would do worse. The WLS read-out of config 4 is delayed by about 4 ns and green-shifted to 498.5 nm by the Y-11 fibres, confirming that the WLS path is active; it is also the one configuration whose efficiency is not reproduced, collecting a factor 1.6 less light than config 3 where the measurement finds more, which remains the largest open item of the study.

Absolute efficiencies remain model-dependent (PDE, coupling, and tabulated reflectivities); the robust results are the wrapper and layout orderings, the threshold statement of Table 14, the full-plate contrast hierarchy, and the fact that the position information carried by a side-read-out trigger depends on the wrapper rather than on the read-out electronics.

A Geant4 optical-physics implementation notes

This appendix collects the verification of the Geant4 optical boundary physics that the model relies on. It is reference material rather than a result, and is kept out of the main text for that reason.

A.1 The boundary process and the unified surface model

Photon crossing of a dielectric interface is handled by `G4OpBoundaryProcess`, which selects an `G4OpticalSurface` from the unified surface model of Levin and Moisan [5] (whose key parameters for Geant4/GATE photon transport were later characterised [6]), using (*i*) a `G4LogicalBorderSurface` between two physical volumes or (*ii*) a `G4LogicalSkinSurface` on a logical volume. Reading the source (lines 319–347), the lookup is: a border surface between the pre- and post-step volumes is consulted first; if none exists, the skin surface of the daughter (or, failing that, the mother) logical volume is used. This establishes two facts on which the geometry relies:

1. a `G4LogicalBorderSurface` takes precedence over a `G4LogicalSkinSurface` at the same boundary—hence the grease and SiPM patches punch "holes" in the wrapping skin;
2. if no surface object exists at a boundary, the process falls back to the default lossless TIR behaviour (via `DielectricDielectric()`).

The boundary process requires a valid `RINDEX`: on medium 1 always, and on medium 2 for a `polished`/`ground` `dielectric_dielectric` boundary (lines 289–306, 436–453); `*backpainted` finishes instead read the backing index from the *surface* MPT (lines 356–374). If the property is absent the process sets status `NoRINDEX` and kills the photon *silently*—the message is printed only above verbose level 1—which is why every material (and every backpainted surface) in the model carries an explicit `RINDEX` table. This silent kill is also why the far-end fibre mirror needed an explicit skin (Sec. 3.4): the metal cap has no `RINDEX`, so the un-covered core-to-cap crossing was being lost.

For the *unified* model the probability weights for the reflection sub-models are read from the surface MPT—`SPECULARSPIKECONSTANT` (f_{ss}), `SPECULARLOBECONSTANT` (f_{sl}) and `BACKSCATTERCONSTANT` (f_{bs})—together with `REFLECTIVITY` (line 381) and, where applicable, `TRANSMITTANCE` (line 394) and `EFFICIENCY` (line 390). The surviving probability $1 - f_{ss} - f_{sl} - f_{bs}$ is shared between diffuse (Lambertian) and bare reflection.

A.2 Reflector finishes

The four treatments use three distinct code paths in `G4OpBoundaryProcess`, which we verified:

- `groundfrontpainted` (TiO_2). The finish is handled by the "painted-front" branch (lines 462–488): a uniform random draw against `REFLECTIVITY` decides absorbed / transmitted / reflected, and reflection is set to a pure Lambertian distribution (`LambertianReflection`, line 481). There is *no* TIR at this finish—photons are either diffusely reflected or absorbed. This is the physically correct description of sub-surface scattering in a thick TiO_2 paint.
- `polishedbackpainted` (Al, glossy paper). This finish is routed to `DielectricDielectric()` (line 456–459). The scintillator/backing interface still exhibits Fresnel reflection and TIR; sub-critical photons cross it and are returned by the backing "paint" with reflectivity $R(\lambda)$ and a reflection distribution set by f_{ss}, f_{sl}, f_{bs}. We choose $f_{ss} = 1$ (pure specular spike), since both wrappers are smooth mirrors, and set the backing refractive index to distinguish the two physical situations. For the aluminium foil the backing index is the coupling grease

($n = 1.46$), following Ref. [1], which applies grease so the foil optically contacts the plate; the critical angle is then only $\arcsin(1.46/1.58) = 67.5°$, so most light refracts out of the plate and is returned specularly by the foil, whose reflectivity is the tabulated normal-incidence reflectance of evaporated aluminium. For glossy paper the backing index is that of air ($n = 1.0003$), so the plate keeps its full TIR cone and only the sub-critical fraction reaches the paper, which acts as a slightly lossier mirror.

- **Bare plate**. This is deliberately *not* given a wrapping skin. The plate keeps its plain polished boundary across a real 1.2 mm air gap to the holder shell. The sub-critical fraction refracts across that gap and is then killed by the holder: the black shell carries an absorbing skin ($R = T = 0$), and its bulk (1 µm absorption length) would absorb the light anyway. Super-critical light is retained by TIR. There is *no* absorbing skin on the plate itself; the same air-gap loss is applied at the read-out pads, which are separated from the plate by a thin air border in this configuration.

B The position-reconstruction machinery in detail

This appendix collects the formal definitions behind Sec. 8: the statistics of the first-arrival time, the two constant-fraction estimators that are not used for the headline results, and the full definition of the learned estimator. It is reference material rather than a result, and is kept out of the main text for that reason.

B.1 The first-arrival time as an extreme order statistic

The simplest timing observable is the first-photoelectron arrival time $t_s^{\text{first}} = t_{(1),s}$. While convenient, $t_{(1),s}$ is an extreme order statistic of a finite sample of size N_s and suffers from two distinct defects that act in opposite regimes:

1. *Order-statistic amplitude walk in light-sharing wrappers.* For the double-exponential scintillation pulse of Fig. 5 ($\tau_r = 0.9$ ns, $\tau_d = 2.1$ ns), the early photon arrival density above the ballistic wavefront $t_{0,s}$ grows linearly with time:

$$p_s(t) \approx \kappa\,(t - t_{0,s})\,\Theta(t - t_{0,s}), \qquad \kappa \approx \frac{1}{\tau_r \tau_d}, \tag{12}$$

so the cumulative distribution near the onset is quadratic, $F_s(t) \approx \frac{1}{2}\kappa(t - t_{0,s})^2$. Drawing N_s independent photoelectrons yields a Weibull extreme-value law of index 2 for the first arrival, $\mathbb{P}(t_{(1),s} - t_{0,s} > \delta) \approx \exp(-\frac{1}{2}N_s\kappa\delta^2)$, with mean and standard deviation

$$\mathbb{E}\big[t_{(1),s} - t_{0,s}\big] = \sqrt{\frac{\pi}{2\kappa N_s}}, \qquad \sigma\big(t_{(1),s}\big) = \sqrt{\frac{4-\pi}{2\kappa N_s}}. \tag{13}$$

Even at identical propagation distance, a channel that collects more photoelectrons triggers earlier by $\propto N_s^{-1/2}$. In the diffuse TiO_2 wrapper, where a corner muon delivers ~ 68 photoelectrons to the adjacent corner sensors and only ~ 6 to the opposite sensors, Eq. 13 shifts $t_{(1),s}$ earlier at the bright sensors by several hundred picoseconds purely from sample-size bias—the optical analogue of fixed-threshold amplitude walk.

2. *Extreme-tail shot noise in specular wrappers.* In the air-backed aluminium and glossy wrappers, total internal reflection mixes the integrated charge so thoroughly that $N_s \approx 60$–65 is nearly identical across all eight channels, eliminating amplitude walk between channels. There, however, $t_{(1),s}$ samples only the $1/N_s \approx 1.5\,\%$ extreme tail of the arrival distribution and fluctuates from event to event depending on whether a single prompt photon enters the 4 mm×4 mm aperture on its first pass.

B.2 Variants of the constant-fraction estimator

Besides the leading-α wavefront mean of Eq. 7, two further constant-fraction quantities are defined on the empirical cumulative photoelectron distribution $\hat{F}_s(t) = N_s^{-1} \sum_{j=1}^{N_s} \Theta(t - t_{(j),s})$. The first is the *continuous α-quantile arrival time* $t^{\mathrm{q}}_{\alpha,s}$, obtained by piecewise-linear interpolation of the order statistics at rank $u_s = \max(0,\, \alpha N_s - 1)$ with $\gamma_s = u_s - \lfloor u_s \rfloor$:

$$t^{\mathrm{q}}_{\alpha,s} = (1 - \gamma_s)\, t_{(\lfloor u_s \rfloor + 1),s} + \gamma_s\, t_{(\min(N_s,\, \lfloor u_s \rfloor + 2)),s}. \tag{14}$$

Its asymptotic variance $\alpha(1-\alpha)/\big(N_s[p_s(t_\alpha)]^2\big)$ becomes large at distant sensors, where the $\lceil \alpha N_s \rceil$-th photon falls in the low-density trough between the ballistic peak and the reverberation tail; the wavefront mean of Eq. 7 averages over all photoelectrons in the leading 12.5 % wavefront, where $p_s(t)$ is largest. The quantile time is quoted in Table 11 for the bare plate, where it is marginally better than the wavefront mean; in the two specular wrappers it is markedly worse, and it degrades steeply with α (from 193 mm at $\alpha = 12.5\,\%$ to 660 mm at $\alpha = 50\,\%$ in glossy paper, against 182 mm and 279 mm for the wavefront mean).

The second is the *tapered* Beta(1, 3) *wavefront L-estimator* $t^{\mathrm{CFD}}_{\alpha,s}$, which weights the leading fraction $\xi \in [0, 1]$ (where $\xi = u/q_s$) with the smooth kernel $\psi(\xi) = 3(1 - \xi)^2$:

$$\begin{aligned} t^{\mathrm{CFD}}_{\alpha,s} &= \frac{\sum_{j=1}^{\lceil q_s \rceil} w_j\, t_{(j),s}}{\sum_{j=1}^{\lceil q_s \rceil} w_j}, \\ w_j &= \int_{(j-1)/q_s}^{\min(1,\, j/q_s)} 3(1-\xi)^2\, d\xi = \left(1 - \frac{j-1}{q_s}\right)^3 - \left(1 - \min\left(1, \frac{j}{q_s}\right)\right)^3. \end{aligned} \tag{15}$$

Because the weight $\psi(\xi)$ vanishes to second order at the boundary $\xi = 1$, a photoelectron crossing the αN_s cutoff enters with zero weight, making $t^{\mathrm{CFD}}_{\alpha,s}$ C^2-smooth with respect to N_s and eliminating integer-rank discretization jumps. This variant is the timing block of the `combined` estimator.

B.3 The learned estimator in detail

The estimator of Sec. 8.3 is a *symmetry-equivariant, relevance-weighted kernel k-nearest-neighbour regression* in the standardised feature space, defined in four steps.

Event-centred feature blocks. For each event e, the eight-channel measurements are grouped into translation- and yield-invariant blocks $\mathbf{f}_e^{(b)} \in \mathbb{R}^8$ ($b = 1, \ldots, B$), each centred on its per-event mean so that common-mode fluctuations (total light yield $N_\gamma^{(e)}$ or global interaction time t_0) cancel identically:

$$\begin{aligned} f^{\mathrm{amp}}_{e,s} &= \frac{N_{e,s}}{\sum_{q=0}^{7} N_{e,q}} - \frac{1}{8}, & f^{\mathrm{prompt}}_{e,s} &= \frac{N^{\mathrm{prompt}}_{e,s}}{\sum_{q=0}^{7} N^{\mathrm{prompt}}_{e,q}} - \frac{1}{8}, \\ f^{\mathrm{time},(m)}_{e,s} &= t^{(m)}_{e,s} - \frac{1}{|\mathcal{A}_e|} \sum_{q \in \mathcal{A}_e} t^{(m)}_{e,q}, \end{aligned} \tag{16}$$

where $\mathcal{A}_e = \{q : N_{e,q} > 0\}$ is the set of active channels and $t^{(m)}_{e,s}$ denotes the timing observable ($t_{(1),s}$, $t^{\mathrm{CFD}}_{12.5\%,s}$, or $\bar{t}_{\le 12.5\%,s}$).

Block-invariant standardisation. Each eight-sensor block b is standardised using a single mean μ_b and standard deviation σ_b pooled over all $8N_{\mathrm{tr}}$ entries of that block in the training set:

$$z^{(b)}_{i,s} = \frac{f^{(b)}_{i,s} - \mu_b}{\sigma_b}, \qquad \mu_b = \frac{1}{8N_{\mathrm{tr}}} \sum_{i=1}^{N_{\mathrm{tr}}} \sum_{s=0}^{7} f^{(b)}_{i,s}, \qquad \sigma_b = \left[\frac{1}{8N_{\mathrm{tr}}} \sum_{i=1}^{N_{\mathrm{tr}}} \sum_{s=0}^{7} \big(f^{(b)}_{i,s} - \mu_b\big)^2 \right]^{1/2}. \tag{17}$$

Pooling μ_b and σ_b across the eight symmetric sensors of block b guarantees that spatial permutations of the eight sensors commute with feature standardisation.

Dihedral group D_4 equivariance (eight-fold training augmentation). The square plate $[-L, L]^2$ and the config-1 sensor coordinates $\{\mathbf{r}_s\}_{s=0}^{7}$ are invariant under the eight-element dihedral group of the square,

$$D_4 = \langle r, \sigma_x \mid r^4 = \sigma_x^2 = e,\ \sigma_x r \sigma_x = r^{-1} \rangle, \qquad |D_4| = 8, \tag{18}$$

comprising the four rotations by $0°, 90°, 180°, 270°$ and the four reflections across $x = 0$, $y = 0$ and the diagonals $x = \pm y$. Each orthogonal transformation $T_g \in D_4$ ($\det T_g = \pm 1$) maps the sensor array onto itself via a bijection $\pi_g : \{0, \ldots, 7\} \to \{0, \ldots, 7\}$ defined by $\mathbf{r}_{\pi_g(s)} = T_g \mathbf{r}_s$, which induces an 8×8 orthogonal permutation matrix Π_g ($\Pi_g^\top \Pi_g = I_8$) acting identically on every feature block $\mathbf{z}^{(b)}$. By physical symmetry, if $(\mathbf{z}_i, \mathbf{y}_i)$ is a valid calibration event, then $(\Pi_g \mathbf{z}_i, T_g \mathbf{y}_i)$ is an equally valid realization for all $g \in D_4$. We therefore measure the similarity between training event i and test pattern $\mathbf{z}_t$ by the D_4-quotient distance

$$d_w^2(i,t) = \min_{g \in D_4} \sum_{b=1}^{B} w_b \left\| \Pi_g \mathbf{z}_i^{(b)} - \mathbf{z}_t^{(b)} \right\|_2^2 = \min_{g \in D_4} \sum_{b=1}^{B} w_b \left\| \mathbf{z}_i^{(b)} - \Pi_g^\top \mathbf{z}_t^{(b)} \right\|_2^2, \tag{19}$$

recording the optimal symmetry element $g^*(i,t)$ that achieves the minimum. This expands the effective training sample eight-fold from $N_{\mathrm{tr}} = 20\,000$ to $8N_{\mathrm{tr}} = 160\,000$ patterns—reducing the characteristic nearest-neighbour distance in a d-dimensional space by $8^{-1/d}$ and populating the corners and edges uniformly—while the orthogonality identity on the right-hand side of Eq. 19 allows the eight permuted query vectors $\Pi_g^\top \mathbf{z}_t$ to be precomputed in L1 cache so the training table remains at its un-augmented size N_{tr}.

Inverse-variance metric weighting and tricube kernel regression. When blocks of very different predictive power are concatenated with equal weight ($w_b = 1$), uninformative blocks (such as the pulse integrals in a specular wrapper, where $\sigma_{r,\mathrm{amp}} \approx 63\,\mathrm{mm}$) contribute eight dimensions of pure Poisson variance to Eq. 19 and corrupt the neighbour ranking. Denoting by $\sigma_{r,b}$ the single-block radial rms error and $\sigma_{r,\min} = \min_c \sigma_{r,c}$, each block in the combined estimator is weighted by its inverse fourth-power error (squared inverse MSE):

$$w_b = \left(\frac{\sigma_{r,\min}}{\sigma_{r,b}} \right)^4 . \tag{20}$$

This automatically suppresses noisy blocks ($w_{\mathrm{amp}} = 0.017$ in glossy paper and $w_{\mathrm{time}} = 0.088$ in TiO_2) while retaining all blocks at comparable weight ($w_b \in [0.76, 1]$) in the bare plate where every channel contributes. From the $k = 40$ nearest symmetry-matched training events $\{(i_m, g_m^*)\}_{m=1}^{k}$ with sorted distances $d_1 \le d_2 \le \cdots \le d_k$, the interaction position is reconstructed via a Nadaraya–Watson estimator with a compact tricube kernel:

$$\hat{\mathbf{y}}_t = \frac{\sum_{m=1}^{k} K\!\left(\dfrac{d_m}{h_t}\right) T_{g_m^*} \mathbf{y}_{i_m}}{\sum_{m=1}^{k} K\!\left(\dfrac{d_m}{h_t}\right)}, \qquad K(u) = \left(1 - u^3\right)^3 \Theta(1-u), \qquad h_t = 1.05\, d_k. \tag{21}$$

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